Canon

X-07

HAND HELD COMPUTER

■ TECHNICAL GUIDE

■ PARTS CATALOG

ENGLISH

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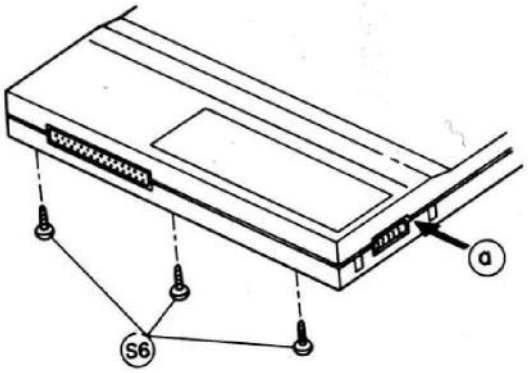
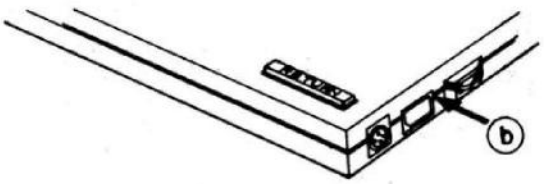
1. RELEASE of COVER

(1) Loosen 3 screws S6

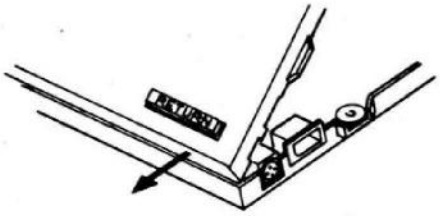
(2) Push ⓐ part of Lower cover and release the hook or Upper cover.



(3) Push ⓑ part of Lower cover.

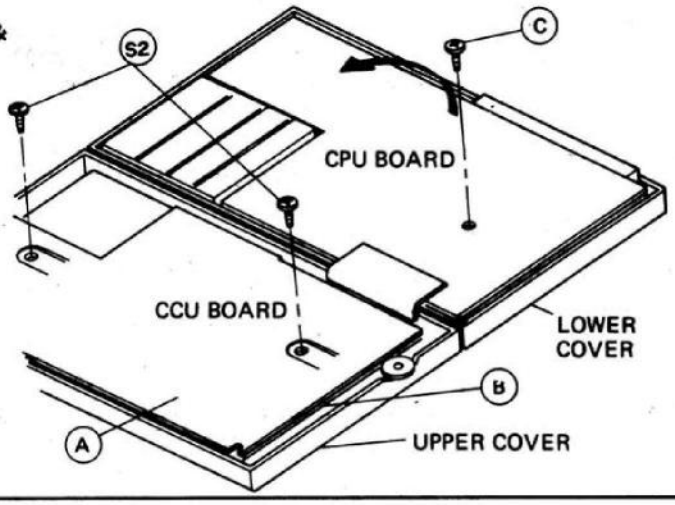


(4) Slide Upper cover.

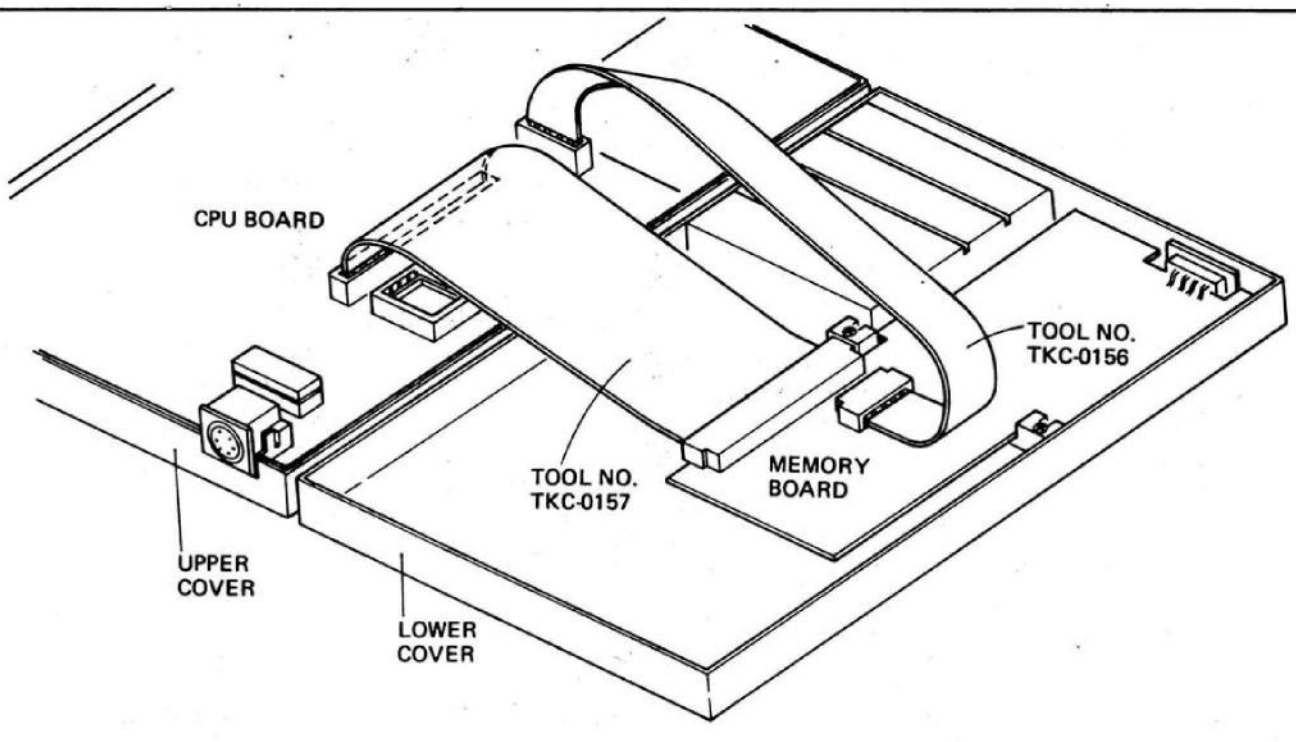


(5) Loosen two screws S2 and remove insulation plate Ⓐ & shield plate Ⓑ of CCU board.

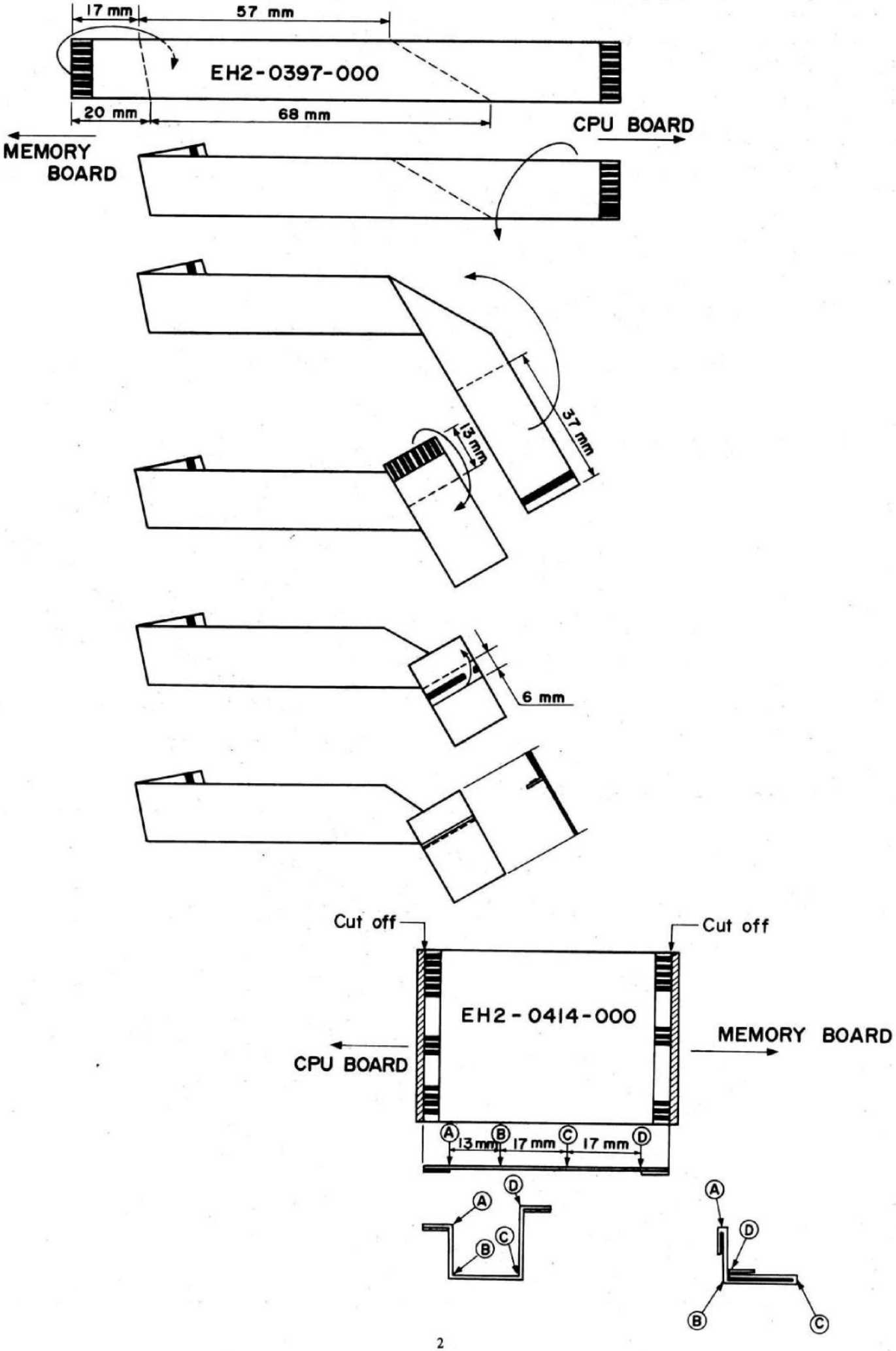
(6) Loosen screw Ⓒ and lift CPU board.



2. ATTACH the FLAT CABLES for REPAIR

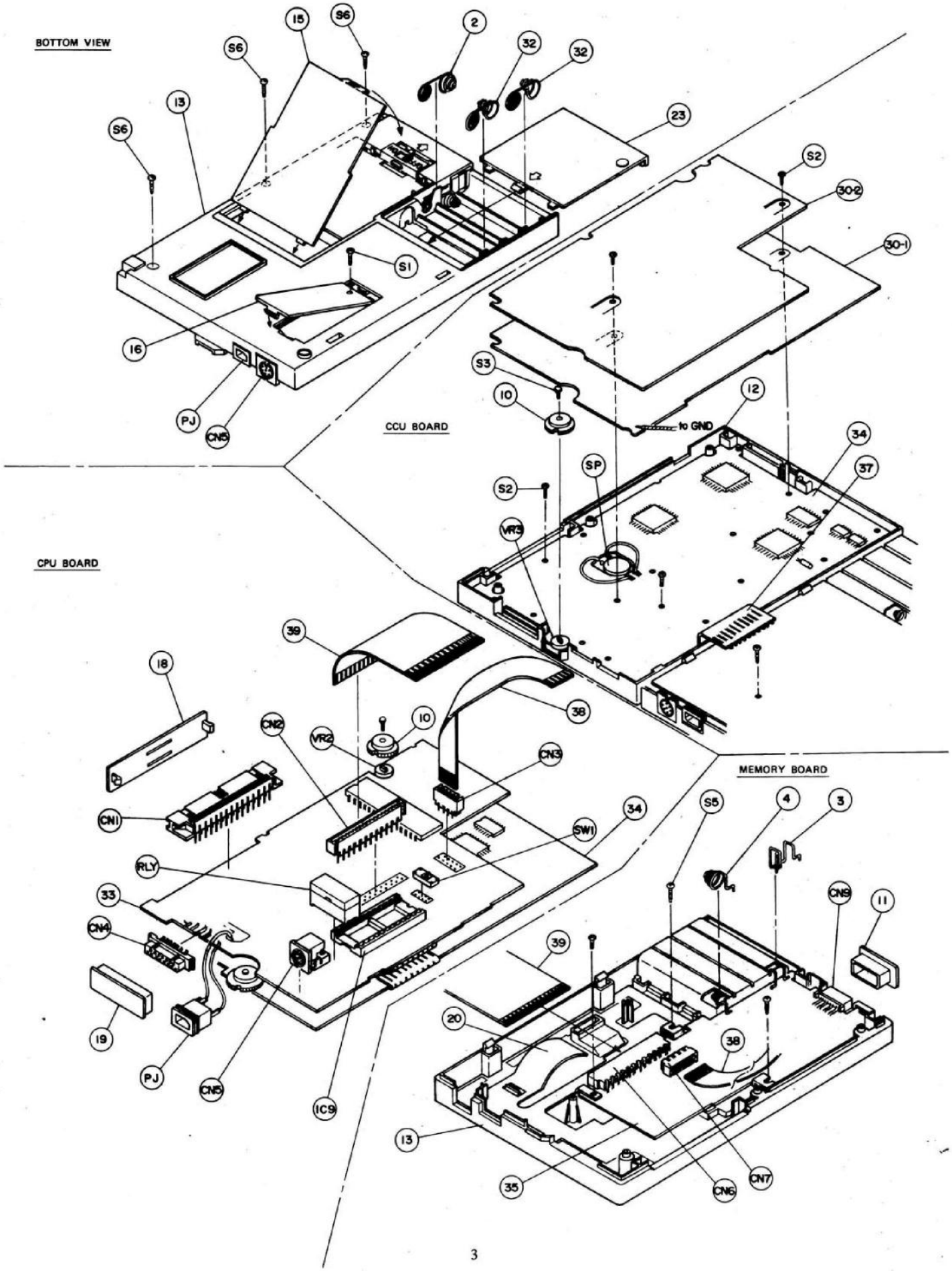


3. FOLDING OF FLAT CABLE

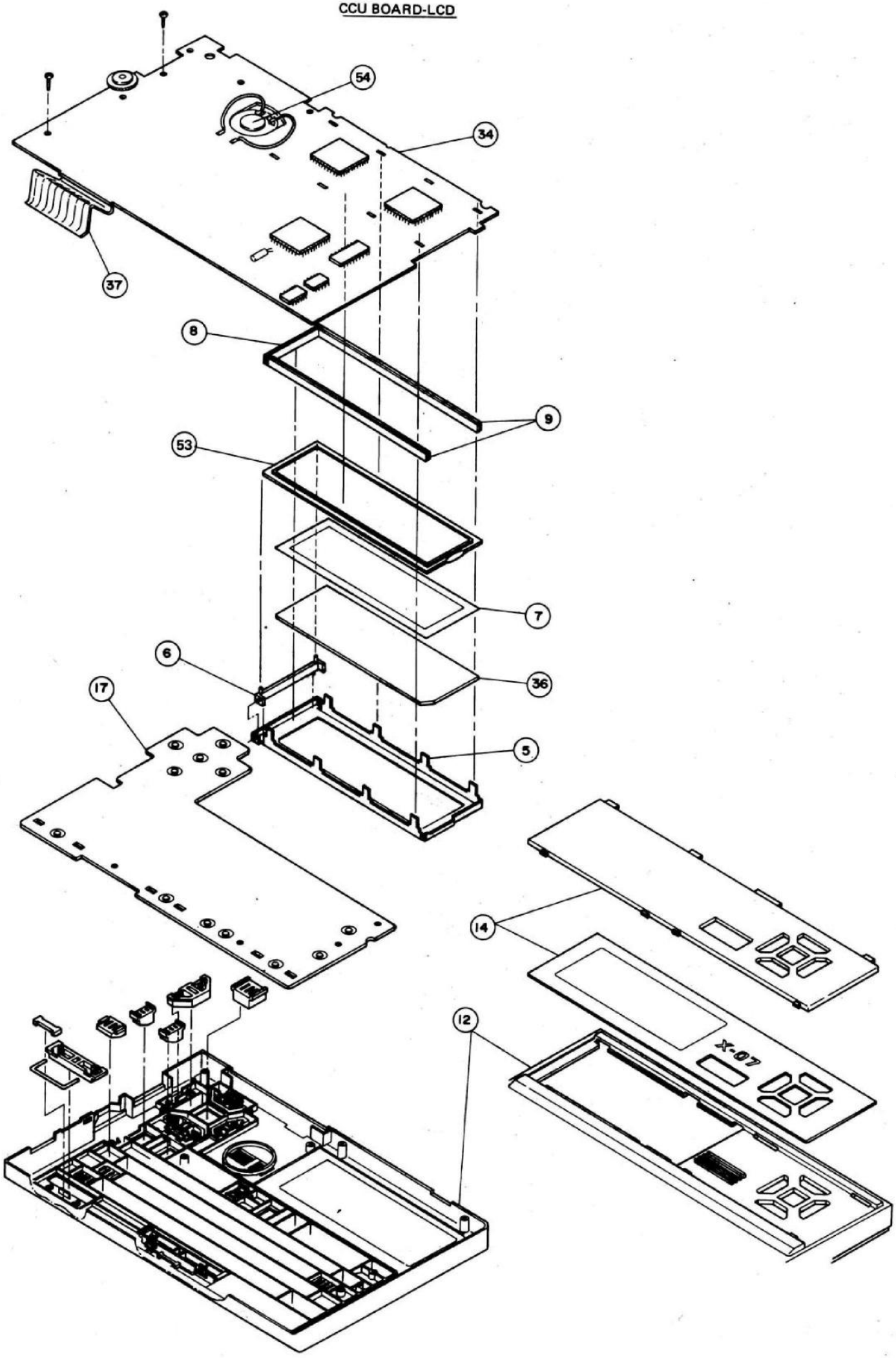


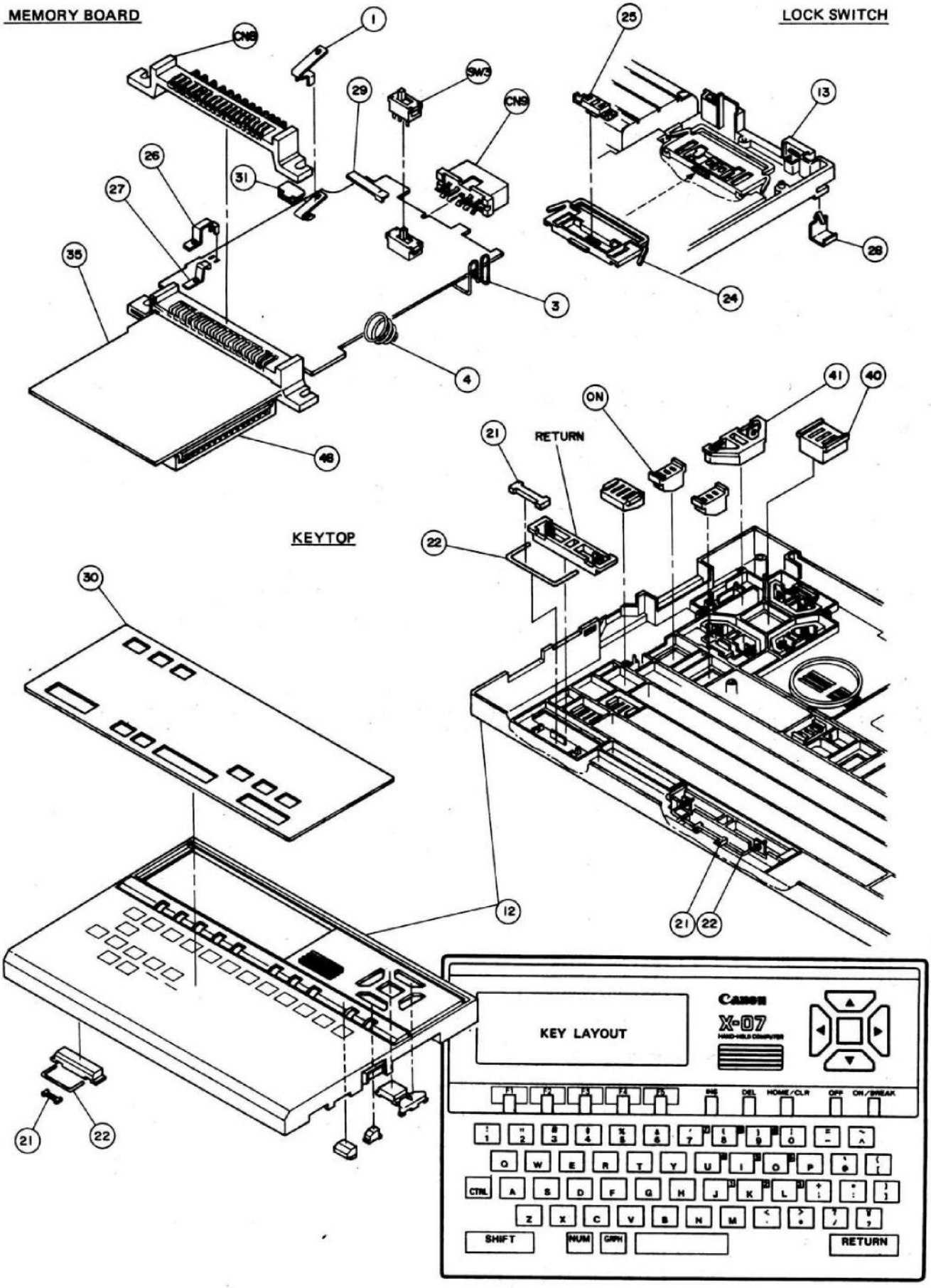
4. PARTS CATALOG

EXPLODED VIEW



CCU BOARD-LCD

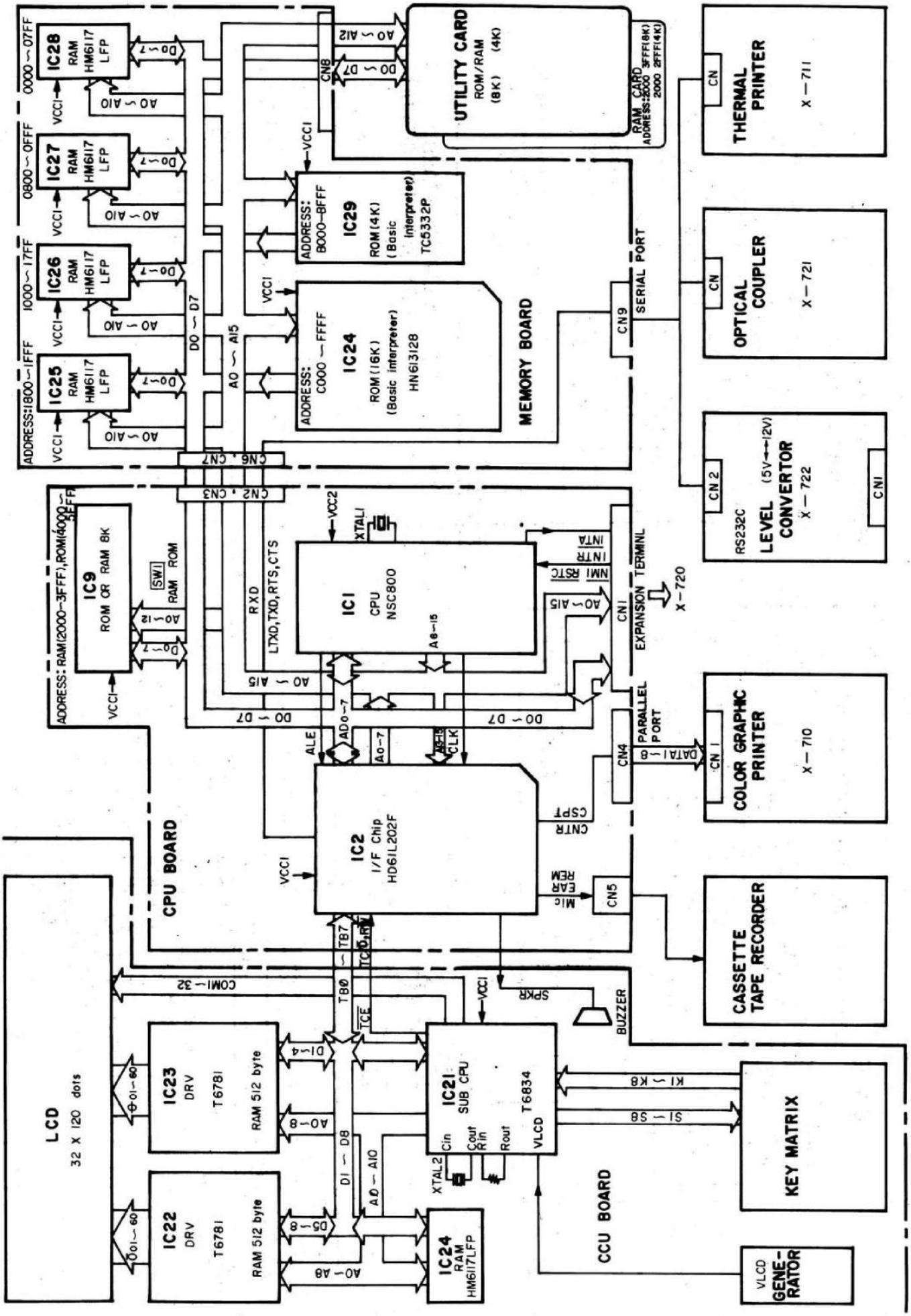



MEMORY BOARD
LOCK SWITCH
KEYTOP
RETURN
ON
KEY LAYOUT
Canon
X-07
SHIFT
RETURN

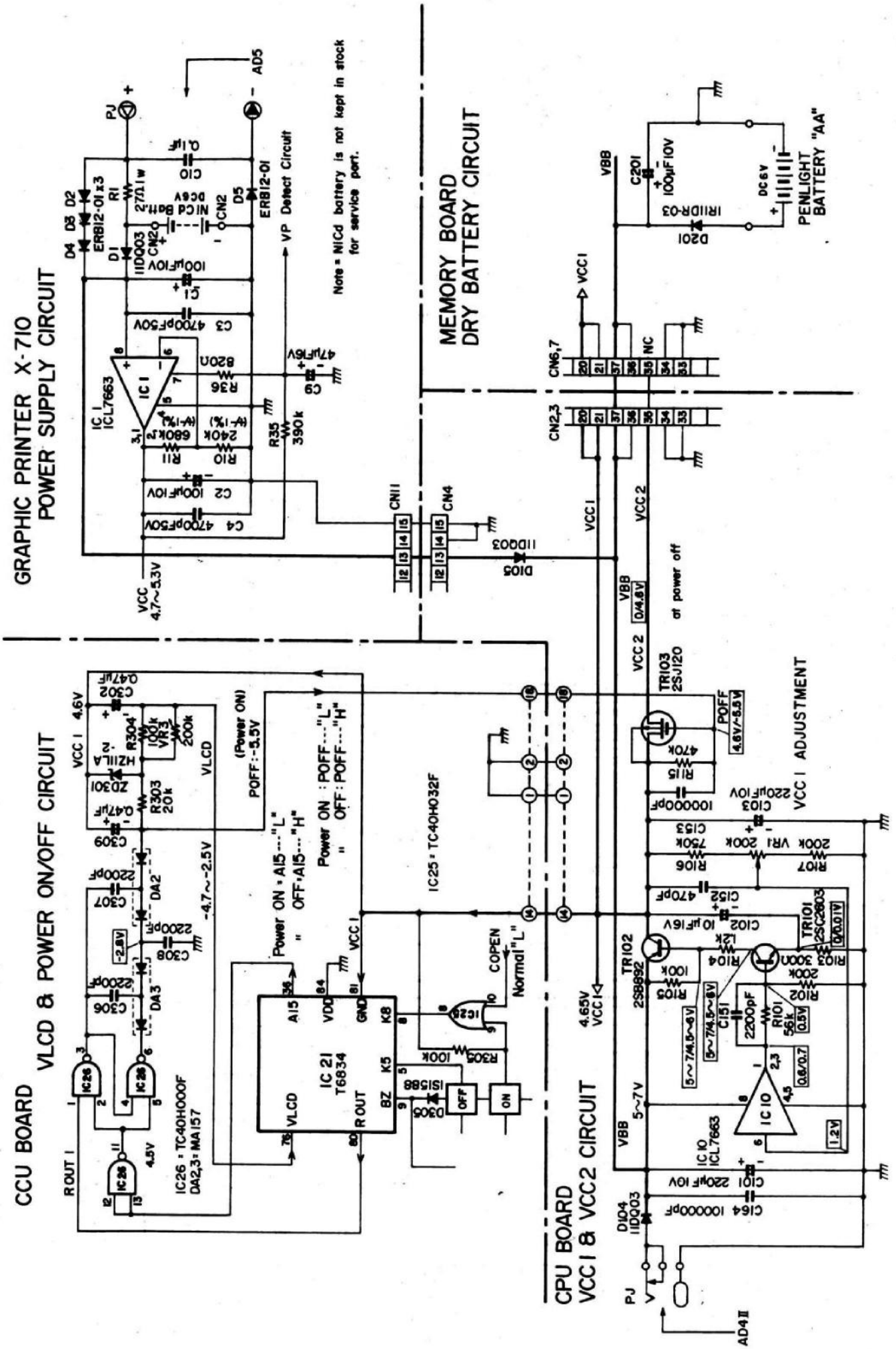
N	KEY NO.	PART NO.	Q'TY	DESCRIPTION		REMARKS	PRICE	
	1	EA1-2731-000	1	SPRING,RESET				
	2	EA1-3750-000	1	TERMINAL,BATTERY, C1				
	3	EA1-4560-000	1	TERMINAL (+)				
	4	EA1-4561-000	1	TERMINAL (-)				
	5	EA1-4858-000	1	HOLDER,LCD				
	6	EA1-4861-000	1	BAR,SET LCD				
	7	EA1-4862-000	1	SPACER,LCD				
	8	EA1-4863-000	1	CONNECTOR (A),CONDUCTIVE RUBBER				
	9	EA1-4864-000	2	CONNECTOR (B),CONDUCTIVE RUBBER				
	10	EA1-4865-000	2	KNOB,VARIABLE RESISTOR				
	11	EA1-4866-000	1	CAP,CONNECTOR				
*	12	EA1-5038-000	1	UPPER COVER				
*	13	EA1-5040-000	1	LOWER COVER				
*	14	EA1-5041-000	1	ACRYL,INDICATOR (DOM.)				
*	15	EA1-5042-000	1	COVER,UTILITY CARD				
*	16	EA1-5043-000	1	COVER,ROM/RAM SOCKET				
*	17	EA1-5044-000	1	RUBBER PLATE, CONDUCTIVE	KEYBOARD			
*	18	EA1-5045-000	1	CAP(A),CONNECTOR				
*	19	EA1-5046-000	1	CAP(B),CONNECTOR				
*	20	EA1-5047-000	1	PLATE,SHIELD				
*	21	EA1-5048-000	3	KEY,ARM, SHIFT,SPACE,RETURN				
*	22	EA1-5051-000	3	LEVER(B)				
*	23	EA1-5052-000	1	COVER,BATTERY				
*	24	EA1-5116-000	1	KNOB,LOCK				
*	25	EA1-5117-000	1	KNOB,SWITCH				
*	26	EA1-5120-000	1	LEAF SPRING(A)				
*	27	EA1-5121-000	1	LEAF SPRING(B)				
*	28	EA1-5122-000	1	CONTACT(A),GND				
*	29	EA1-5123-000	1	CONTACT(B),GND				
*	30	EA1-5156-000	1	PLATE,KEYBOARD				
*	30-1	EA1-5335-000	1	PLATE,SHIELD,CCU BOARD				
*	30-2	EA1-5336-000	1	PLATE,INSULATION,CCU BOARD				
*	30-3	EA1-5340-000	1	ACRYL,INDICATOR (EXPORT)				

N	KEY NO.	PART NO.	Q'TY	DESCRIPTION		REMARKS	PRICE	
*	31	EA1-5342-000	1	CONTACT				
*		EA1-8082-000	1	HARD CASE,X-07				
	32	EA9-0119-000	2	TERMINAL,BATTERY(C2)				
*	33	EH1-0669-000	1	PCB,CPU BOARD				
*	34	EH1-0670-000	1	PCB,CCU BOARD(KEYBOARD)				
*	35	EH1-0671-000	1	PCB,MEMORY BOARD				
	36	EH2-0387-000	1	ACRYL,FILTER,LCD				
*	37	EH2-0395-000	1	FLAT CABLE,19P				
*	38	EH2-0397-000	1	FLAT CABLE,12P				
*	39	EH2-0414-000	1	FLAT CABLE,34P				
*	40	ES2-2185-000	1	KEYTOP,GRAY,F6				
*		ES2-2186-000	1	KEYTOP,SPACE				
*		ES2-2187-000	1	KEYTOP,SHIFT				
*		ES2-2188-000	1	KEYTOP,RETURN				
*	41	ES2-2189-000	4	KEYTOP,CURSOLE				
*		ES2-2190-000	5	KEYTOP,BLUE,F1～F5				
*		ES2-2266-000	4	KEYTOP,LIGHT GRAY,INS,OFF,DEL,CLR				
*		ES2-2267-000	1	KEYTOP,BROWN,ON/BRK				
*	A	ES9-5202-000	1	KEYTOP SET(EXP.),1,2,3,4,5,6,A BCDEFGNQRSTVWXYZ,CTRL				
*	B	ES9-5203-000	1	KEYTOP SET(EXP.),7,8,9,0,HIJKLMOPU ?,+,*,=,<,>, ¥				
*		ES9-5204-000	1	KEYTOP SET(EXP.),NUMx12,GRPHx12				
*	↑	ES9-5205-000	1	KEYTOP SET(DOM.),1,2,3,4,5,6,A BCDEFGNQRSTVWXYZ,CTRL				
*	□	ES9-5206-000	1	KEYTOP SET(DOM.),7,8,9,0,HIJKLMOPU ?,+,*,=,<,>, ¥				
*		ES9-5207-000	1	KEYTOP SET(DOM.),カナx12,GRPHx12				
	S1	XA9-0113-000	1	SCREW, TAPPING M2x5, +/-				
	S2	XA9-0154-000	19	SCREW, TAPPING M2x4				
	S3	XA1-7140-305	2	SCREW 1.4x3				
	S4	XB3-1260-607	2	SCREW, TAPPING 2.6x6				
	S5	XB3-1261-007	2	SCREW, TAPPING 2.6x10				
	S6	XB3-1261-207	3	SCREW, TAPPING 2.6x12				

5. BLOCK DIAGRAM

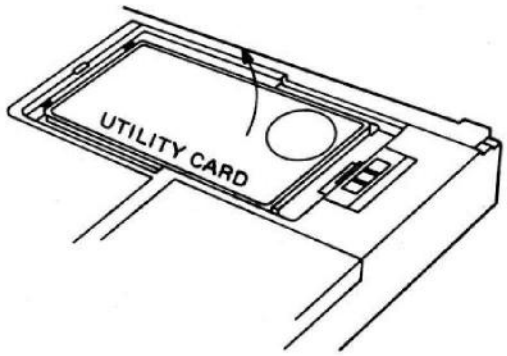


6. POWER SUPPLY CIRCUIT

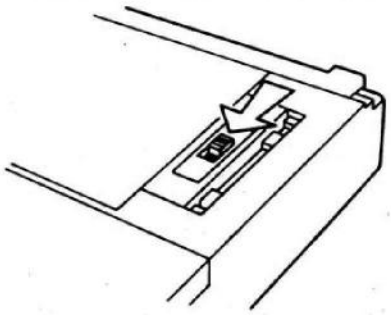


7. SELF-CHECK OF CCU

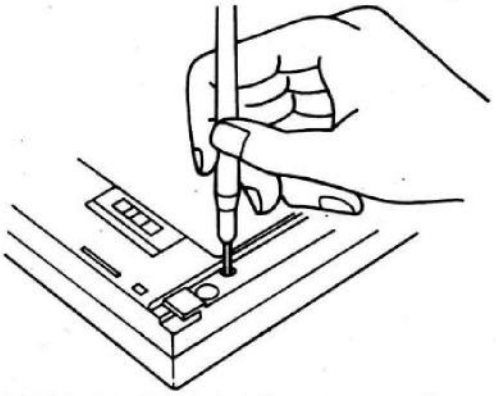
1. Remove the utility card.



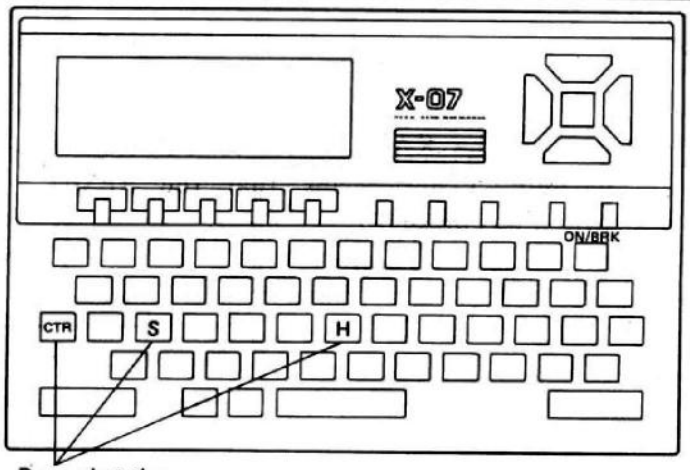
2. Set the LOCK switch of utility card to ON.



3. Press the RESET switch.



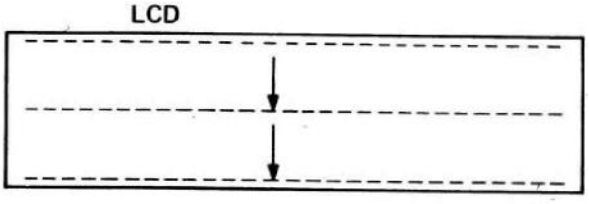
4. Press and hold down the CTRL , S , and H keys, then press the ON key. (If the ON key is pressed too long, an SRES signal is output and initial clearing is performed, disabling self-check.)



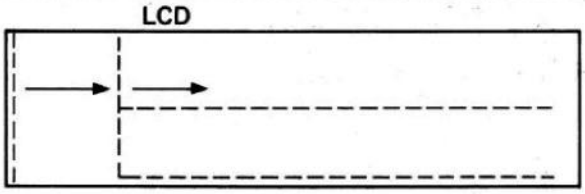
Pressed at the same time.

Press the ON/BREAK key for a little while, while holding the CTRL , S , and H keys down.

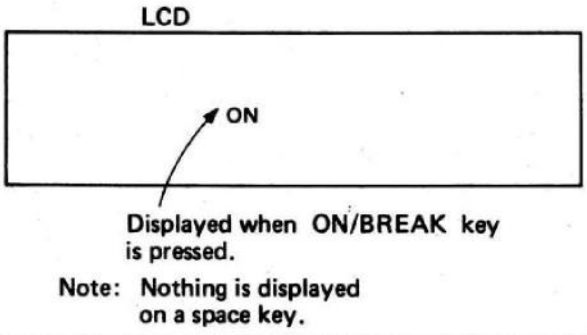
5. Check the display. 1
   Check of sub-CPU on CCU board, driver, and display is performed automatically.



6. Check of display 2



7. Nothing is displayed on the LCD after the display check is completed.
   * Check key input.



8. Self-check of the CCU board is released when the RESET switch on the bottom of the body is pressed.

9. When self-check is possible, the CCU board is not faulty. If self-check is impossible, repair the CCU board referring to the CCU BOARD flowchart.

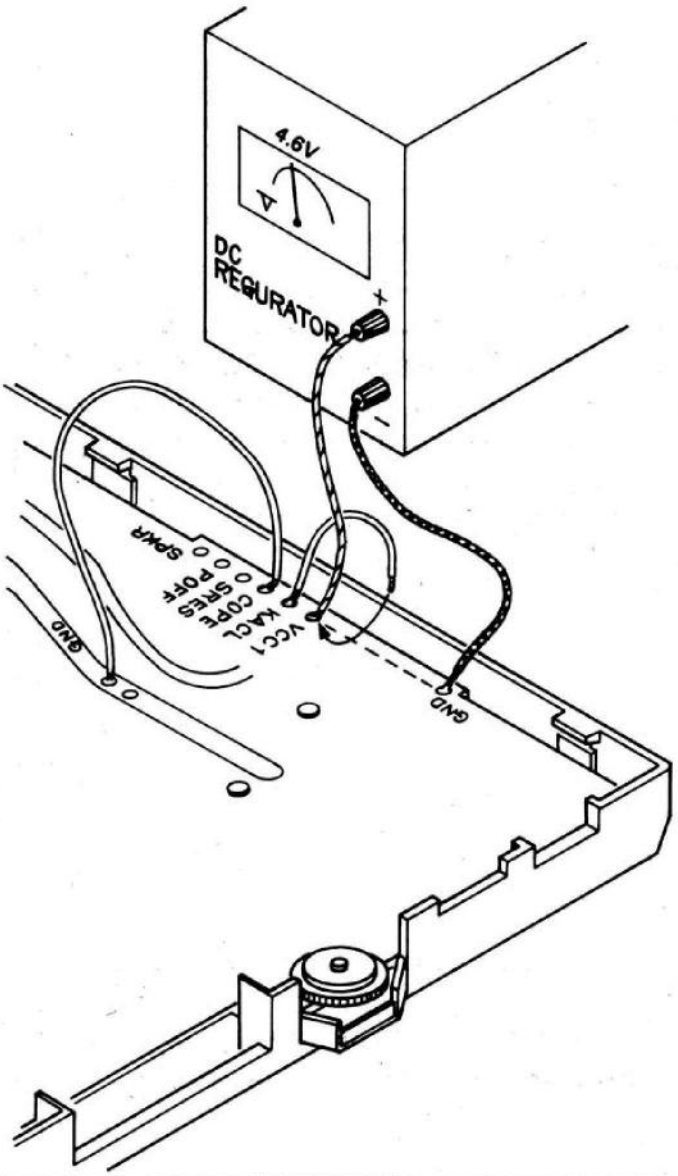
[Independent self-check of CCU board]

(1) Apply +4.6 V to the VCC1 terminal.

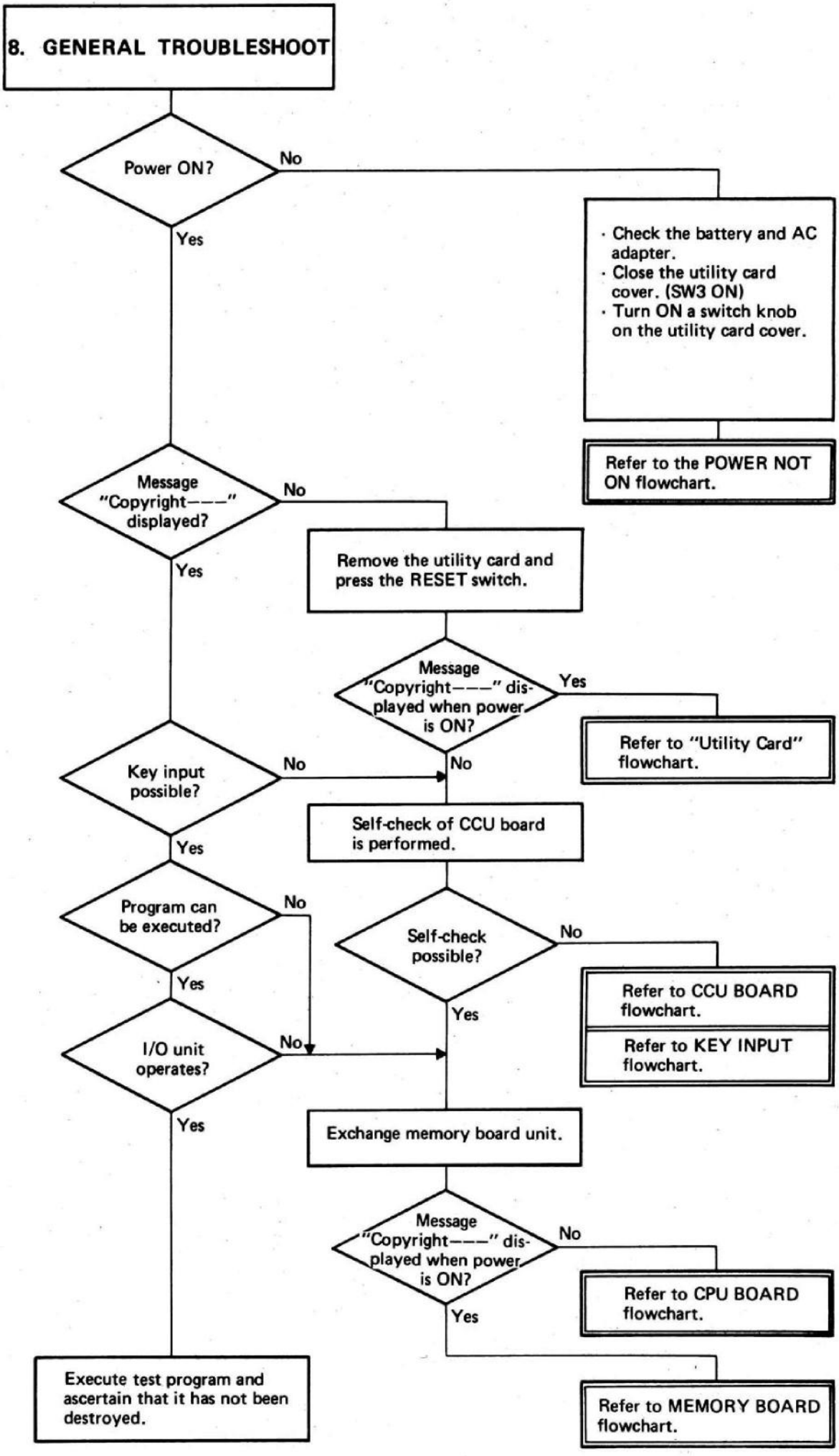
(2) Connect the COPEN signal to GND.

(3) Contact the KACL signal to VCC1, then separate. (This means that the RESET switch was pressed.)

(4) Execute items 4~9 of the self-check successively.



8. GENERAL TROUBLESHOOT



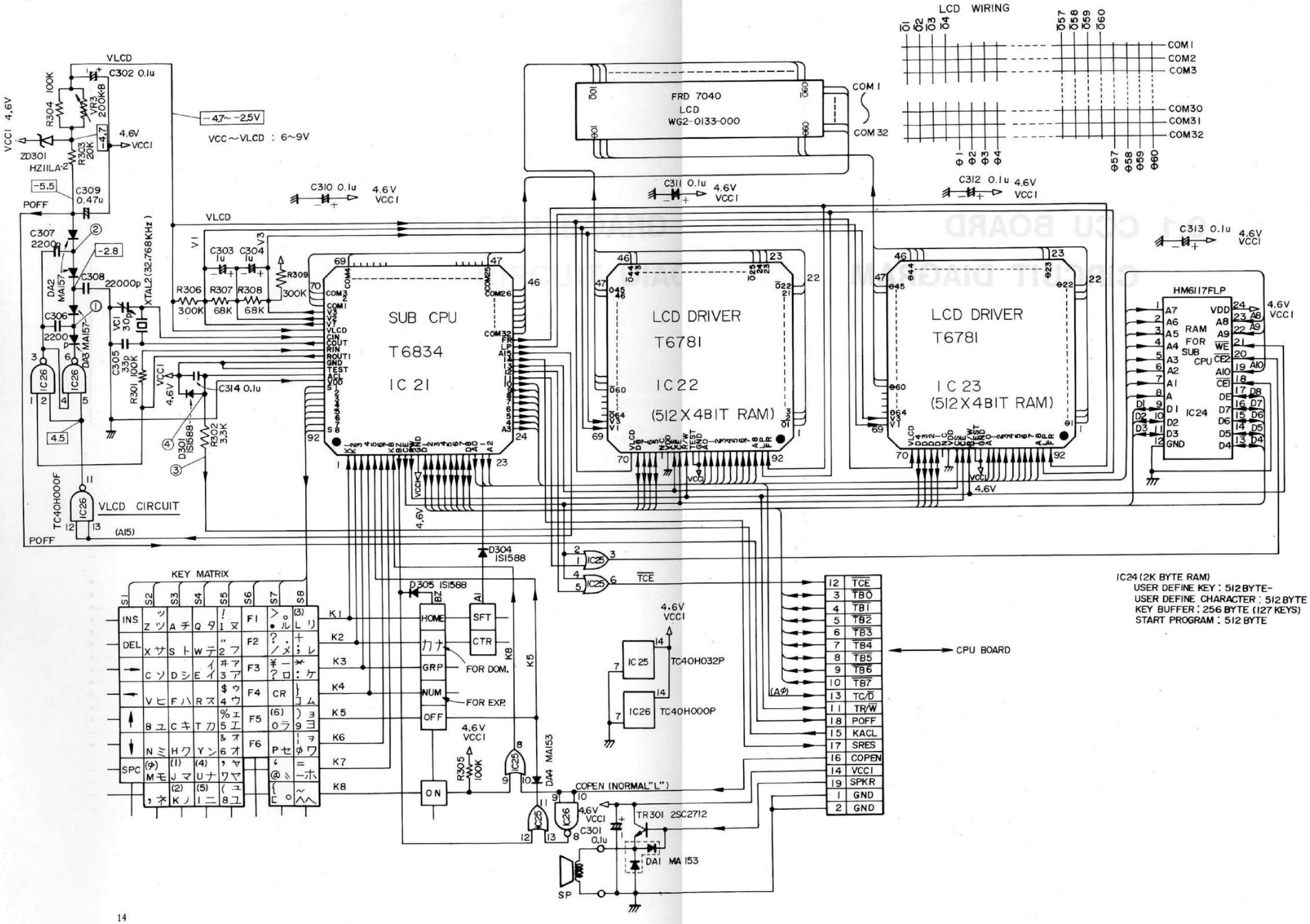
9. X-07

CIRCUIT DIAGRAM
TROUBLESHOOTING

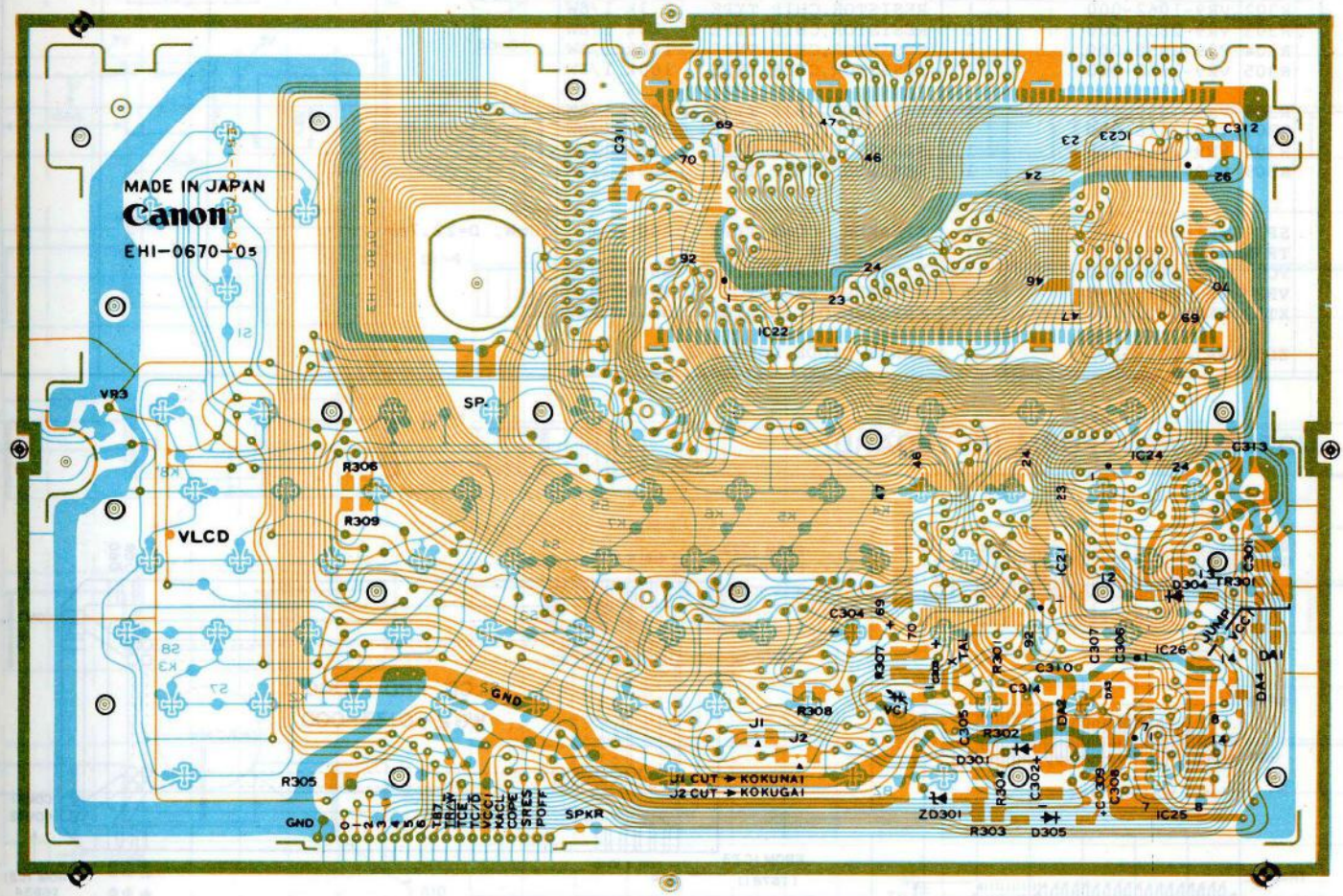
9.1 CCU BOARD CIRCUIT DIAGRAM

9. X-07 CIRCUIT DIAGRAM

9-1) CCU Board



CARD LAYOUT



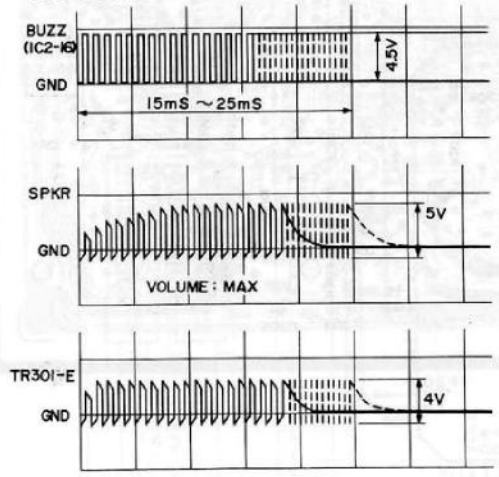
CCU BOARD

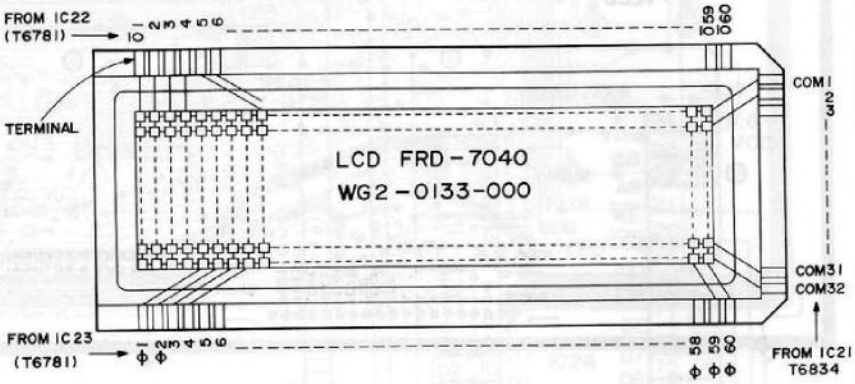
N	KEY NO.	PART NO.	Q'TY	DESCRIPTION		REMARKS	PRICE	
	C301	VC9-0532-000	1	CERAMIC CAPACITOR	0.1uF 25V			
	C302	VC9-1119-000	1	TANTALUM CAPACITOR	0.47uF 25V			
	C303	VC9-1120-000	1	TANTALUM CAPACITOR	1uF 16V			
	C304	VC9-1120-000	1	TANTALUM CAPACITOR	1uF 16V			
	C305	VC9-1122-000	1	CERAMIC CAPACITOR	33pF 25V			
*	C306	VC9-1225-000	1	CERAMIC CAPACITOR	2200pF 25V			
*	C307	VC9-1225-000	1	CERAMIC CAPACITOR	2200pF 25V			
*	C308	VC9-1224-000	1	CERAMIC CAPACITOR	22000pF 25V			
	C309	VC9-1119-000	1	TANTALUM CAPACITOR	0.47uF 25V			
	C310	VC9-0532-000	1	CERAMIC CAPACITOR	0.1uF 25V			
	C311	VC9-0532-000	1	CERAMIC CAPACITOR	0.1uF 25V			
	C312	VC9-0532-000	1	CERAMIC CAPACITOR	0.1uF 25V			
	C313	VC9-0532-000	1	CERAMIC CAPACITOR	0.1uF 25V			
	C314	VC9-0532-000	1	CERAMIC CAPACITOR	0.1uF 25V			
	D301	X65-5032-000	1	DIODE 1S1588				
	D302							
	D303							
	D304	X65-5032-000	1	DIODE 1S1588				
	D305	X65-5032-000	1	DIODE 1S1588				
	DA1	WA1-0375-000	1	DIODE MA153, CHIP TYPE				
	DA2	WA1-0380-000	1	DIODE MA157, CHIP TYPE				
	DA3	WA1-0380-000	1	DIODE MA157, CHIP TYPE				
	DA4	WA1-0357-000	1	DIODE MA153, CHIP TYPE				
*	IC21	WA3-9171-000	1	CMOS LSI T6834, SUB CPU				
	IC22	WA3-0791-000	1	CMOS LSI T6781, DRIVER				
	IC23	WA3-0791-000	1	CMOS LSI T6781, DRIVER				
	IC24	WA3-0955-000	1	CMOS LSI HM6117LFP-4, RAM	2k-byte			
	IC25	WA3-0963-000	1	CMOS IC TC40H032F, OR-GATE				
	IC26	WA3-0800-000	1	CMOS IC TC40H000F, NAND-GATE				
	LCD	WG2-0133-000	1	LIQUID CRYSTAL DISPLAY FRD7040				

N	KEY NO.	PART NO.	Q'TY	DESCRIPTION		REMARKS	PRICE	
	R301	VR9-1058-000	1	RESISTOR,CHIP TYPE	100k 1/8W			
	R302	VR9-1062-000	1	RESISTOR,CHIP TYPE	3.3k 1/8W			
	R303	VR9-2835-000	1	RESISTOR,CHIP TYPE	20k 1/8W			
	R304	VR9-1058-000	1	RESISTOR,CHIP TYPE	100k 1/8W			
	R305	VR9-1058-000	1	RESISTOR,CHIP TYPE	100k 1/8W			
	R306	VR9-1059-000	1	RESISTOR,CHIP TYPE	300k 1/8W			
	R307	VR9-2418-000	1	RESISTOR,CHIP TYPE	68k 1/8W			
	R308	VR9-2418-000	1	RESISTOR,CHIP TYPE	68k 1/8W			
	R309	VR9-1059-000	1	RESISTOR,CHIP TYPE	300k 1/8W			
	SP	WR1-0031-000	1	SPEAKER, CONE TYPE	200 ohm	30mW, D=22.7mm		
	TR301	WA2-0343-000	1	TRANSISTOR 2SC2712				
	VC1	VC9-0467-000	1	VARIABLE CAPACITOR	30pF			
	VR3	VR9-2618-000	1	VARIABLE RESISTOR	200k 50mW			
	XTL2	WK2-0002-000	1	CRYSTAL OACILLATOR	32.768kHz			
	ZD301	WA1-0308-000	1	ZENER DIODE HZ11LA-2				

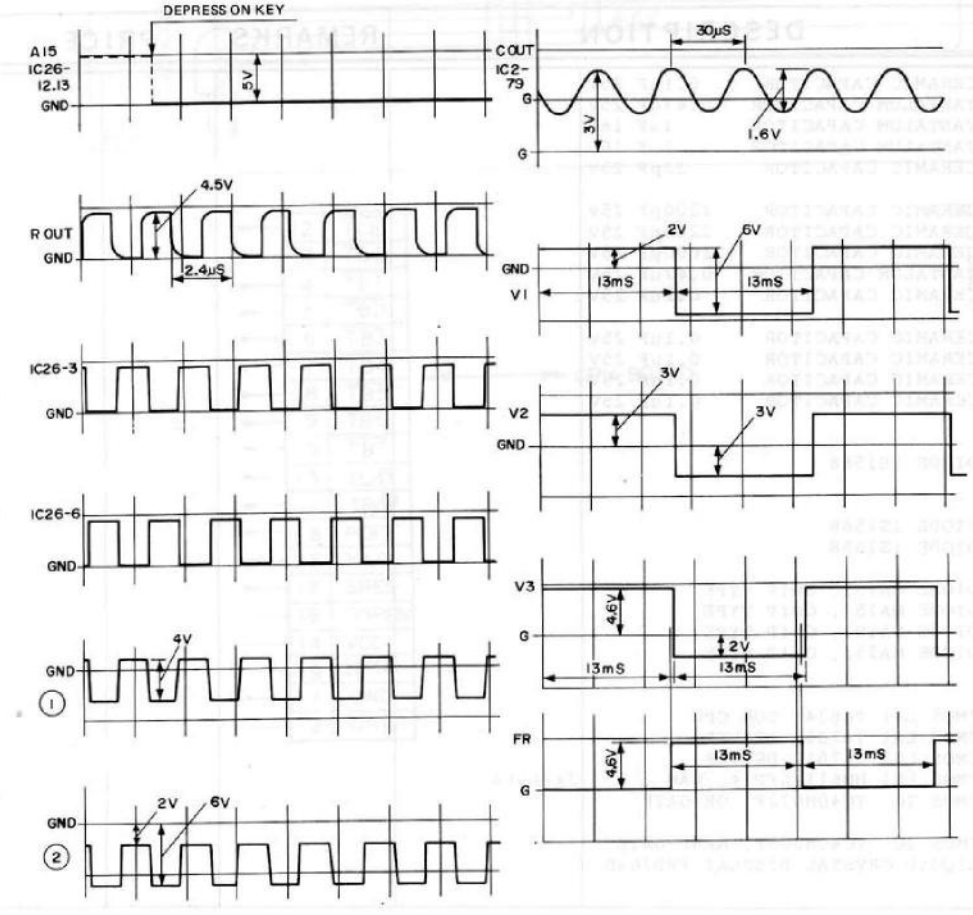
NORMAL WAVEFORM (KEYBOARD)

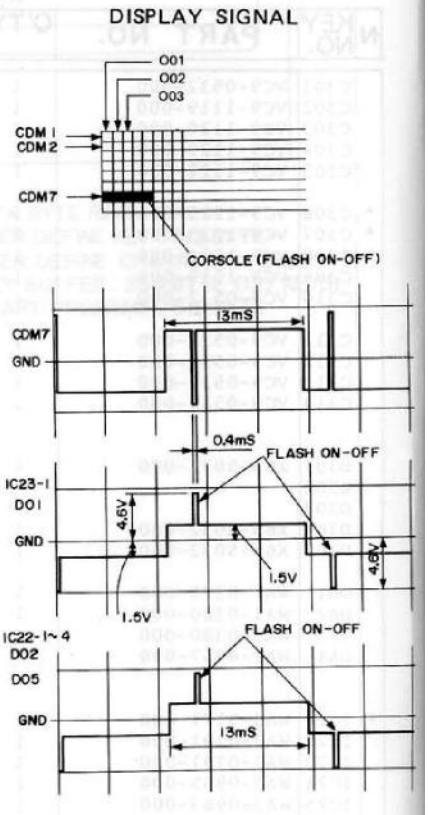
BUZZER

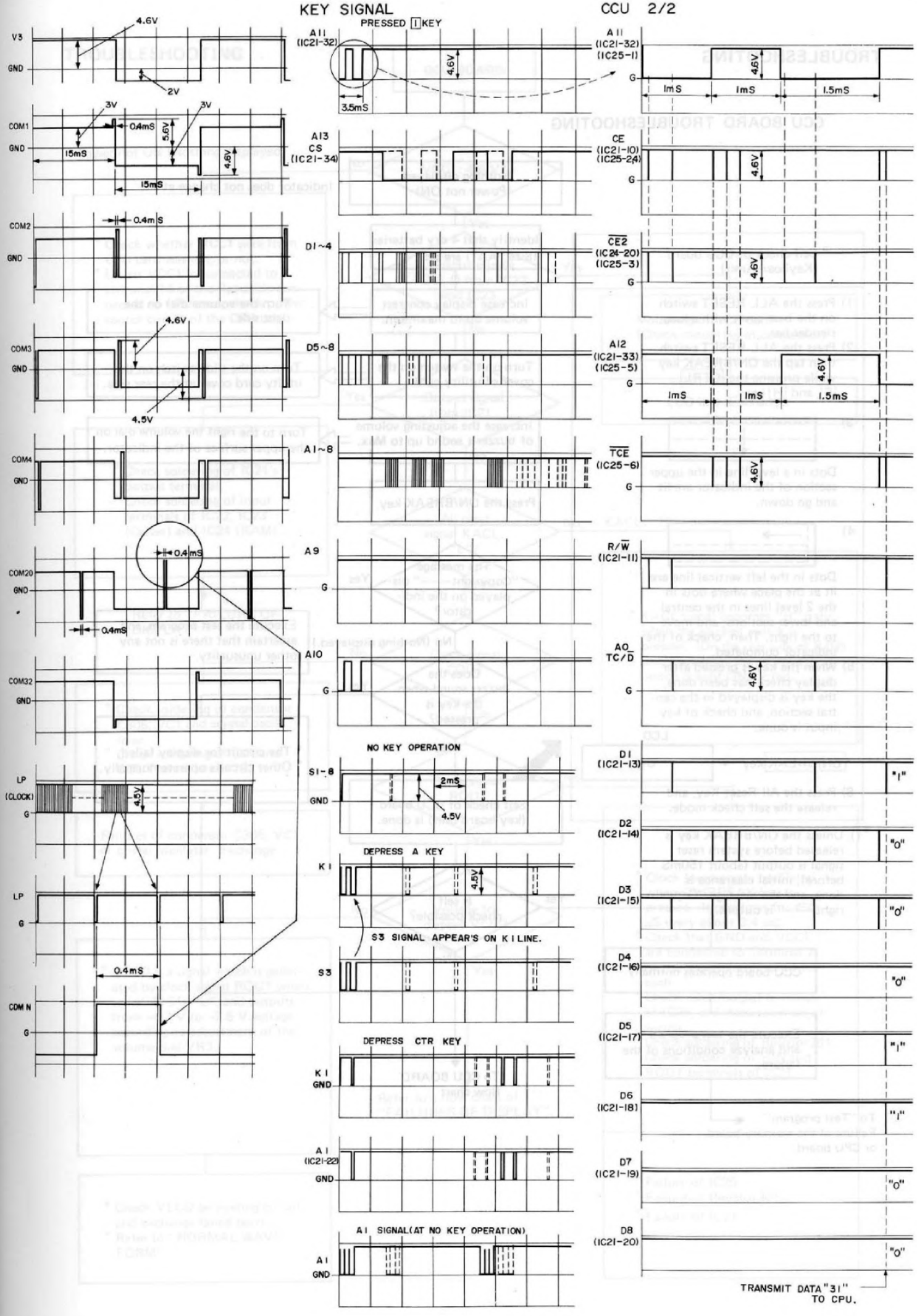




1. VLCD Circuit

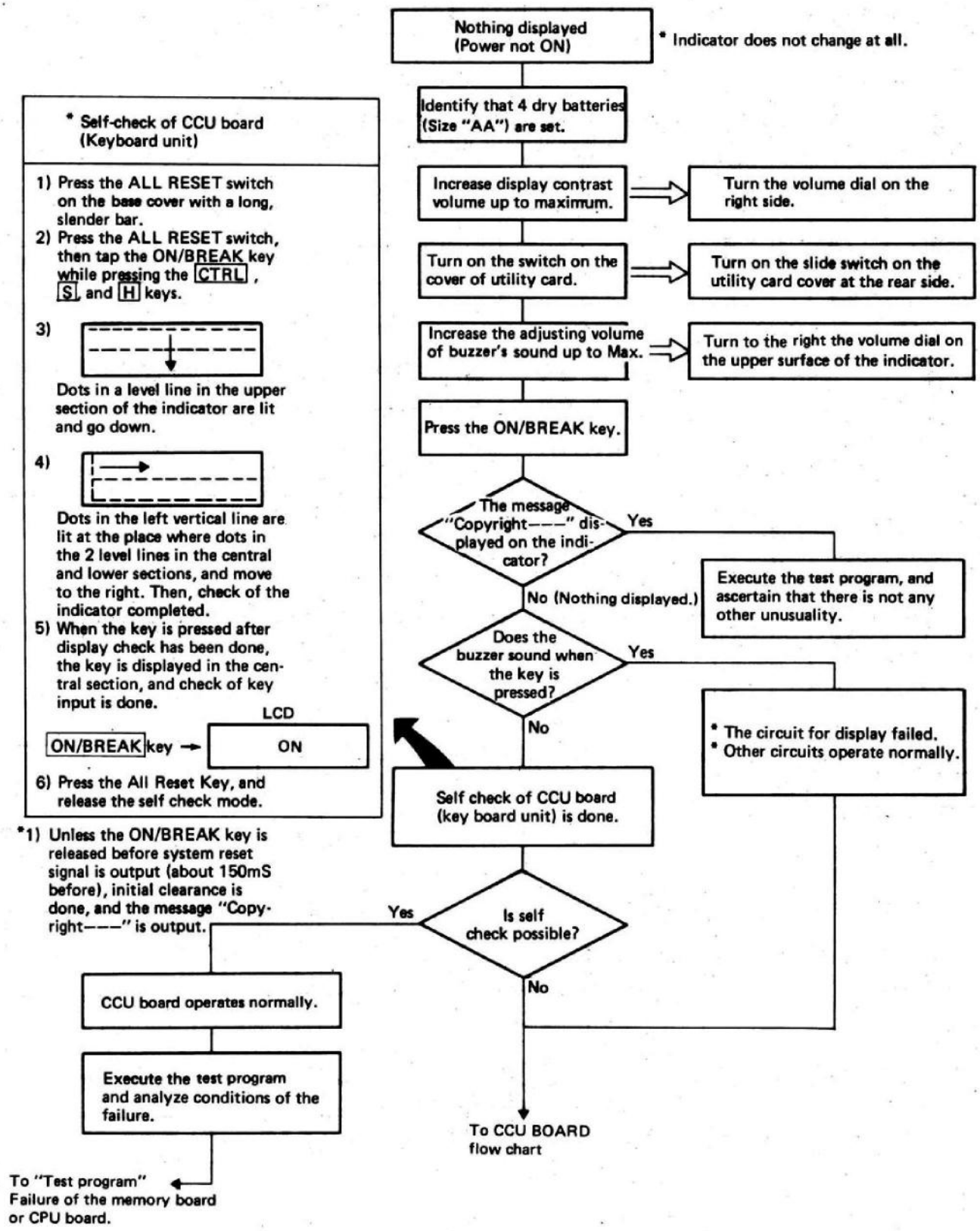




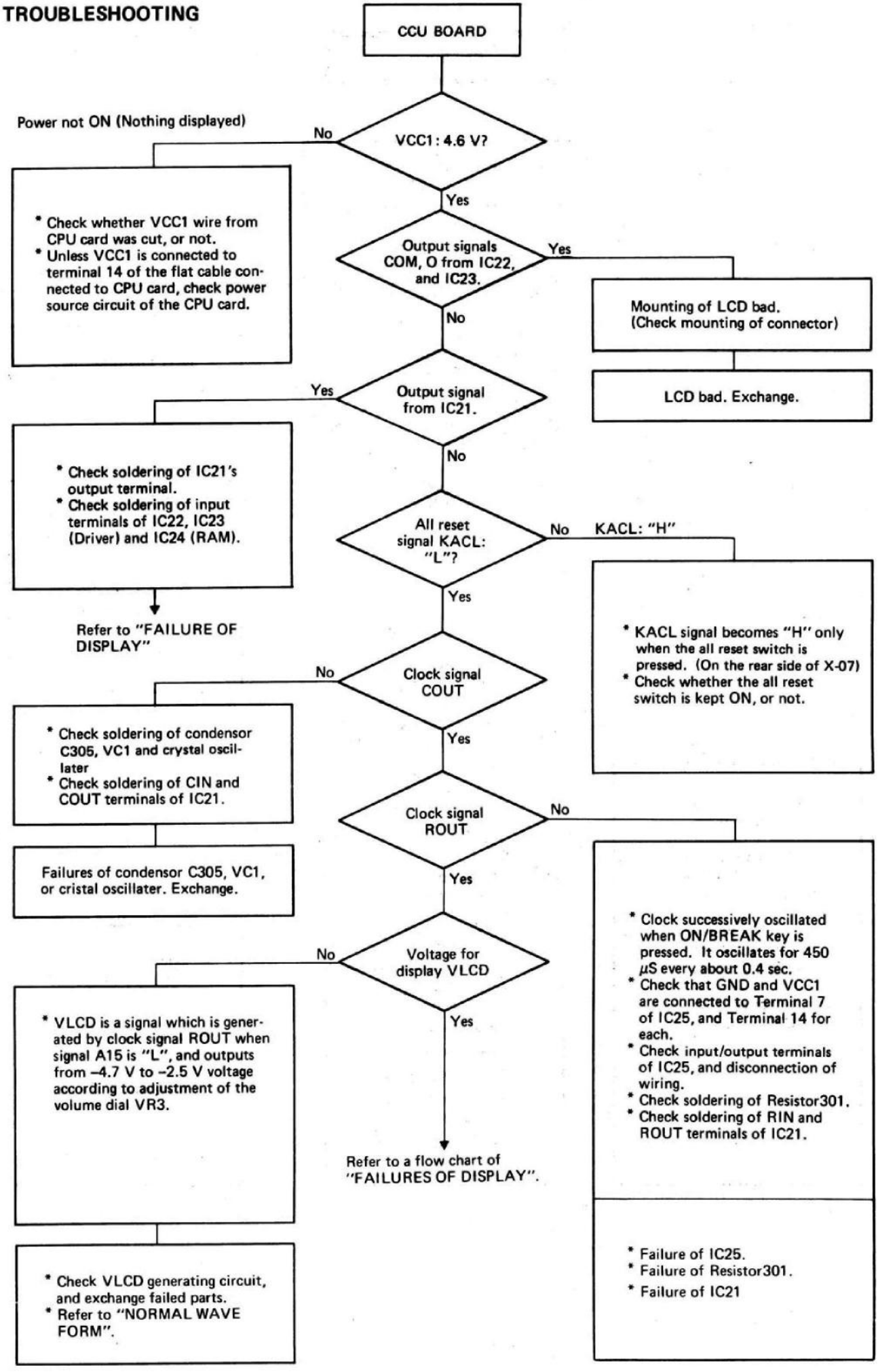

PRESSED [1] KEY
V3
GND
4.6V
2V
COM1
3V
0.4mS
5.6V
15mS
4.6V
COM2
COM3
4.5V
COM4
COM20
COM32
LP
(CLOCK)
COM N
A11 (IC21-32)
3.5mS
A13 CS (IC21-34)
D1~4
D5~8
A1~8
A9
A10
NO KEY OPERATION
S1~8
2mS
DEPRESS A KEY
K1
S3 SIGNAL APPEAR'S ON K1 LINE.
S3
DEPRESS CTR KEY
A1 (IC21-22)
A1 SIGNAL(AT NO KEY OPERATION)
A11 (IC21-32) (IC25-1)
1mS
1.5mS
CE (IC21-10) (IC25-24)
CE2 (IC24-20) (IC25-3)
A12 (IC21-33) (IC25-5)
TCE (IC25-6)
R/W (IC21-11)
AO TC/D
D1 (IC21-13) "1"
D2 (IC21-14) "0"
D3 (IC21-15) "0"
D4 (IC21-16) "0"
D5 (IC21-17) "1"
D6 (IC21-18) "1"
D7 (IC21-19) "0"
D8 (IC21-20) "0"
TRANSMIT DATA "31" TO CPU.

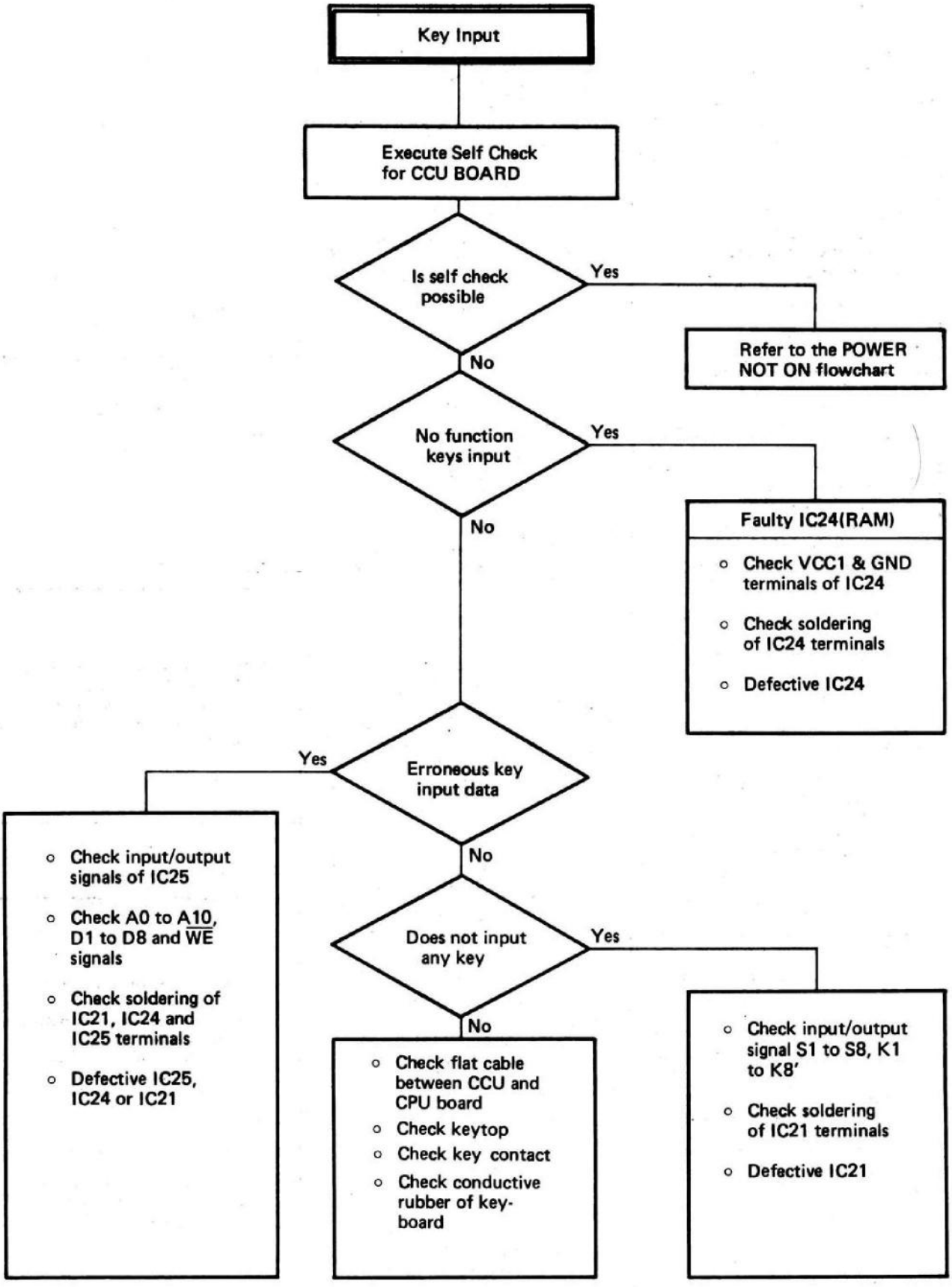
TROUBLESHOOTING

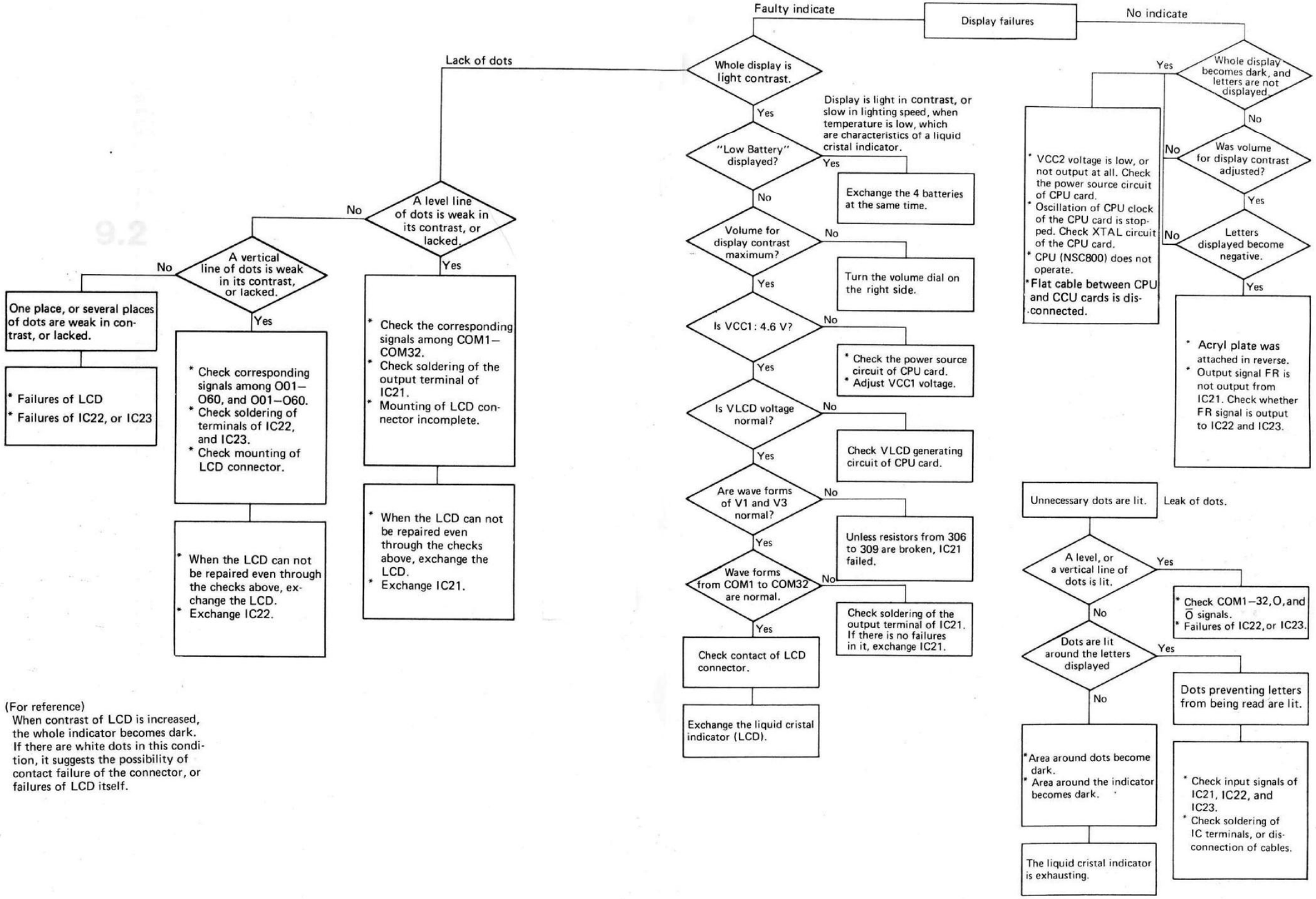
CCU BOARD TROUBLESHOOTING



TROUBLESHOOTING



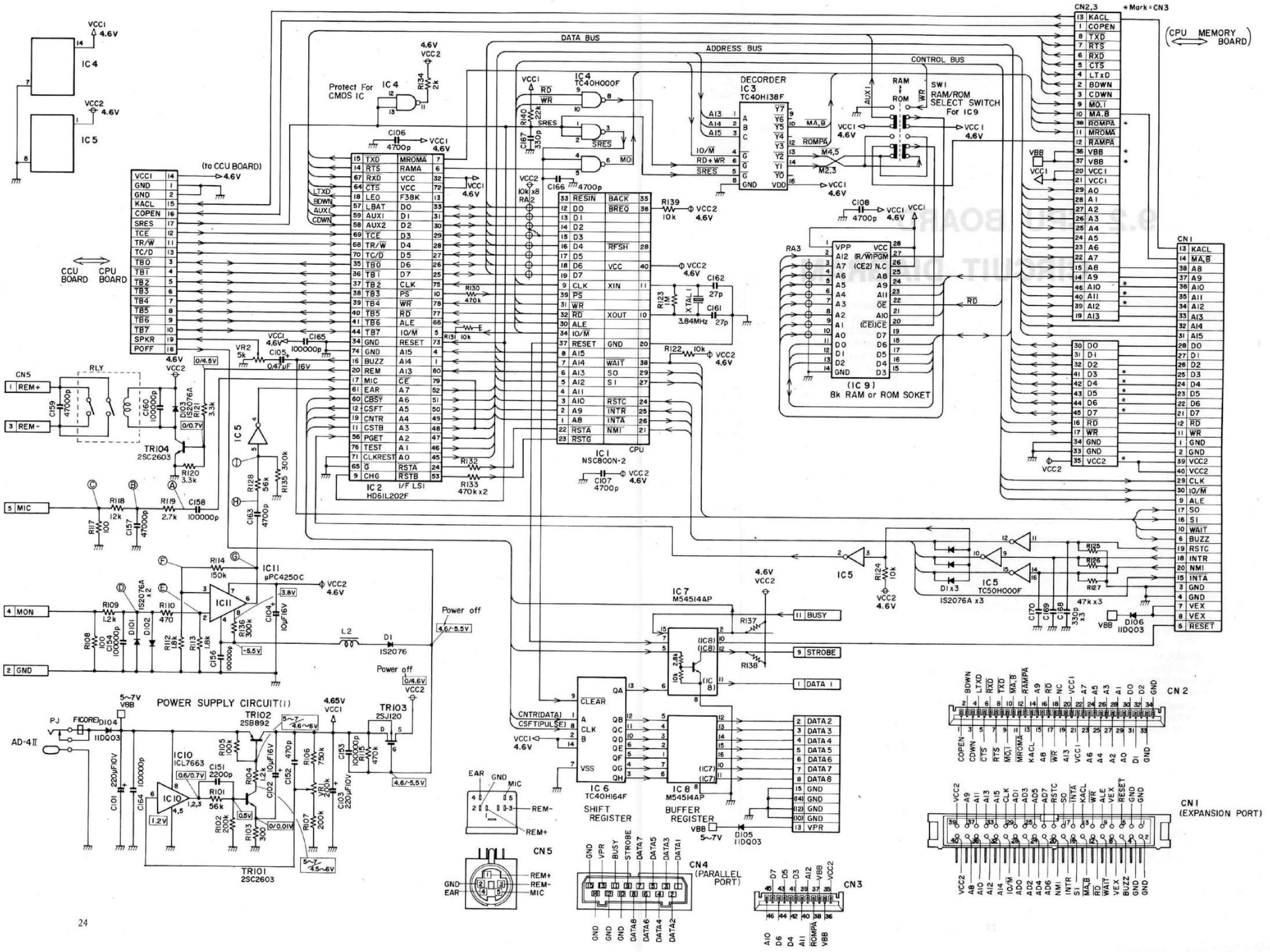

Key Input
Execute Self Check for CCU BOARD
Is self check possible
Yes
Refer to the POWER NOT ON flowchart
No
No function keys input
Yes
Faulty IC24(RAM)
o Check VCC1 & GND terminals of IC24
o Check soldering of IC24 terminals
o Defective IC24
No
Erroneous key input data
Yes
o Check input/output signals of IC25
o Check A0 to A10, D1 to D8 and WE signals
o Check soldering of IC21, IC24 and IC25 terminals
o Defective IC25, IC24 or IC21
No
Does not input any key
Yes
o Check input/output signal S1 to S8, K1 to K8'
o Check soldering of IC21 terminals
o Defective IC21
No
o Check flat cable between CCU and CPU board
o Check keytop
o Check key contact
o Check conductive rubber of key-board



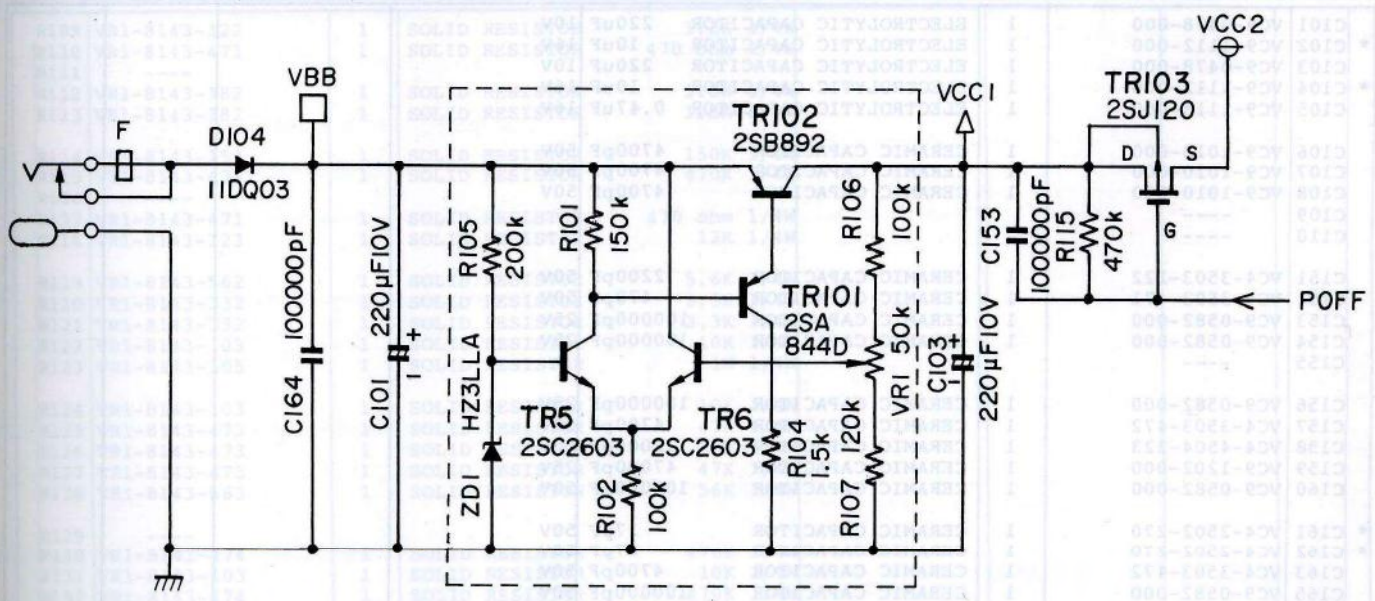
(For reference)
When contrast of LCD is increased, the whole indicator becomes dark. If there are white dots in this condition, it suggests the possibility of contact failure of the connector, or failures of LCD itself.

9.2 CPU BOARD CIRCUIT DIAGRAM

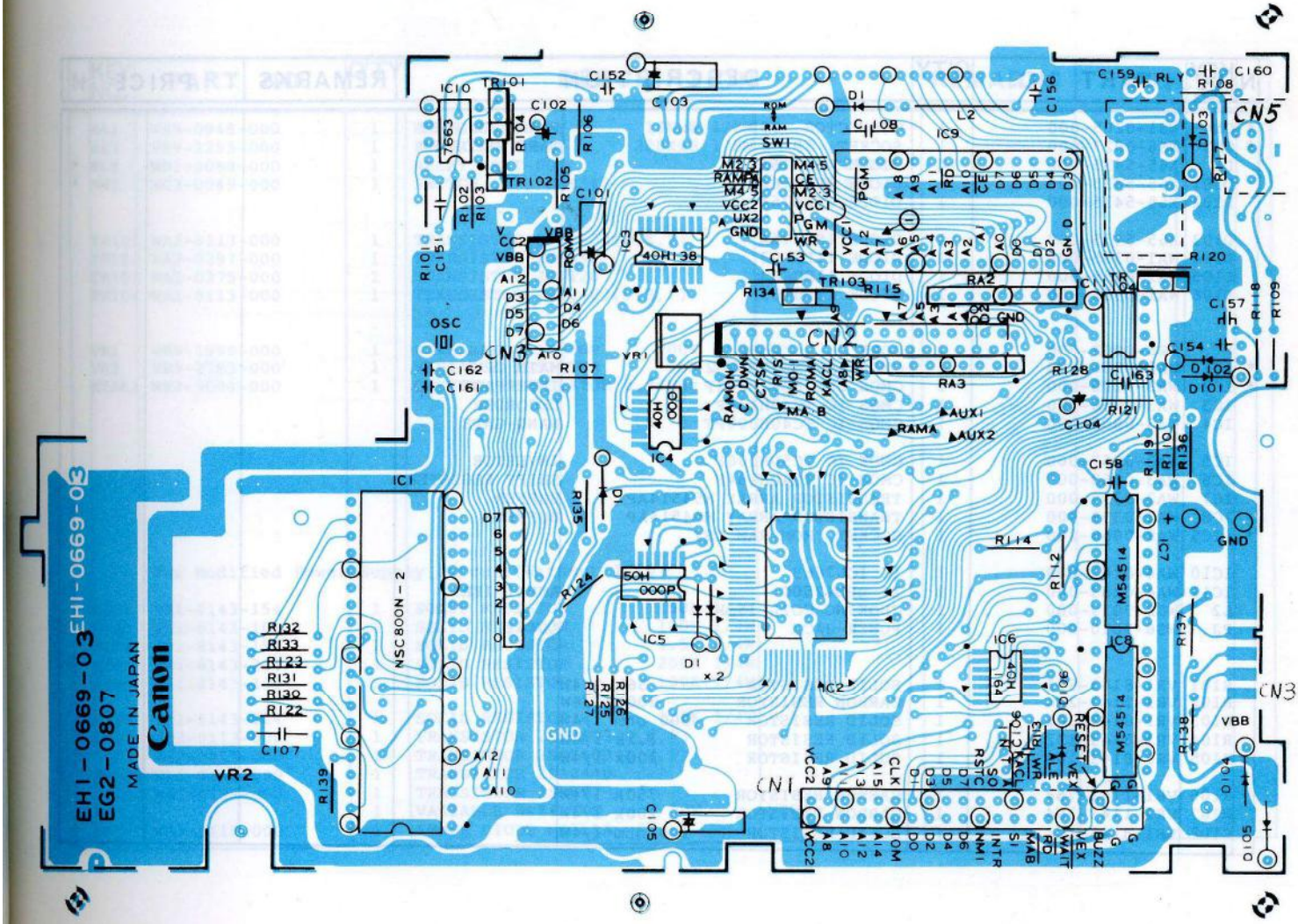
9-2) CPU BOARD CIRCUIT DIAGRAM



MODIFIED POWER SUPPLY CIRCUIT (WITHOUT IC10)



CARD LAYOUT



シルク図

N	KEY NO.	PART NO.	Q'TY	DESCRIPTION		REMARKS	PRICE	
	C101	VC9-0478-000	1	ELECTROLYTIC CAPACITOR	220uF 10V			
*	C102	VC9-1112-000	1	ELECTROLYTIC CAPACITOR	10uF 16V			
	C103	VC9-0478-000	1	ELECTROLYTIC CAPACITOR	220uF 10V			
*	C104	VC9-1112-000	1	ELECTROLYTIC CAPACITOR	10uF 16V			
	C105	VC9-1118-000	1	ELECTROLYTIC CAPACITOR	0.47uF 16V			
	C106	VC9-1010-000	1	CERAMIC CAPACITOR	4700pF 50V			
	C107	VC9-1010-000	1	CERAMIC CAPACITOR	4700pF 50V			
	C108	VC9-1010-000	1	CERAMIC CAPACITOR	4700pF 50V			
	C109	----						
	C110	----						
	C151	VC4-3503-222	1	CERAMIC CAPACITOR	2200pF 50V			
	C152	VC4-3503-471	1	CERAMIC CAPACITOR	470pF 50V			
	C153	VC9-0582-000	1	CERAMIC CAPACITOR	100000pF 25V			
	C154	VC9-0582-000	1	CERAMIC CAPACITOR	100000pF 25V			
	C155	----						
	C156	VC9-0582-000	1	CERAMIC CAPACITOR	100000pF 25V			
	C157	VC4-3503-472	1	CERAMIC CAPACITOR	4700pF 50V			
	C158	VC4-4504-223	1	CERAMIC CAPACITOR	22000pF 50V			
	C159	VC9-1202-000	1	CERAMIC CAPACITOR	47000pF 25V			
	C160	VC9-0582-000	1	CERAMIC CAPACITOR	100000pF 50V			
*	C161	VC4-2502-270	1	CERAMIC CAPACITOR	27pF 50V			
*	C162	VC4-2502-270	1	CERAMIC CAPACITOR	27pF 50V			
	C163	VC4-3503-472	1	CERAMIC CAPACITOR	4700pF 50V			
	C165	VC9-0582-000	1	CERAMIC CAPACITOR	100000pF 50V			
	C166	VC4-3503-472	1	CERAMIC CAPACITOR	4700pF 50V			
	C167	VC4-3503-331	1	CERAMIC CAPACITOR	330pF 50V			
	C168	VC4-3503-331	1	CERAMIC CAPACITOR	330pF 50V			
	C169	VC4-3503-331	1	CERAMIC CAPACITOR	330pF 50V			
	C170	VC4-3503-331	1	CERAMIC CAPACITOR	330pF 50V			
*	CN1	WS1-0211-000	1	CONNECTOR, 40P		EXPANSION PORT		
*	CN2	VS1-0148-034	1	CONNECTOR, 34P		CPU-MEMORY PCB		
*	CN3	VS1-0148-012	1	CONNECTOR, 12P		CPU-MEMORY PCB		

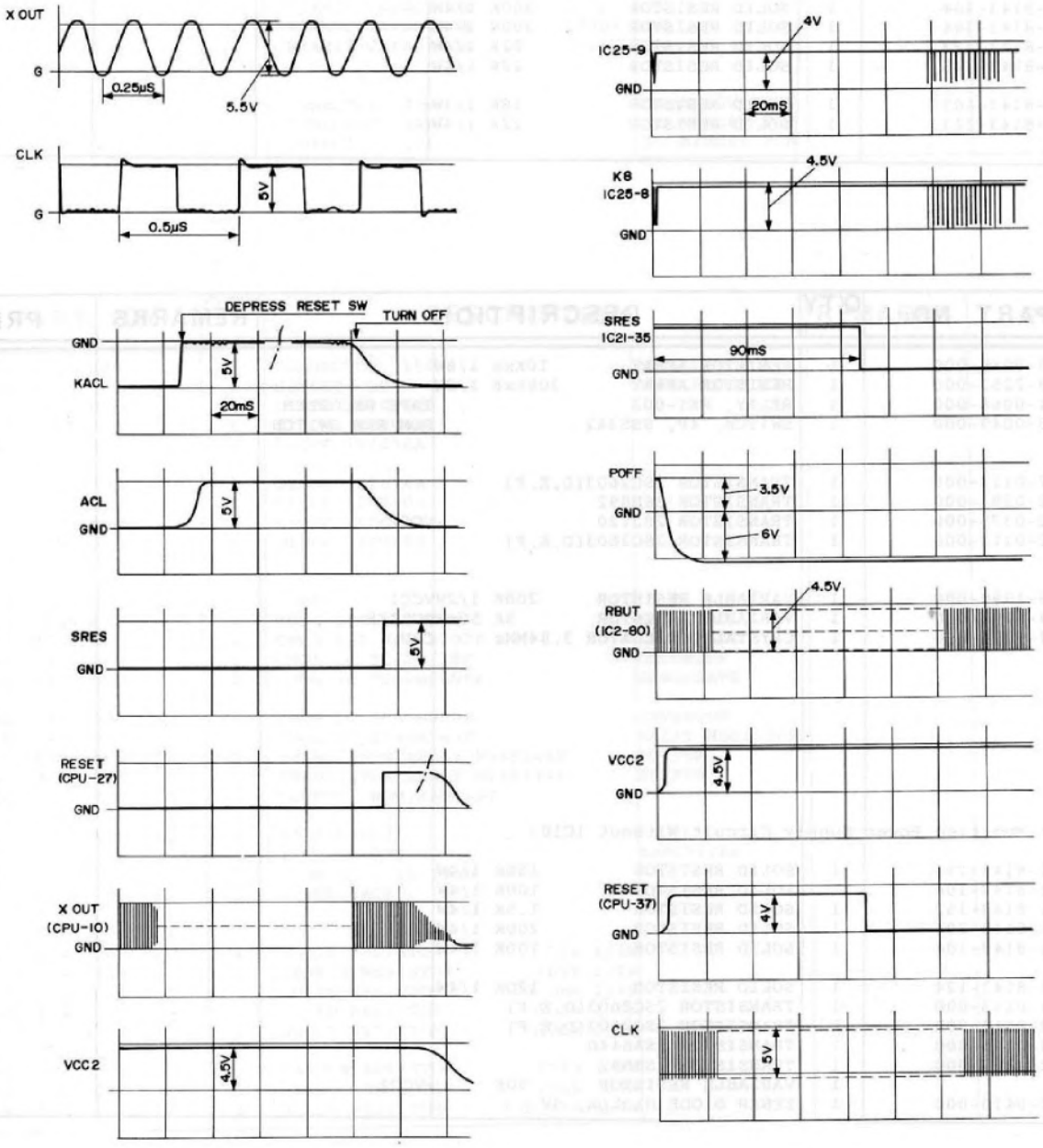
N	KEY NO.	PART NO.	Q'TY	DESCRIPTION		REMARKS	PRICE	
*	CN4	WS1-0205-000	1	CONNECTOR, 15P		PARALLEL PORT		
*	CN5	WS6-0027-000	1	SOCKET, 5P		TAPE RECORDER		
	D1	X65-5435-000	4	DIODE 1S2076A				
	D101	X65-5435-000	1	DIODE 1S2076A				
	D102	X65-5435-000	1	DIODE 1S2076A				
	D103	X65-5435-000	1	DIODE 1S2076A				
	D104	WA1-0355-000	1	DIODE 11D0-03				
	D105	WA1-0355-000	1	DIODE 11D0-03				
	D106	WA1-0355-000	1	DIODE 11D0-03				
	F	WE8-0002-000	1	CORE				
*	IC1	WA3-0956-000	1	CMOS LSI NSC800N-2		MAIN CPU		
*	IC2	WA3-9170-000	1	CMOS LSI HD61L202F		INTERFACE		
	IC3	WA3-0958-000	1	CMOS IC TC40H138F		DECORDER		
	IC4	WA3-0800-000	1	CMOS IC TC40H000PF		NAND-GATE		
	IC5	WA3-0957-000	1	CMOS IC TC50H000F		INVERTOR		
	IC6	WA3-0960-000	1	CMOS IC TC40H164F		SHIFT REGISTER		
	IC7	WA2-0399-000	1	TRANSISTOR ARRAY M54514AP		BUFFER		
	IC8	WA2-0399-000	1	TRANSISTOR ARRAY M54514AP		BUFFER		
*	(IC9)	WA9-0058-000	1	SOCKET, ROM/RAM LSI				
	IC10	WA4-0261-000	1	IC ICL7663				
	IC11	WA4-0283-000	1	IC uPD4250C		AMPLIFIER		
	L2	WE2-9019-000	1	CHOKING COIL 15mH				
	PJ	WS6-9010-000	1	POWER JACK		AD4-II		
	R101	VR1-8143-563	1	SOLID RESISTOR	56k 1/4W			
	R102	VR1-1143-204	1	CARBON RESISTOR	200k 1/4W			
	R103	VR1-8143-301	1	SOLID RESISTOR	300 ohm 1/4W			
	R104	VR1-8143-122	1	SOLID RESISTOR	1.2k 1/4W			
	R105	VR1-8143-104	1	SOLID RESISTOR	100k 1/4W			
	R106	VR1-1143-754	1	CARBON RESISTOR	750k 1/4W			
	R107	VR1-1143-204	1	CARBON RESISTOR	200k 1/4W			
	R108	VR1-8143-101	1	SOLID RESISTOR	100 ohm 1/4W			

N	KEY NO.	PART NO.	Q'TY	DESCRIPTION		REMARKS	PRICE	
	R109	VR1-8143-122	1	SOLID RESISTOR	1.2K 1/4W			
	R110	VR1-8143-471	1	SOLID RESISTOR	470 ohm 1/4W			
	R111	----						
	R112	VR1-8143-182	1	SOLID RESISTOR	1.8K 1/4W			
	R113	VR1-8143-182	1	SOLID RESISTOR	1.8K 1/4W			
	R114	VR1-8143-154	1	SOLID RESISTOR	150K 1/4W			
	R115	VR1-8143-474	1	SOLID RESISTOR	470K 1/4W			
	R116	----						
	R117	VR1-8143-471	1	SOLID RESISTOR	470 ohm 1/4W			
	R118	VR1-8143-123	1	SOLID RESISTOR	12K 1/4W			
	R119	VR1-8143-562	1	SOLID RESISTOR	5.6K 1/4W			
	R120	VR1-8143-332	1	SOLID RESISTOR	3.3K 1/4W			
	R121	VR1-8143-332	1	SOLID RESISTOR	3.3K 1/4W			
	R122	VR1-8143-103	1	SOLID RESISTOR	10K 1/4W			
	R123	VR1-8143-105	1	SOLID RESISTOR	1M 1/4W			
	R124	VR1-8143-103	1	SOLID RESISTOR	10K 1/4W			
	R125	VR1-8143-473	1	SOLID RESISTOR	47K 1/4W			
	R126	VR1-8143-473	1	SOLID RESISTOR	47K 1/4W			
	R127	VR1-8143-473	1	SOLID RESISTOR	47K 1/4W			
	R128	VR1-8143-563	1	SOLID RESISTOR	56K 1/4W			
	R129	----						
	R130	VR1-8143-474	1	SOLID RESISTOR	470K 1/4W			
	R131	VR1-8143-103	1	SOLID RESISTOR	10K 1/4W			
	R132	VR1-8143-474	1	SOLID RESISTOR	470K 1/4W			
	R133	VR1-8143-474	1	SOLID RESISTOR	470K 1/4W			
	R134	VR1-8143-202	1	SOLID RESISTOR	2K 1/4W			
	R135	VR1-8143-304	1	SOLID RESISTOR	300K 1/4W			
	R136	VR1-8143-304	1	SOLID RESISTOR	300K 1/4W			
	R137	VR1-8143-223	1	SOLID RESISTOR	22K 1/4W			
	R138	VR1-8143-223	1	SOLID RESISTOR	22K 1/4W			
	R139	VR1-8143-103	1	SOLID RESISTOR	10K 1/4W			
	R140	VR1-8143-223	1	SOLID RESISTOR	22K 1/4W			

N	KEY NO.	PART NO.	Q'TY	DESCRIPTION		REMARKS	PRICE	
	RA2	VR9-0948-000	1	RESISTOR ARRAY	10Kx8 1/8W			
	RA3	VR9-2253-000	1	RESISTOR ARRAY	300Kx8 1/8W			
*	RLY	WB1-0068-000	1	RELAY, RKT-002		TAPE RECORDER		
*	SW1	WC3-0049-000	1	SWITCH, 4P, SSS342		ROM/RAM SWITCH		
	TR101	WA2-0113-000	1	TRANSISTOR 2SC2603(D,E,F)				
	TR102	WA2-0297-000	1	TRANSISTOR 2SB892				
	TR103	WA2-0375-000	1	TRANSISTOR 2SJ120		VCC2		
	TR104	WA2-0113-000	1	TRANSISTOR 2SC2603(D,E,F)				
	VR1	VR9-1999-000	1	VARIABLE RESISTOR	200K 1/2W	VCC1		
	VR2	VR9-2783-000	1	VARIABLE RESISTOR	5K 50mW	BUZZER		
	XTAL1	WK2-9004-000	1	CRYSTAL OSCILLATOR	3.84MHz	CPU		
		For Modified Power Supply Circuit(Without IC10)						
	R101	VR1-8143-154	1	SOLID RESISTOR	150K 1/4W			
	R102	VR1-8143-104	1	SOLID RESISTOR	100K 1/4W			
	R104	VR1-8143-152	1	SOLID RESISTOR	1.5K 1/4W			
	R105	VR1-8143-204	1	SOLID RESISTOR	200K 1/4W			
	R106	VR1-8143-104	1	SOLID RESISTOR	100K 1/4W			
	R107	VR1-8143-124	1	SOLID RESISTOR	120K 1/4W			
	TR5	WA2-0113-000	1	TRANSISTOR 2SC2603(D,E,F)				
	TR6	WA2-0113-000	1	TRANSISTOR 2SC2603(D,E,F)				
	TR101	WA2-0195-000	1	TRANSISTOR 2SA844D				
	TR102	WA2-0297-000	1	TRANSISTOR 2SB892				
	VR1		1	VARIABLE RESISTOR	50K	VCC1		
	ZD1	WA1-0410-000	1	ZENER DIODE HZ3LLA, 3V				

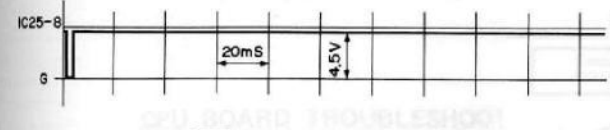
MEMO

NORMAL WAVEFORM

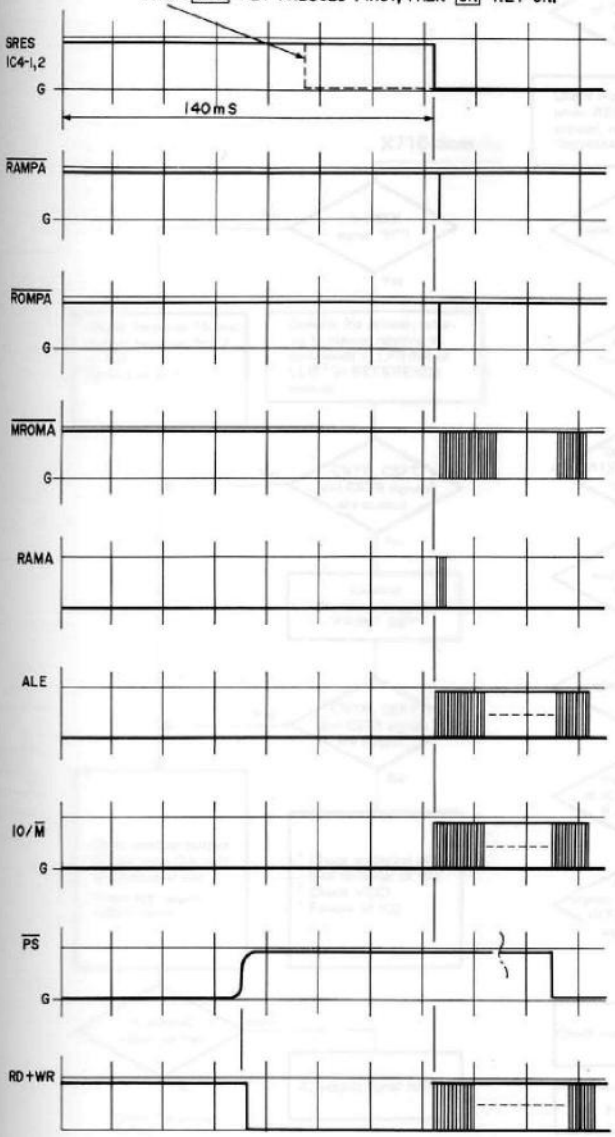


RAMPA, ROMPA, RAMA & $\overline{PS}$.

FIST PRESS RESET SW, THEN [ON] KEY IS PRESSED.

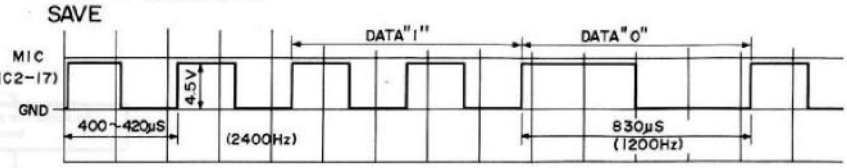


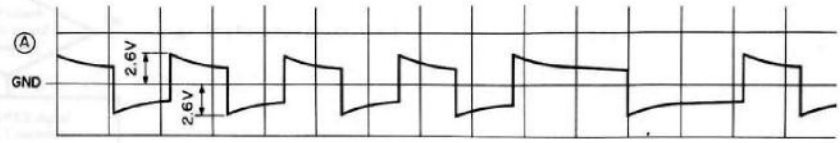
WHEN [OFF] KEY PRESSED FIRST, THEN [ON] KEY ON.

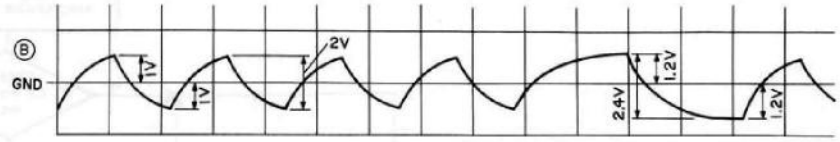


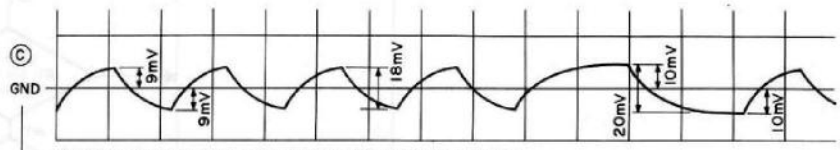
TAPE RECORDER

SAVE

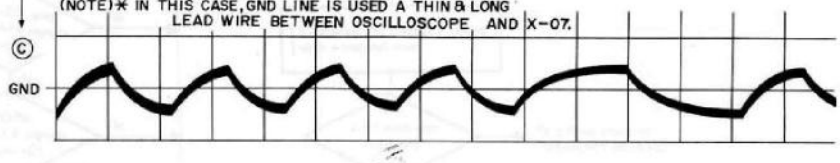




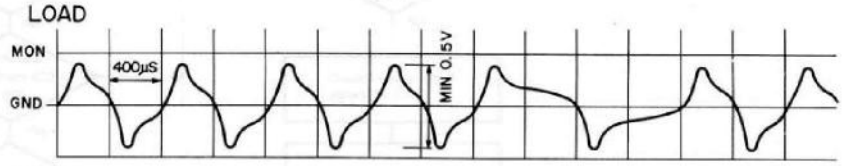


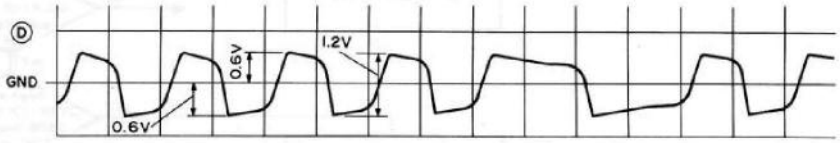


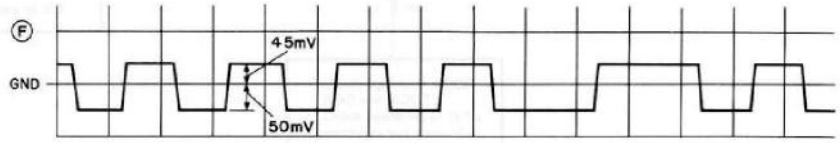
(NOTE)* IN THIS CASE, GND LINE IS USED A THIN & LONG LEAD WIRE BETWEEN OSCILLOSCOPE AND X-07.

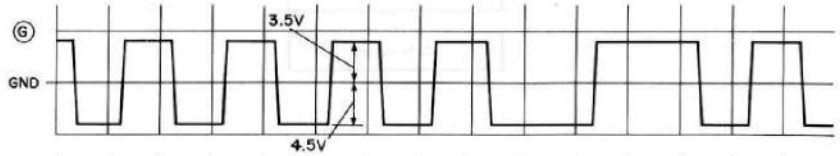


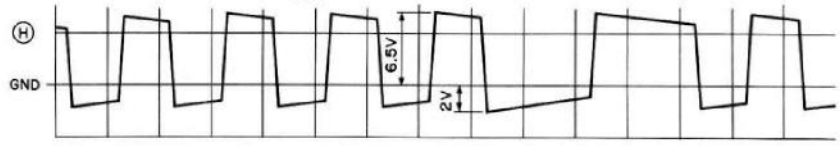
LOAD

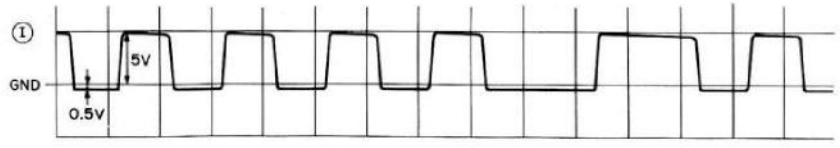


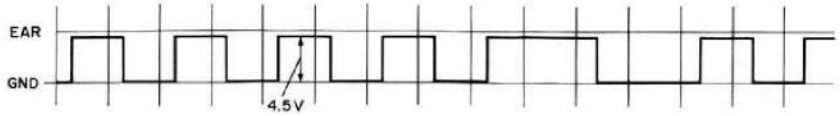




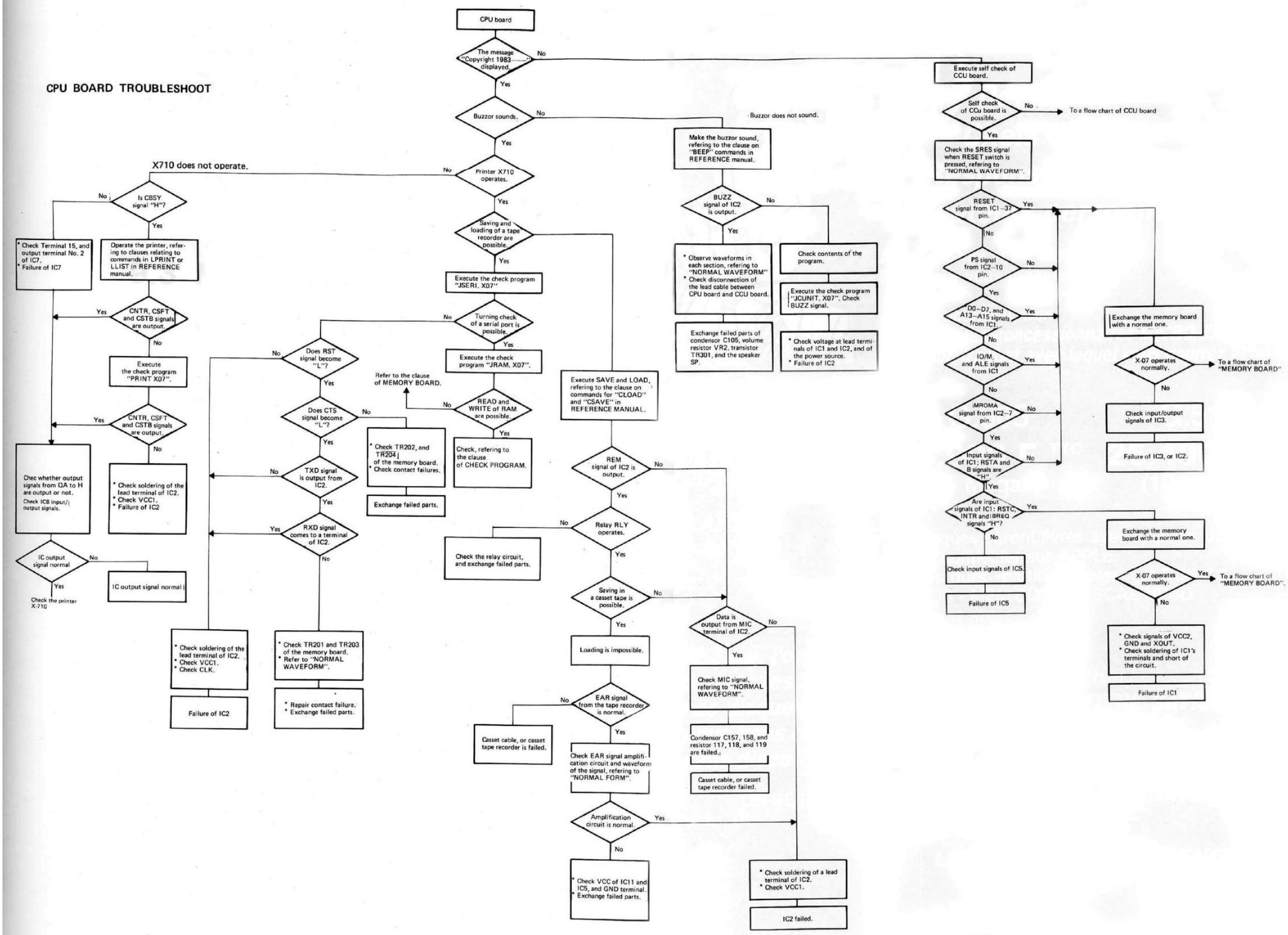




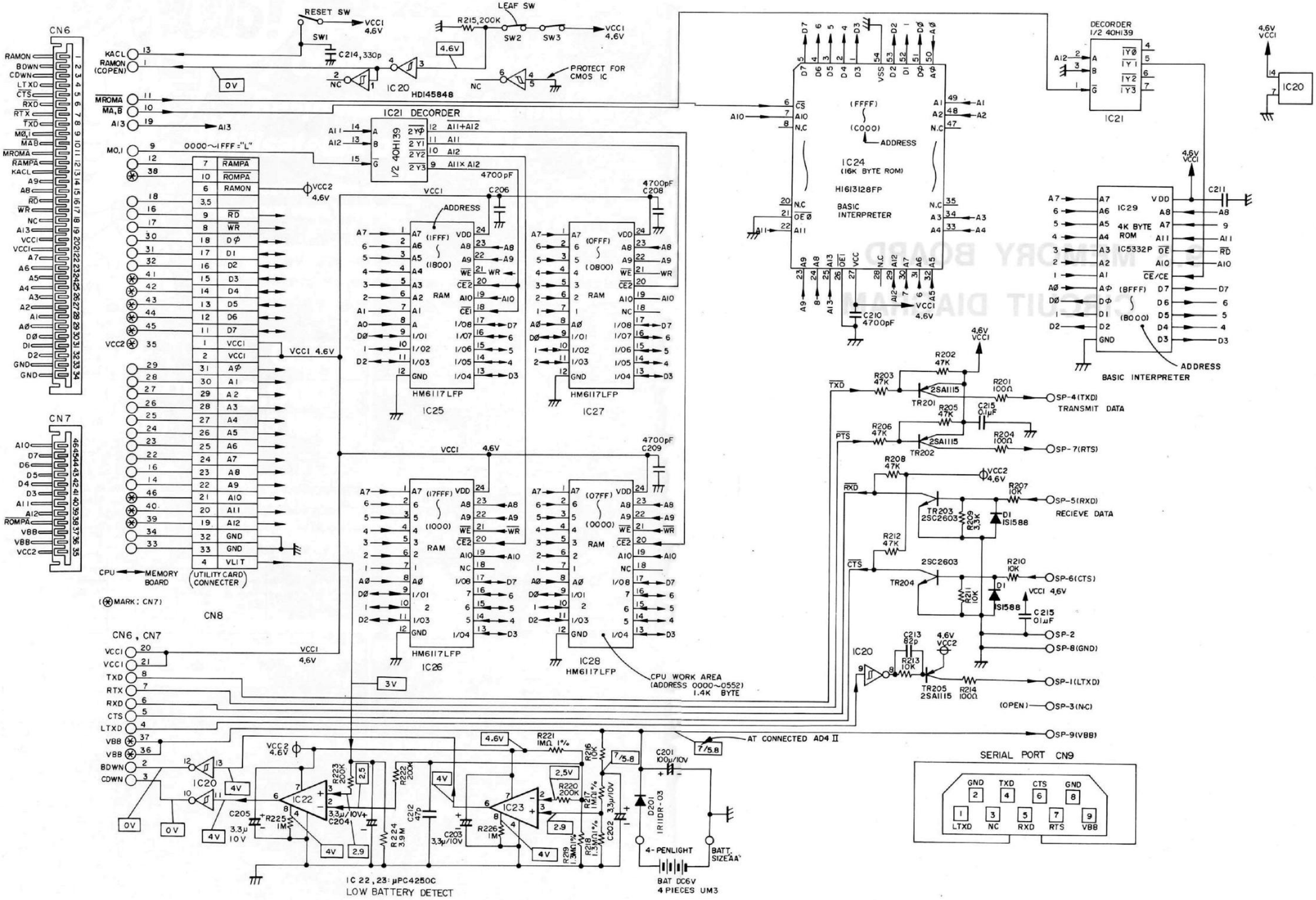




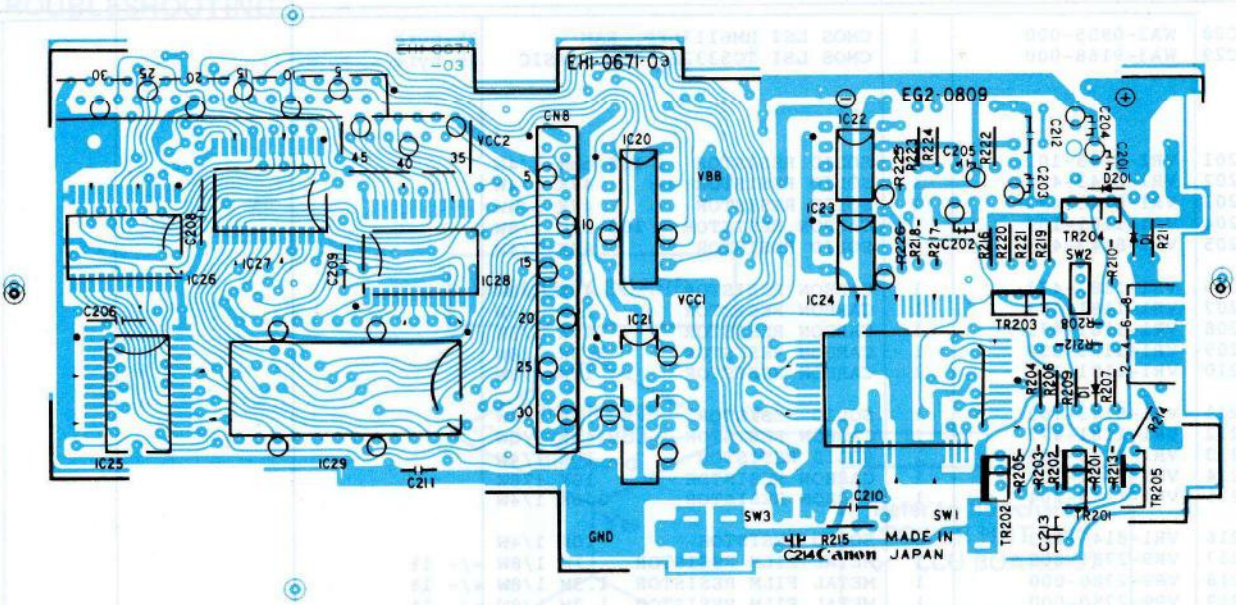
CPU BOARD TROUBLESHOOT



9.3 MEMORY BOARD CIRCUIT DIAGRAM


CN6
RAMON
BDWN
CDWN
LTXD
CTS
RXD
RTX
TXD
MØ,I
MAB
MROMA
RAMPA
KACL
A9
A8
RD
WR
NC
A13
VCCI
A7
A6
A5
A4
A3
A2
A1
AØ
DØ
DI
D2
GND
CN7
A10
D7
D6
D5
D4
D3
A11
A12
ROMPA
VBB
VCC2
RESET SW
SW1
VCC1 4.6V
C214,330p
KACL
RAMON (COPEN)
OV
LEAF SW
R215,200K
SW2
SW3
4.6V
PROTECT FOR CMOS IC
IC20 HD145848
MROMA
MA,B
A13
MO,I
0000~1FFF :"L"
VCC2 4.6V
IC21 DECORDER
1/2 40H139
A11+A12
A11
A12
A11×A12
4700pF
C206
4700pF C208
ADDRESS
(1FFF)
(1800)
(0FFF)
(0800)
RAM
HM6117LFP
IC25
IC27
4700pF C209
(17FF)
(1000)
(07FF)
(0000)
IC26
IC28
CPU WORK AREA
(ADDRESS 0000~0552)
1.4K BYTE
RAMPA
ROMPA
RAMON
RD
WR
DØ
D1
D2
D3
VCC1
AØ
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
A11
A12
VLIT
CPU ← MEMORY BOARD
(UTILITY CARD CONNECTER)
(⊗MARK: CN7)
CN8
CN6, CN7
TXD
RTX
RXD
CTS
LTXD
VBB
BDWN
CDWN
3V
IC22, 23: μPC4250C
LOW BATTERY DETECT
IC24
(16K BYTE ROM)
HI613128FP
BASIC INTERPRETER
(FFFF)
(C000)
ADDRESS
DECORDER
1/2 40H139
IC21
IC29
4K BYTE ROM
IC5332P
(BFFF)
(B000)
BASIC INTERPRETER
C211
C210 4700pF
R202 47K
R203 47K
R205 47K
R206 47K
R208 47K
R212 47K
R201 100Ω
R204 100Ω
R207 10K
R209 3.3K
R210 10K
R211 10K
R213 10K
R214 100Ω
C215 0.1μF
C213 82p
TR201 2SA1115
TR202 2SA1115
TR203 2SC2603
TR204 2SC2603
TR205 2SA1115
D1 1S1588
SP-4(TXD) TRANSMIT DATA
SP-7(RTS)
SP-5(RXD) RECIEVE DATA
SP-6(CTS)
SP-2
SP-8(GND)
SP-1(LTXD)
SP-3(N.C)
SP-9(VBB)
AT CONNECTED AD4 II
SERIAL PORT CN9
GND TXD CTS GND
2 4 6 8
1 3 5 7 9
LTXD NC RXD RTS VBB
R221 1MΩ 1%
R216 10K
R220 200K
R217 1MΩ 1%
R218 1.3MΩ 1%
R219 1.3MΩ 1%
R222 200K
R223 200K
R224 3.9M
R225 1M
R226 1M
C201 100μ/10V
C202 3.3μ/10V
C203 3.3μ/10V
C204 3.3μ/10V
C205 3.3μ 10V
C212 47p
D201 IR1IDR-03
4-PENLIGHT
BATT. SIZE"AA"
BAT DC6V
4 PIECES UM3

CARD LAYOUT


EHI-0671-03
EG2-0809
CN8
IC20
IC22
IC23
IC24
IC21
IC25
IC26
IC27
IC28
IC29
VCC2
VCC1
VBB
GND
C205
C206
C208
C209
C210
C211
C212
C213
C214
C215
C216
C217
C218
C219
C220
C221
C222
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C227
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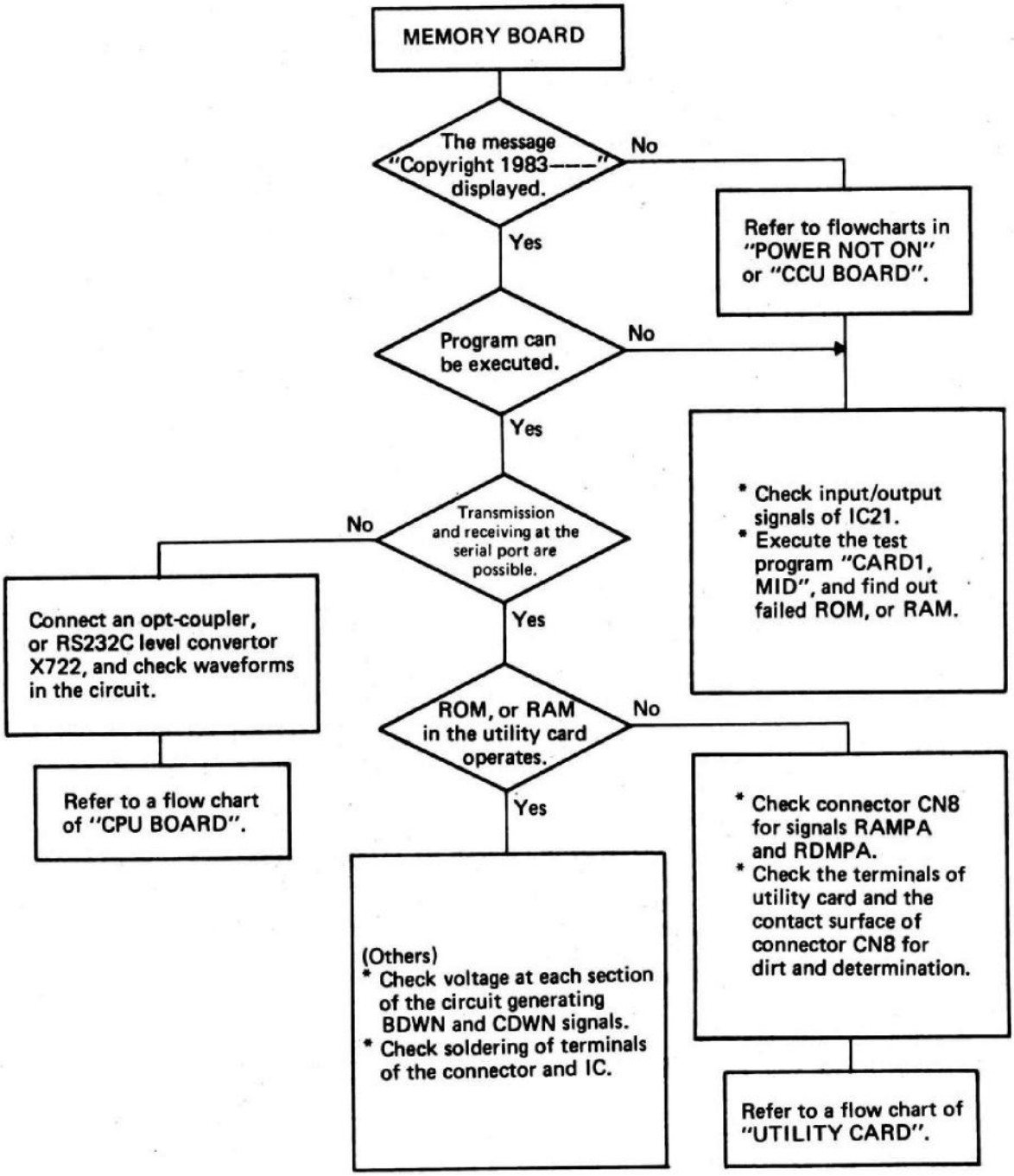
C241
C242
C243
C244
C245
C246
C247

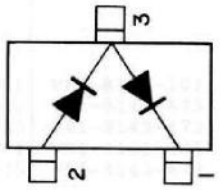
N	KEY NO.	PART NO.	Q'TY	DESCRIPTION		REMARKS	PRICE	
	IC28	WA3-0955-000	1	CMOS LSI HM6117LFP, RAM	2k-byte			
*	IC29	WA3-9168-000	1	CMOS LSI TC5332P, ROM, BASIC	4k-byte			
	R201	VR1-8143-101	1	SOLID RESISTOR	100 ohm 1/4W			
	R202	VR1-8143-473	1	SOLID RESISTOR	47K 1/4W			
	R203	VR1-8143-473	1	SOLID RESISTOR	47K 1/4W			
	R204	VR1-1181-101	1	CARBON RESISTOR	100 ohm 1/8W			
	R205	VR1-8143-473	1	SOLID RESISTOR	47K 1/4W			
	R206	VR1-1181-473	1	CARBON RESISTOR	47K 1/8W			
	R207	VR1-1181-103	1	CARBON RESISTOR	10K 1/8W			
	R208	VR1-1181-473	1	CARBON RESISTOR	47K 1/8W			
	R209	VR1-1181-332	1	CARBON RESISTOR	3.3K 1/8W			
	R210	VR1-1181-103	1	CARBON RESISTOR	10K 1/8W			
	R211	VR1-8143-103	1	SOLID RESISTOR	10K 1/4W			
	R212	VR1-1181-473	1	CARBON RESISTOR	47K 1/8W			
	R213	VR1-8143-103	1	SOLID RESISTOR	10K 1/4W			
	R214	VR1-1181-103	1	CARBON RESISTOR	10K 1/8W			
	R215	VR1-1143-204	1	CARBON RESISTOR	200K 1/4W			
	R216	VR1-8143-103	1	SOLID RESISTOR	10K 1/4W			
	R217	VR9-2781-000	1	METAL FILM RESISTOR	1M 1/8W	+/- 1%		
	R218	VR9-2780-000	1	METAL FILM RESISTOR	1.3M 1/8W	+/- 1%		
	R219	VR9-2780-000	1	METAL FILM RESISTOR	1.3M 1/8W	+/- 1%		
	R220	VR1-1143-204	1	CARBON RESISTOR	200K 1/4W			
	R221	VR9-2781-000	1	METAL FILM RESISTOR	1M 1/8W	+/- 1%		
	R222	VR1-1143-204	1	CARBON RESISTOR	200K 1/4W			
	R223	VR1-1143-204	1	CARBON RESISTOR	200K 1/4W			
	R224	VR1-1143-395	1	CARBON RESISTOR	3.9M 1/4W			
*	SW3	WC3-0039-000	1	SWITCH, SSS312		LOCK SWITCH		
	TR201	WA2-0240-000	1	TRANSISTOR 2SA1115(F)				
	TR202	WA2-0240-000	1	TRANSISTOR 2SA1115(F)				

N	KEY NO.	PART NO.	Q'TY	DESCRIPTION	REMARKS	PRICE	
	TR203	WA2-0113-000	1	TRANSISTOR 2SC2603(D,E,F)			
	TR204	WA2-0113-000	1	TRANSISTOR 2SC2603(D,E,F)			
	TR205	WA2-0240-000	1	TRANSISTOR 2SA1115(F)			


MEMORY BOARD
The message "Copyright 1983----" displayed.
No
Refer to flowcharts in "POWER NOT ON" or "CCU BOARD".
Yes
Program can be executed.
No
Yes
* Check input/output signals of IC21.
* Execute the test program "CARD1, MID", and find out failed ROM, or RAM.
Transmission and receiving at the serial port are possible.
No
Yes
Connect an opt-coupler, or RS232C level convertor X722, and check waveforms in the circuit.
Refer to a flow chart of "CPU BOARD".
ROM, or RAM in the utility card operates.
No
Yes
* Check connector CN8 for signals RAMPA and RDMPA.
* Check the terminals of utility card and the contact surface of connector CN8 for dirt and determination.
Refer to a flow chart of "UTILITY CARD".
(Others)
* Check voltage at each section of the circuit generating BDWN and CDWN signals.
* Check soldering of terminals of the connector and IC.

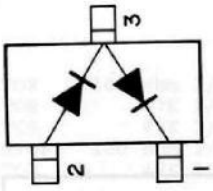
10. IC & TR TERMINALS

WA1-0357-000
DIODE MA153



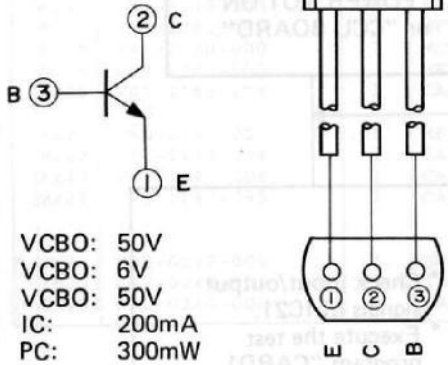
I forward: 0.2A
I rectified: 0.1A
V reverse: 40V

WA1-0380-000
DIODE MA157



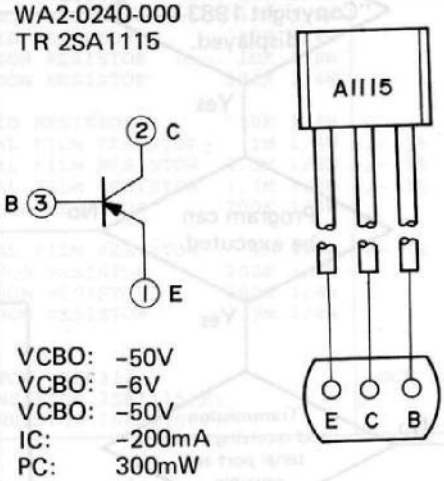
I forward: 0.225A
I rectified: 0.1A
V reverse: 40V

WA2-0113-000
TR 2SC2603



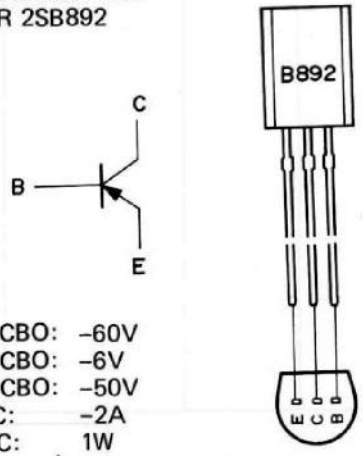
VCBO: 50V
VCBO: 6V
VCBO: 50V
IC: 200mA
PC: 300mW

WA2-0240-000
TR 2SA1115



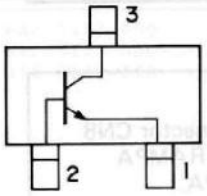
VCBO: –50V
VCBO: –6V
VCBO: –50V
IC: –200mA
PC: 300mW

WA2-0297-000
TR 2SB892



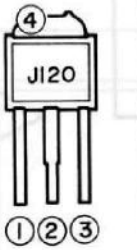
VCBO: –60V
VCBO: –6V
VCBO: –50V
IC: –2A
PC: 1W

WA2-0343-000
TR 2SC2712

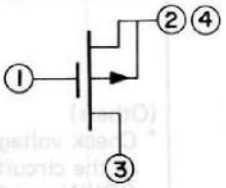


VCBO: 60V, VCBO: 50V
VCBO: 5V, IC: 150mA
PC: 150mW

WA2-0375-000
TR 2SJ120

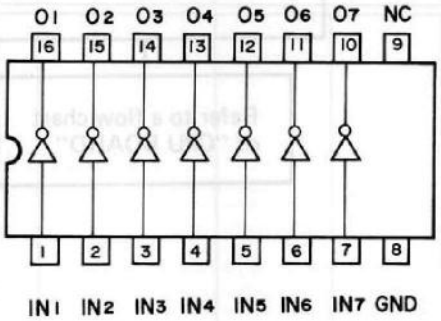


① GATE
②, ④ DRAIN
③ SOURCE

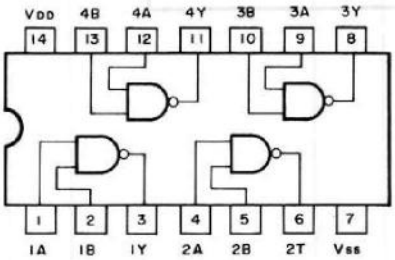


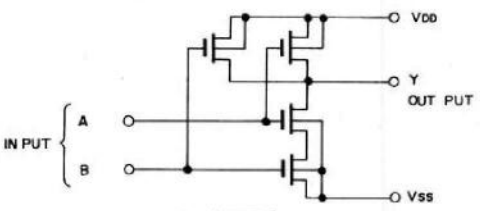
VDSS: –40V
VGSS: ±20V
ID: –2A
Pch: 10W

WA2-0399-000
TR ARRAY M54514AP



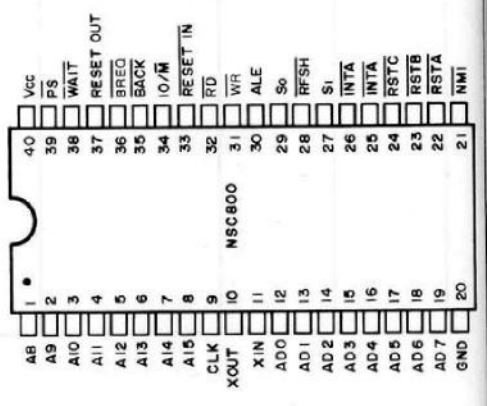
WA3-0800-000 NAND GATE TC40H000F



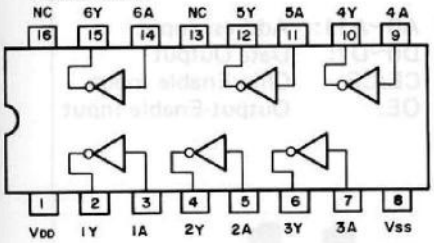


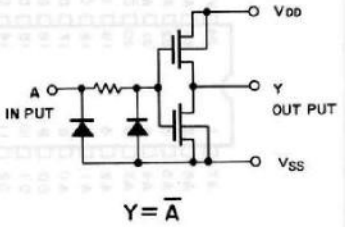
$Y = \overline{A} \times B$

WA3-0956-000
CPU NSC800



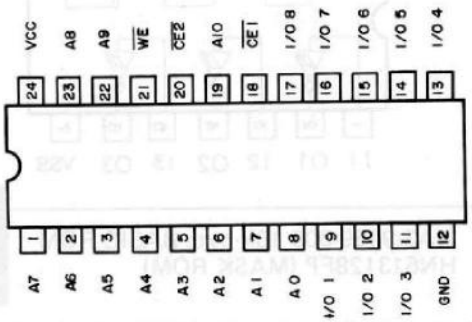
WA3-0957-000 INVERTER
TC50H000F



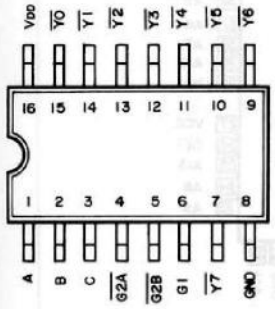


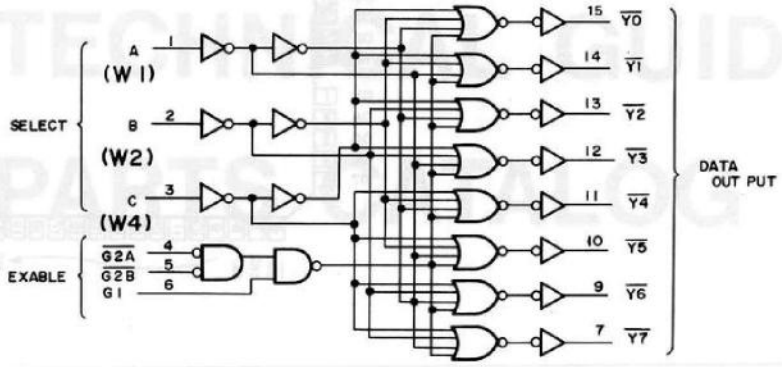
$Y = \overline{A}$

WA3-0955-000
STATIC RAM 2k byte
HM6117LFP-4

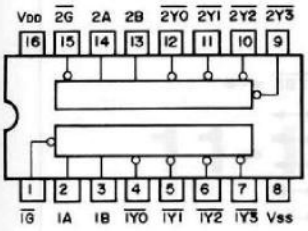


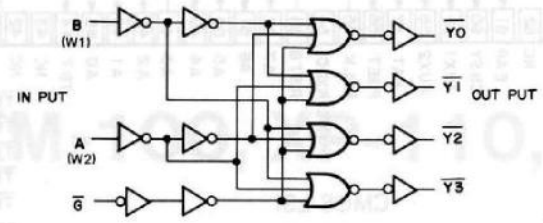
WA3-0958-000 DECODER TC40H138F



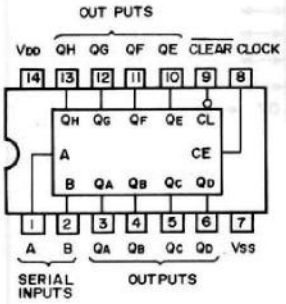


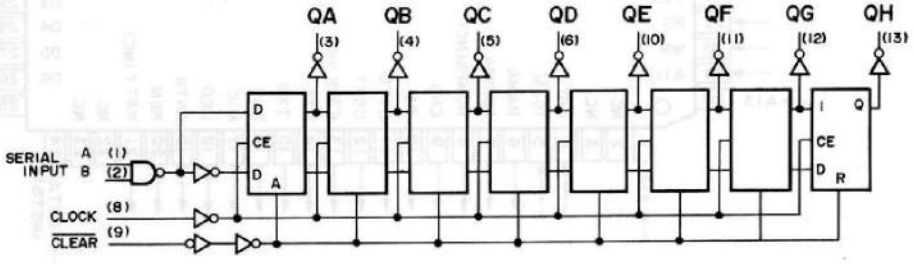
WA3-0959-000 DECODER TC40H139P



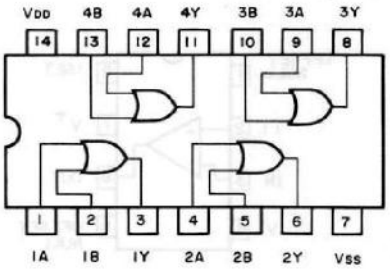


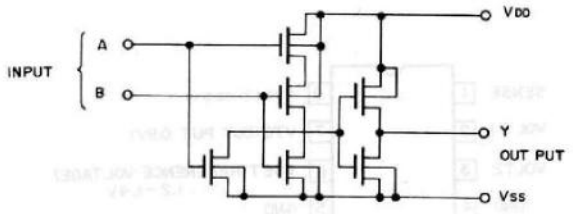
WA3-0960-000 8-bit SHIFT REGISTER (SERIAL-IN, PARALLEL-OUT) TC40H164F





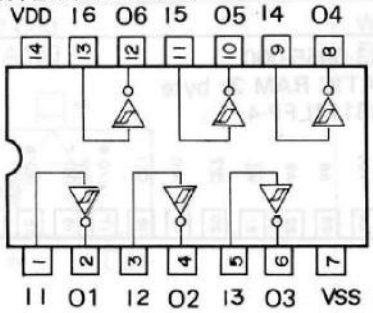
WA3-0963-000 TC40H032F



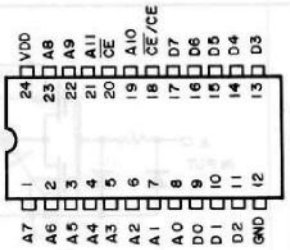


$Y = A + B$

WA3-0976-000
INVERTER HD14584B

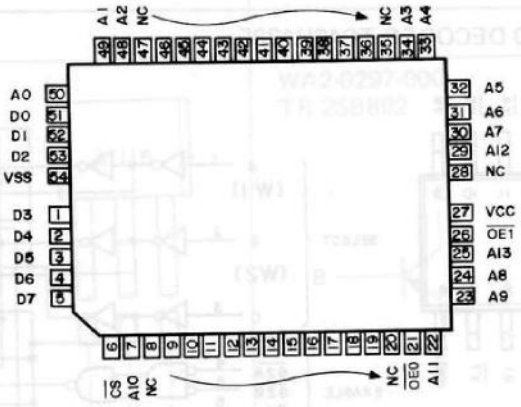


WA3-9168-000 4k-byte BASIC ROM
TC5332P (MASK ROM)

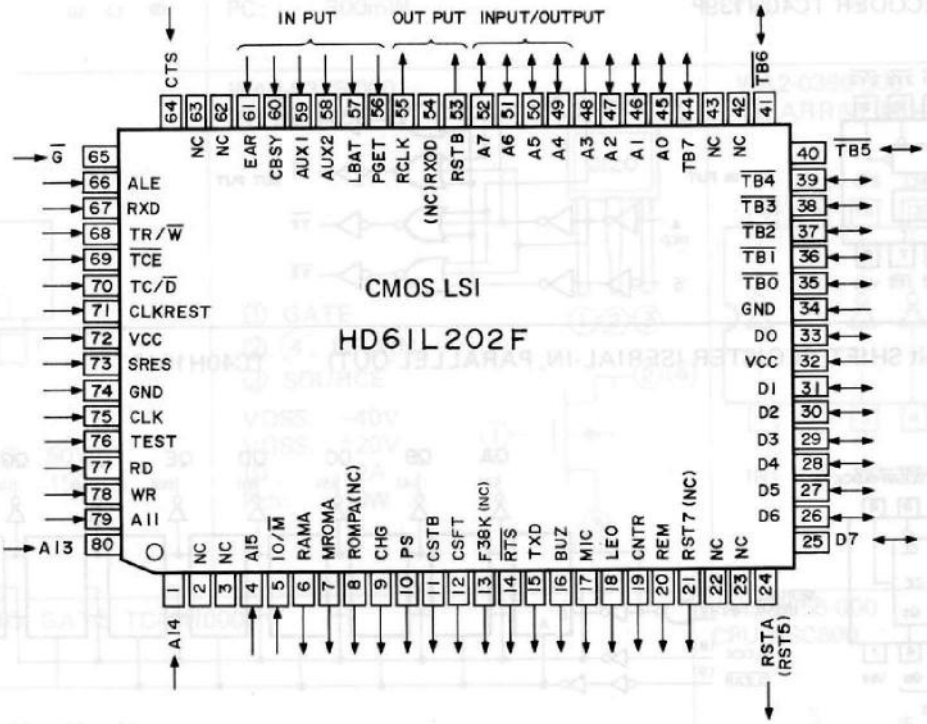


A0~A11: Address Input
D0~D7: Data Output
CE/CE: Chip Enable Input
OE: Output-Enable Input

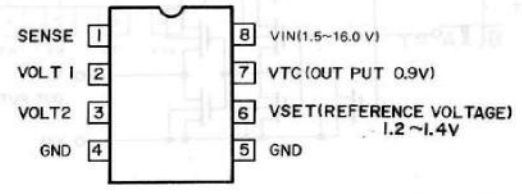
WA3-9169-000 16k-byte BASIC ROM
HN613128FP (MASK ROM)



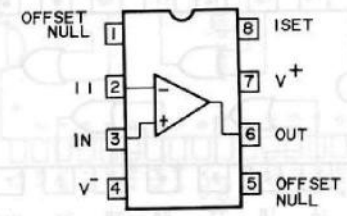
WA3-9170-000 INTERFACE LSI HD61L202F



WA4-0261-000
CMOS VOLTAGE REGULATOR
ICL7663CPA



WA4-0283-000
AMPLIFIER μPC4250C



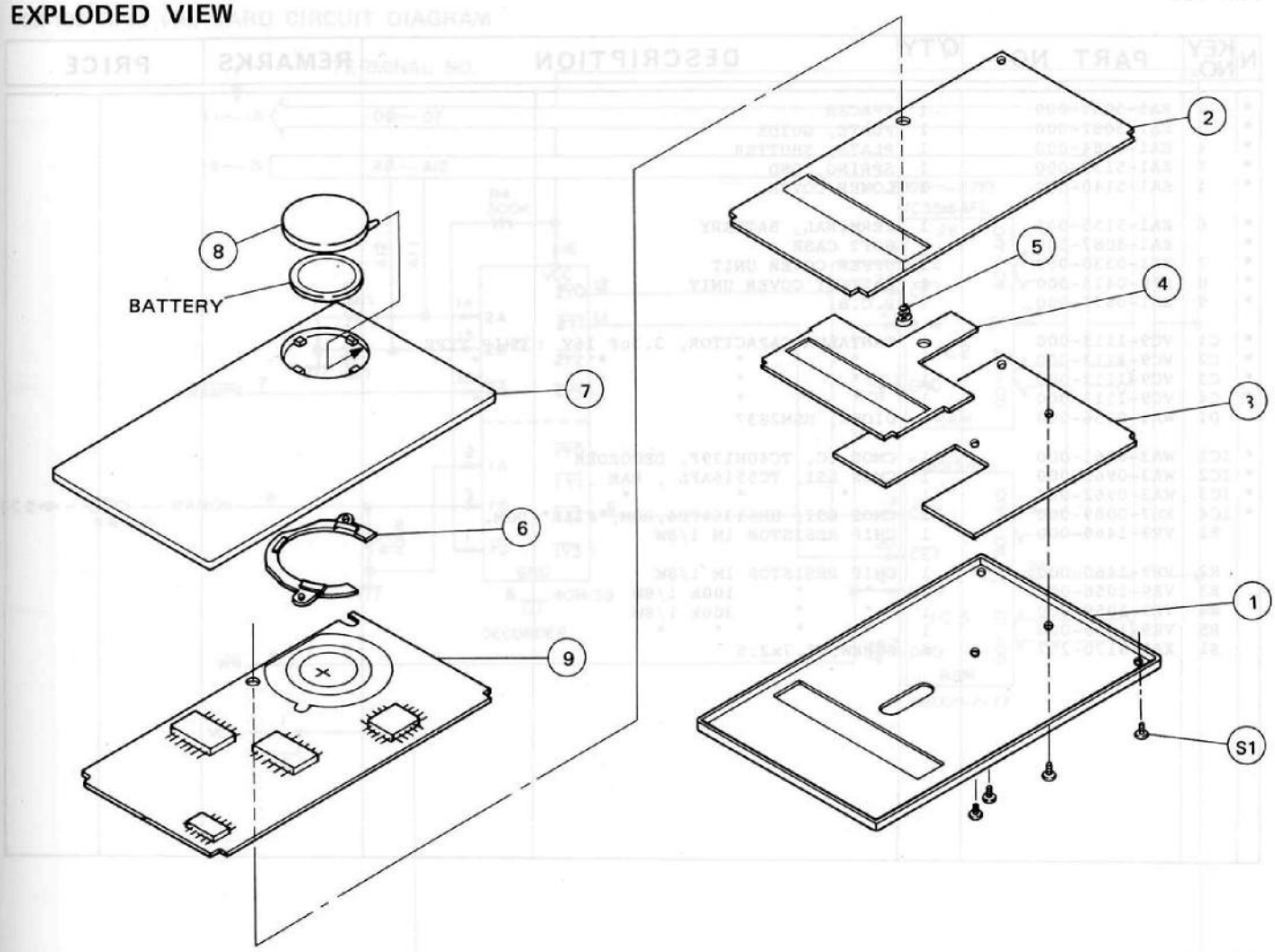
11. OPTION

TECHNICAL GUIDE

PARTS CATALOG

11-1 XM-100, XP-110, XP-120

EXPLODED VIEW



XP-120

N	KEY NO.	PART NO.	Q'TY	DESCRIPTION		REMARKS	PRICE	
*	1	EA1-5080-000	1	LOWER COVER				
*	2	EA1-5081-000	1	SPACER				
*	3	EA1-5082-000	1	PLATE, GUIDE				
*	4	EA1-5084-000	1	PLATE, SHUTTER				
*	5	EA1-5135-000	1	SPRING, GND				
*	6	EA1-5155-000	1	TERMINAL, BATTERY				
*		EA1-8087-000	1	SOFT CASE				
*	7	EF1-0328-000	1	UPPER COVER UNIT				
*	8	EF9-0415-000	1	BATTERY COVER UNIT				
*	9	EH1-0677-000	1	P.C.B.				
*	C1	VC9-1113-000	1	TANTALUM CAPACITOR,3.3uF 16V	CHIP TYPE			
*	C2	VC9-1113-000	1	" " " "	"			
*	C3	VC9-1113-000	1	" " " "	"			
*	C4	VC9-1113-000	1	" " " "	"			
	D1	WA1-0356-000	1	DIODE, HSM2837				
*	IC1	WA3-0961-000	1	CMOS IC, TC40H139F, DECORDER				
*	IC2	WA3-0962-000	1	CMOS LSI, TC5516AFL, RAM				
*	IC3	WA3-0962-000	1	" " "				
*	IC4	EH7-0087-000	1	CMOS LSI,HN61364FP4,ROM,"TABLE"	DOM.			
	R1	VR9-1460-000	1	CHIP RESISTOR, 1M 1/8W				
	R2	VR9-1460-000	1	CHIP RESISTOR, 1M 1/8W				
	R3	VR9-1058-000	1	" , 100k 1/8W				
	R4	VR9-1059-000	1	" , 300k 1/8W				
	R5	VR9-1059-000	1	" , 300k 1/8W				
	S1	XA1-3170-257	6	SCREW, 1.7x2.5				

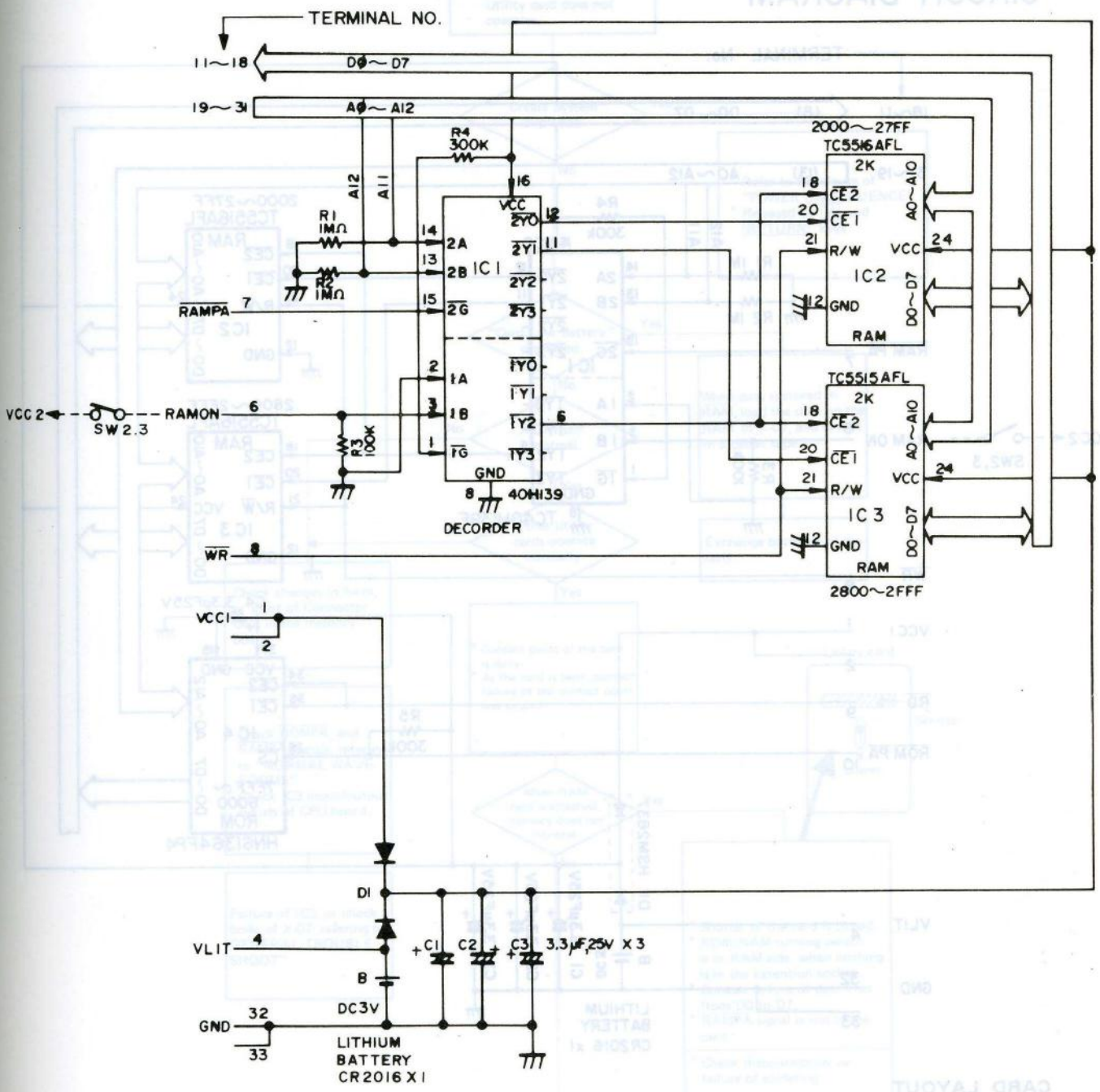
XM-100

N	KEY NO.	PART NO.	Q'TY	DESCRIPTION		REMARKS	PRICE	
*	2	EA1-5081-000	1	SPACER				
*	3	EA1-5082-000	1	PLATE, GUIDE				
*	4	EA1-5084-000	1	PLATE, SHUTTER				
*	5	EA1-5135-000	1	SPRING, GND				
*	1	EA1-5140-000	1	LOWER COVER				
*	6	EA1-5155-000	1	TERMINAL, BATTERY				
*		EA1-8087-000	1	SOFT CASE				
*	7	EF1-0330-000	1	UPPER COVER UNIT				
*	8	EF9-0415-000	1	BATTERY COVER UNIT				
*	9	EH1-0677-000	1	P.C.B.				
*	C1	VC9-1113-000	1	TANTALUM CAPACITOR, 3.3uF 16V	CHIP TYPE			
*	C2	VC9-1113-000	1	" " "	"			
*	C3	VC9-1113-000	1	" " "	"			
*	C4	VC9-1113-000	1	" " "	"			
	D1	WA1-0356-000	1	DIODE, HSM2837				
*	IC1	WA3-0961-000	1	CMOS IC, TC40H139F, DECORDER				
*	IC2	WA3-0962-000	1	CMOS LSI, TC5516AFL , RAM				
*	IC3	WA3-0962-000	1	" " "				
*	IC4	EH7-0089-000	1	CMOS LSI, HN61364FP4,ROM,"FILE"	DOM.			
	R1	VR9-1460-000	1	CHIP RESISTOR 1M 1/8W				
	R2	VR9-1460-000	1	CHIP RESISTOR 1M 1/8W				
	R3	VR9-1058-000	1	" " 100k 1/8W				
	R4	VR9-1059-000	1	" " 300k 1/8W				
	R5	VR9-1059-000	1	" " " "				
	S1	XA1-3170-257	6	SCREW, 1.7x2.5				

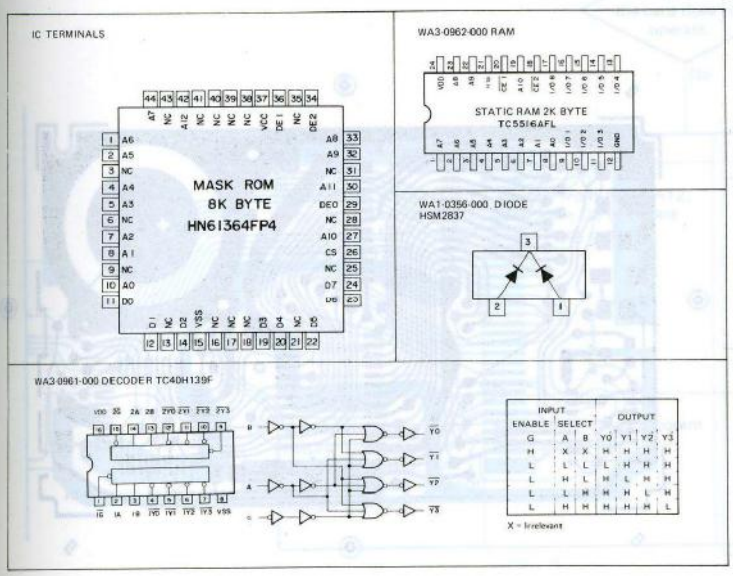
XP-110

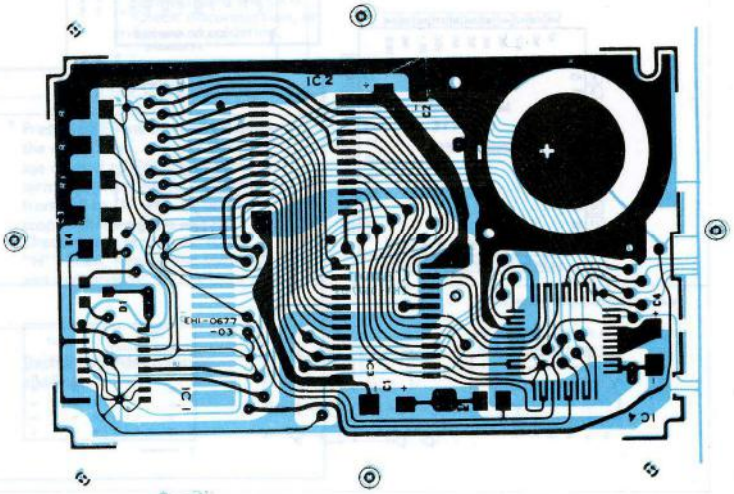
N	KEY NO.	PART NO.	Q'TY	DESCRIPTION		REMARKS	PRICE	
*	2	EA1-5081-000	1	SPACER				
*	3	EA1-5082-000	1	PLATE, GUIDE				
*	4	EA1-5084-000	1	PLATE, SHUTTER				
*	5	EA1-5135-000	1	SPRING, GND				
*	1	EA1-5144-000	1	LOWER COVER				
*	6	EA1-5155-000	1	TERMINAL, BATTERY				
*		EA1-8087-000	1	SOFT CASE				
*	7	EF1-0332-000	1	UPPER COVER UNIT				
*	8	EF9-0415-000	1	BATTERY COVER UNIT				
*	9	EH1-0677-000	1	P.C.B.				
*	C1	VC9-1113-000	1	TANTALUM CAPACITOR, 3.3uF 16V	CHIP TYPE			
*	C2	VC9-1113-000	1	" " "	"			
*	C3	VC9-1113-000	1	" " "	"			
	D1	WA1-0356-000	1	DIODE, HSM2837				
*	IC1	WA3-0961-000	1	CMOS IC, TC40H139F, DECORDER				
*	IC2	WA3-0962-000	1	CMOS LSI, TC5516AFL, RAM				
*	IC3	WA3-0962-000	1	" " "				
	R1	VR9-1460-000	1	CHIP RESISTOR, 1M 1/8W				
	R2	VR9-1460-000	1	" "				
	R3	VR9-1058-000	1	" 100k 1/8W				
	R4	VR9-1059-000	1	CHIP RESISTOR, 300k 1/8W				
	S1	XA1-3170-257	6	SCREW, 1.7x2.5				

XM-100 RAM (4k) CARD CIRCUIT DIAGRAM

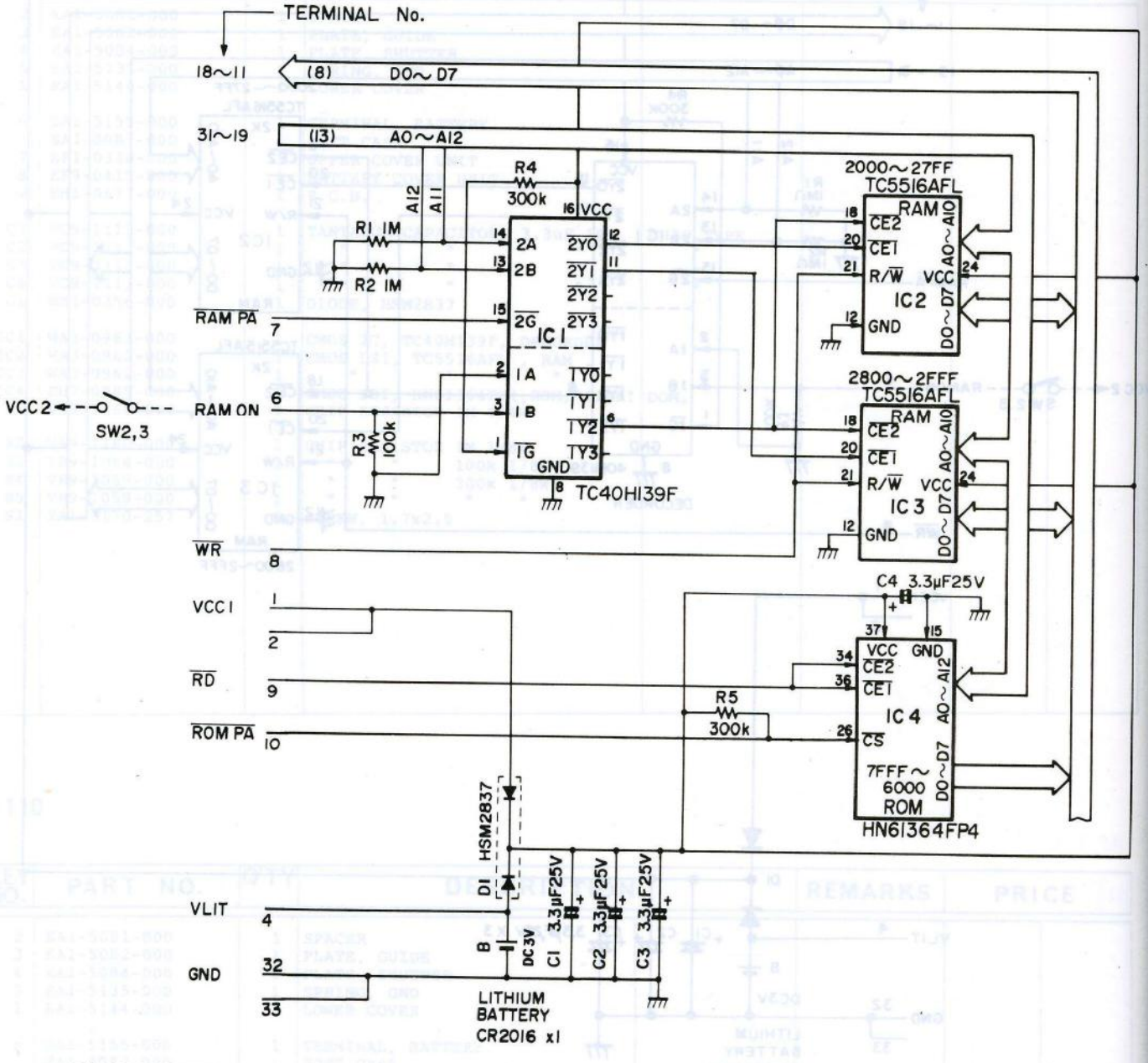


CARD LAYOUT

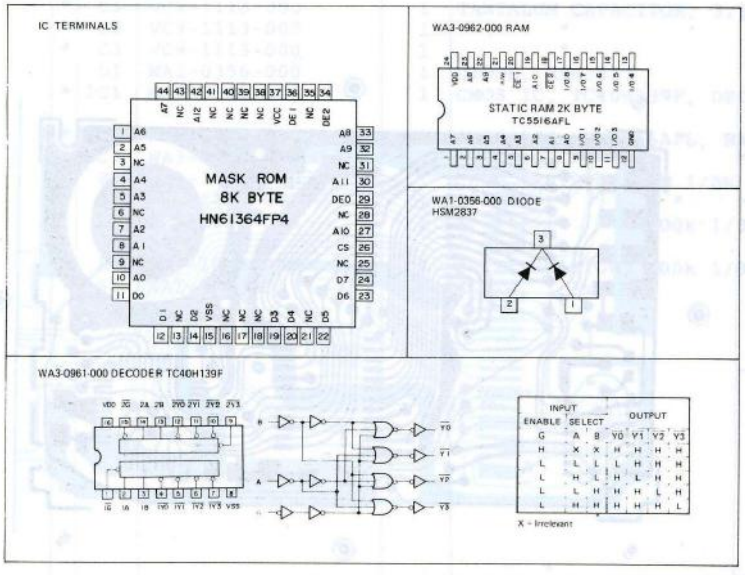


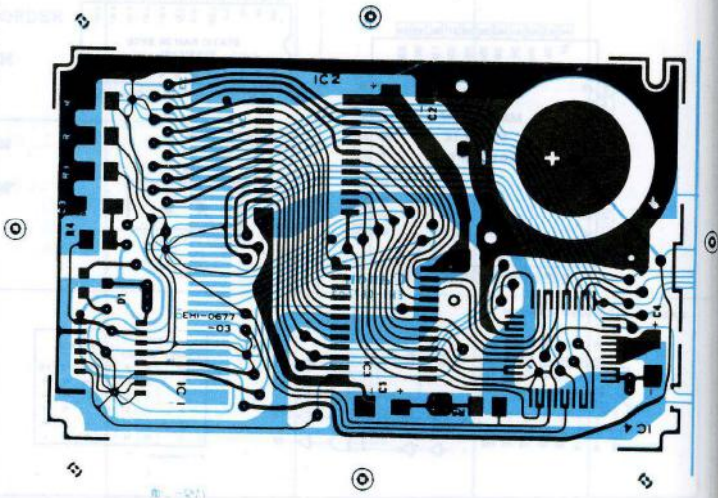


XP-110, XP-120
CIRCUIT DIAGRAM

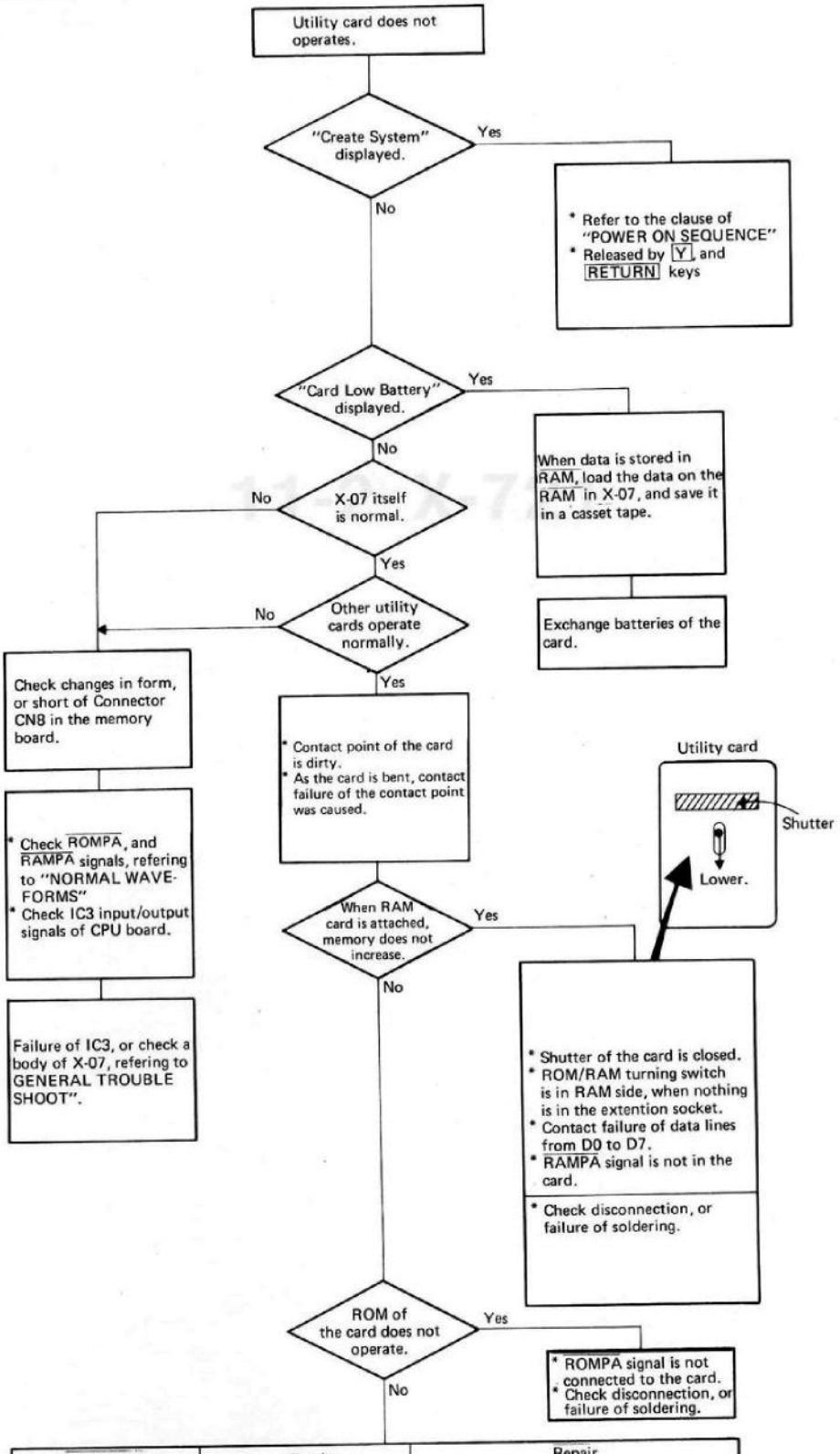


CARD LAYOUT





TROUBLESHOOTING

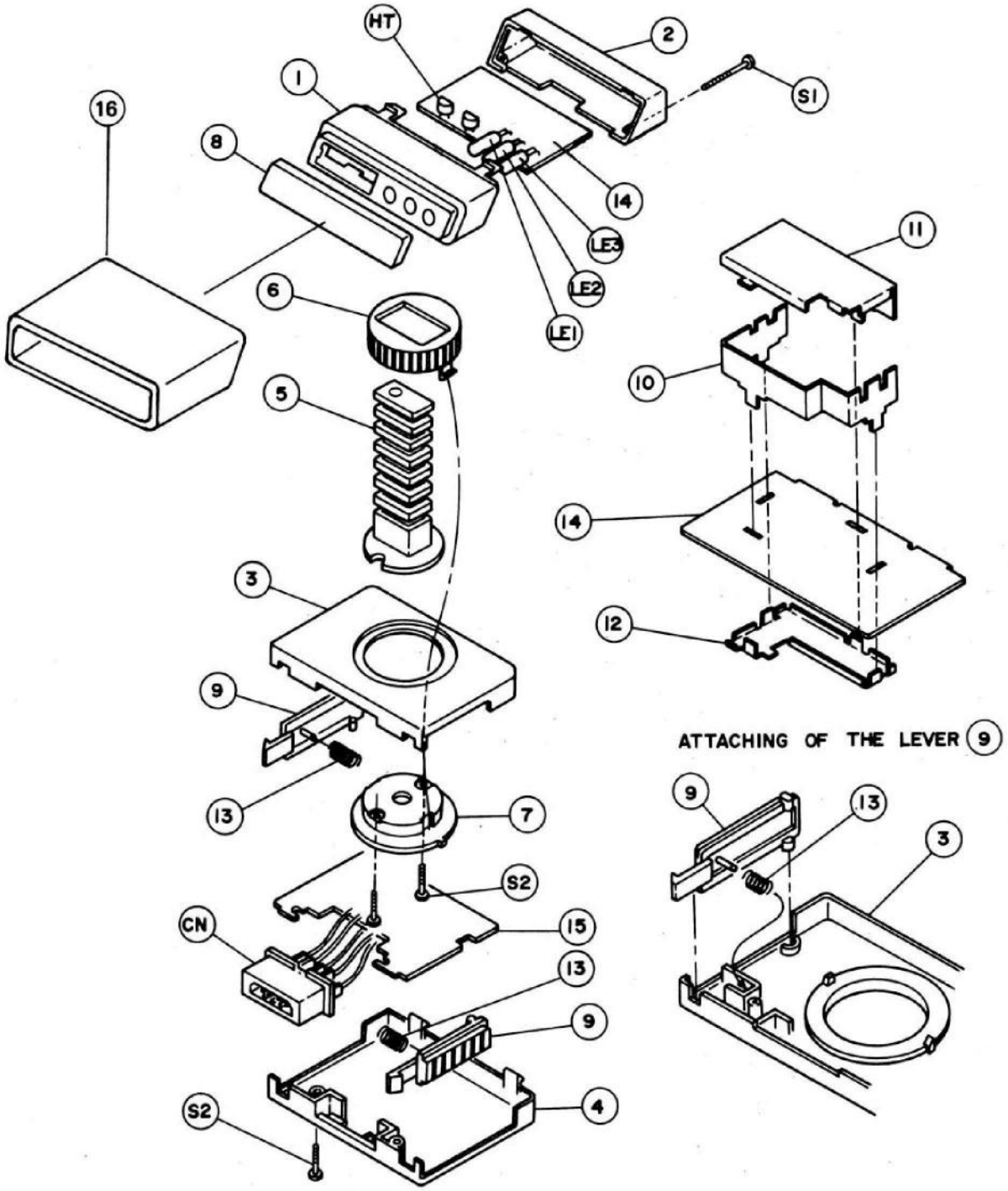


Phenominon	Check	Repair
When power source is on, it can not be switched off by OFF key.(The message "Copyright 1983———" not displayed)	Signals from A0 to A12, and from D0 to D7 are kept being output.	* Press RESET switch and take out the card. Then, connect the package of card to GND, and observe terminals from A0 to A12, and from D0 to D7 with an oscilloscope. Check signals in "H". * Check which terminals' IC are "H". Then, remove the terminals, and exchange IC in "H".
The message "Copyright 1983———" is displayed when power is on, but RAM of the card does not operate normally.	Execute the check program "JMEM. X07".	Exchange RAMs of error addresses.

11-2 X-721

11-2) Optical Coupler X-721

EXPLODED VIEW

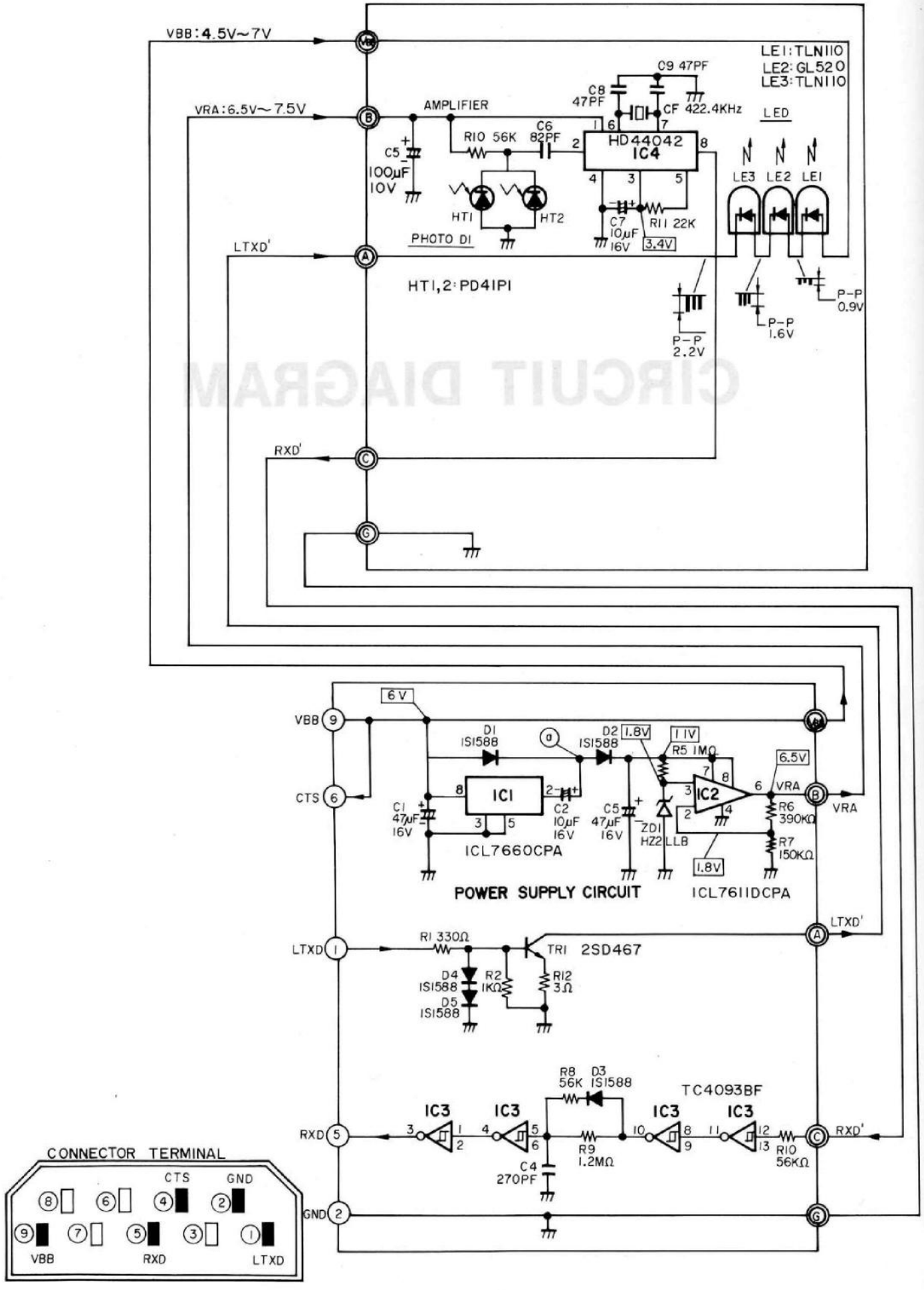


N	KEY NO.	PART NO.	Q'TY	DESCRIPTION		REMARKS	PRICE	
*	1	EA1-5062-000	1	COVER(A)				
*	2	EA1-5063-000	1	COVER(B)				
*	3	EA1-5064-000	1	UPPER COVER,BASE				
*	4	EA1-5065-000	1	LOWER COVER,BASE				
*	5	EA1-5066-000	1	SNAKE SHAFT,RUBBER				
*	6	EA1-5067-000	1	BUSHING(A),ROTATORY				
*	7	EA1-5068-000	1	BUSHING(B),ROTATORY				
*	8	EA1-5069-000	1	ACRYL,WINDOW				
*	9	EA1-5070-000	2	LEVER,ATTACHING				
*	10	EA1-5071-000	1	PLATE,SHIELD				
*	11	EA1-5072-000	1	SHIELD COVER(A)				
*	12	EA1-5073-000	1	SHIELD COVER(B)				
*	13	EA1-5074-000	2	COIL SPRING				
*	14	EH1-0674-000	1	P.C.B, AMPLIFIER				
*	15	EH1-0675-000	1	P.C.B, POWER SUPPLY				
*	16	EA1-5348-000	1	HOOD, RUBBER				
*		EA1-5352-000	1	RUBBER SHEET, CONDUCTIVE				
*	C 1	VC9-1210-000	1	ELECTROLYTIC CAPACITOR	47uF 16V			
*	C 2	VC9-1112-000	1	ELECTROLYTIC CAPACITOR	10uF 16V			
	C 3							
	C 4	VC4-3503-271	1	CERAMIC CAPACITOR	270PF 50V			
*	C 5	VC9-1210-000	1	ELECTROLYTIC CAPACITOR	47uF 16V			
*	C 6	VC4-2502-820	1	CERAMIC CAPACITOR	82PF 50V			
	C 7							
	C 8	VC4-2502-470	1	CERAMIC CAPACITOR	47PF 50V			
	C 9	VC4-2502-470	1	CERAMIC CAPACITOR	47PF 50V			
	CF	WK2-9002-000	1	CERAMIC FILTER 422.4kHz				
*	CN	WS1-0222-000	1	CONNECTOR, 9P				
	D 1	X65-5032-000	1	DIODE, 1S1588				
	D 2	X65-5032-000	1	DIODE, 1S1588				
	D 3	X65-5032-000	1	DIODE, 1S1588				
	D 4	X65-5032-000	1	DIODE, 1S1588				
	D 5	X65-5032-000	1	DIODE, 1S1588				

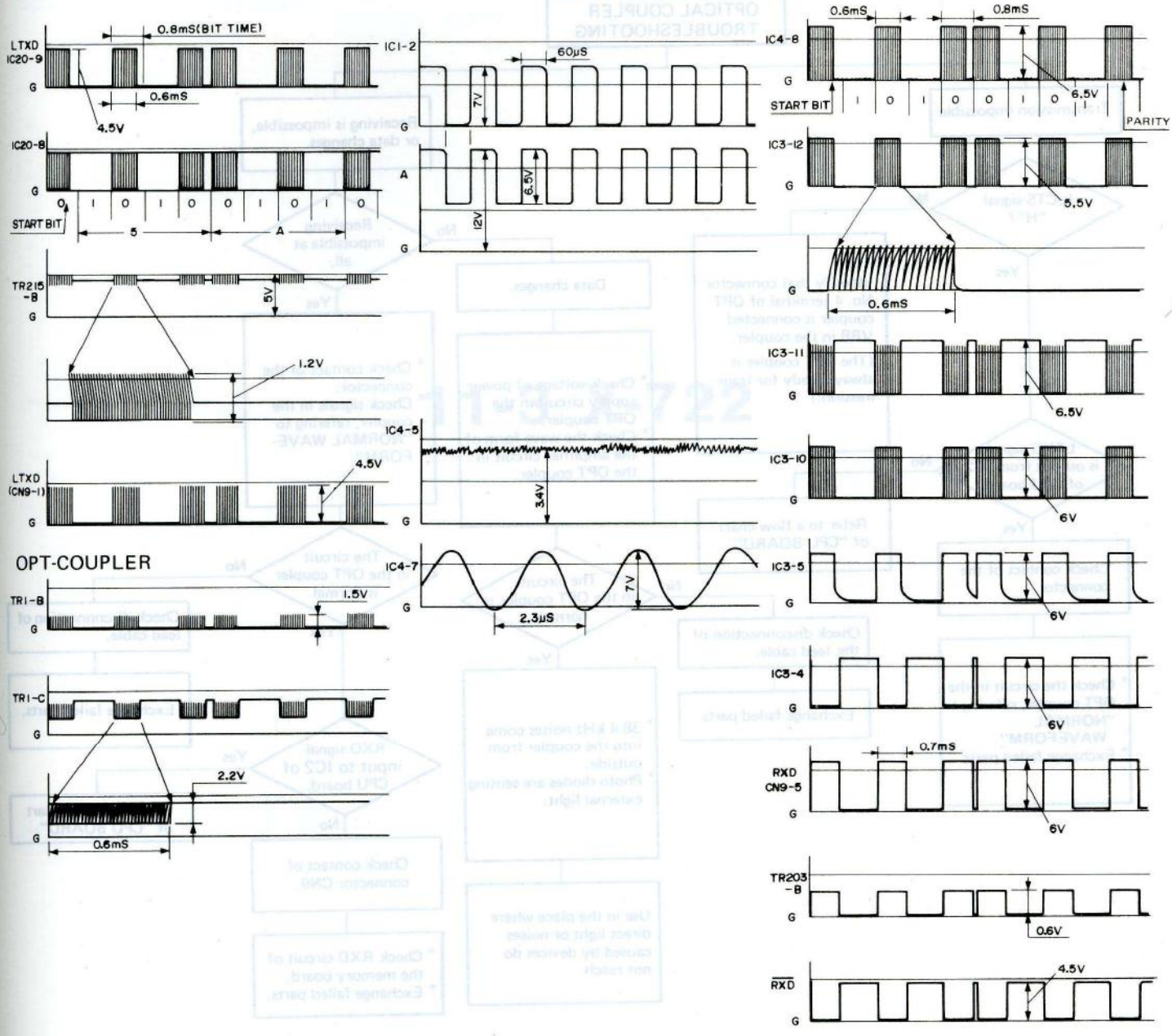
CIRCUIT DIAGRAM

N	KEY NO.	PART NO.	Q'TY	DESCRIPTION		REMARKS	PRICE	
	HT1	WG3-0025-000	1	PHOTO DIODE, PD-41PI				
	HT2	WG3-0025-000	1	PHOTO DIODE, PD-41PI				
	IC1	WA4-0262-000	1	CMOS IC L7660CPA				
	IC2	WA4-0263-000	1	CMOS IC L7611DCPA				
	IC3	WA3-0982-000	1	CMOS IC TC4093BF				
	IC4	WA4-0260-000	1	CMOS IC HD44042				
*	LE1	WG1-0189-000	1	LED TLN110				
	LE2	WG1-0171-000	1	LED GL-520				
*	LE3	WG1-0189-000	1	LED TLN110				
	R 1	VR1-8143-331	1	SOLID RESISTOR	330 ohm 1/4W			
	R 2	VR1-8143-102	1	SOLID RESISTOR	1k 1/4W			
	R 3							
	R 4							
	R 5	VR1-8143-105	1	SOLID RESISTOR	1M 1/4W			
	R 6	VR9-2980-000	1	RESISTOR	390k 1/4W			
	R 7	VR9-2779-000	1	RESISTOR	150k 1/4W			
	R 8	VR1-8143-563	1	SOLID RESISTOR	56k 1/4W			
	R 9	VR1-1143-125	1	CARBON RESISTOR	1.2M 1/4W			
	R10	VR1-8143-563	1	SOLID RESISTOR	56k 1/4W			
	R11	VR9-1037-000	1	RESISTOR	22k 1/4W			
	R12	VR1-1143-309	1	CARBON RESISTOR	3 ohm 1/4W			
	TR1	X65-6369-000	1	TRANSISTOR 2SD467				
	ZD1	WA1-0287-000	1	ZENER DIODE HZ2LLB				
		SCREW						
	S 1	XA9-0183-000	2	SCREW, TAPPING, M2x28				
	S 2	XA3-1201-007	4	SCREW, TAPPING, PH2x10				

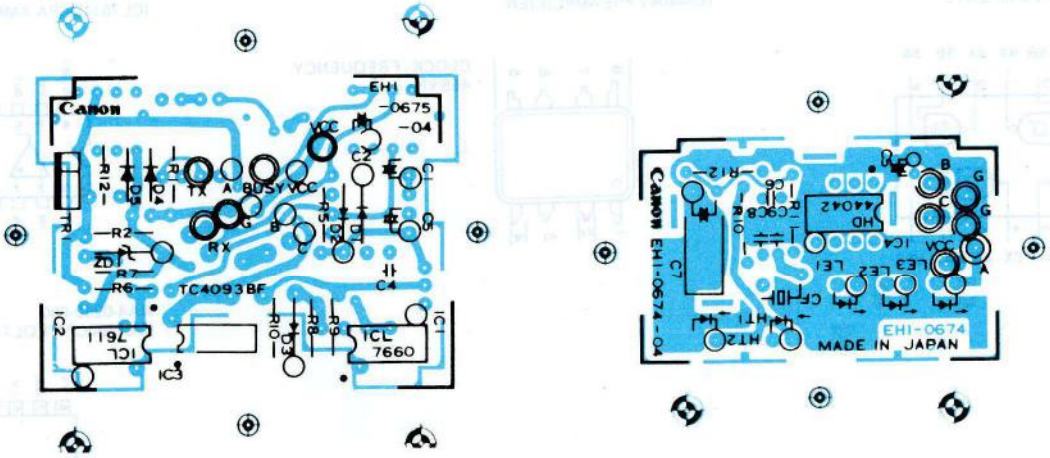
CIRCUIT DIAGRAM


VBB:4.5V~7V
VRA:6.5V~7.5V
LTXD'
RXD'
LE1:TLN110
LE2:GL520
LE3:TLN110
AMPLIFIER
LED
C9 47PF
C8 47PF
CF 422.4KHz
R10 56K
C6 82PF
HD44042
IC4
C5 100μF 10V
HT1
HT2
PHOTO DI
C7 10μF 16V
R11 22K
3.4V
LE3 LE2 LE1
HT1,2:PD41P1
P-P 2.2V
P-P 1.6V
P-P 0.9V
6V
D1 1S1588
D2 1S1588
1.8V
11V
R5 1MΩ
6.5V
IC1
IC2
VRA
C1 47μF 16V
C2 10μF 16V
C5 47μF 16V
ZD1 HZ2LLB
R6 390KΩ
R7 150KΩ
ICL7660CPA
POWER SUPPLY CIRCUIT
ICL7611DCPA
R1 330Ω
TR1 2SD467
D4 1S1588
D5 1S1588
R2 1KΩ
R12 3Ω
R8 56K
D3 1S1588
TC4093BF
IC3
R9 1.2MΩ
C4 270PF
R10 56KΩ
CONNECTOR TERMINAL
CTS
GND
VBB
RXD
LTXD

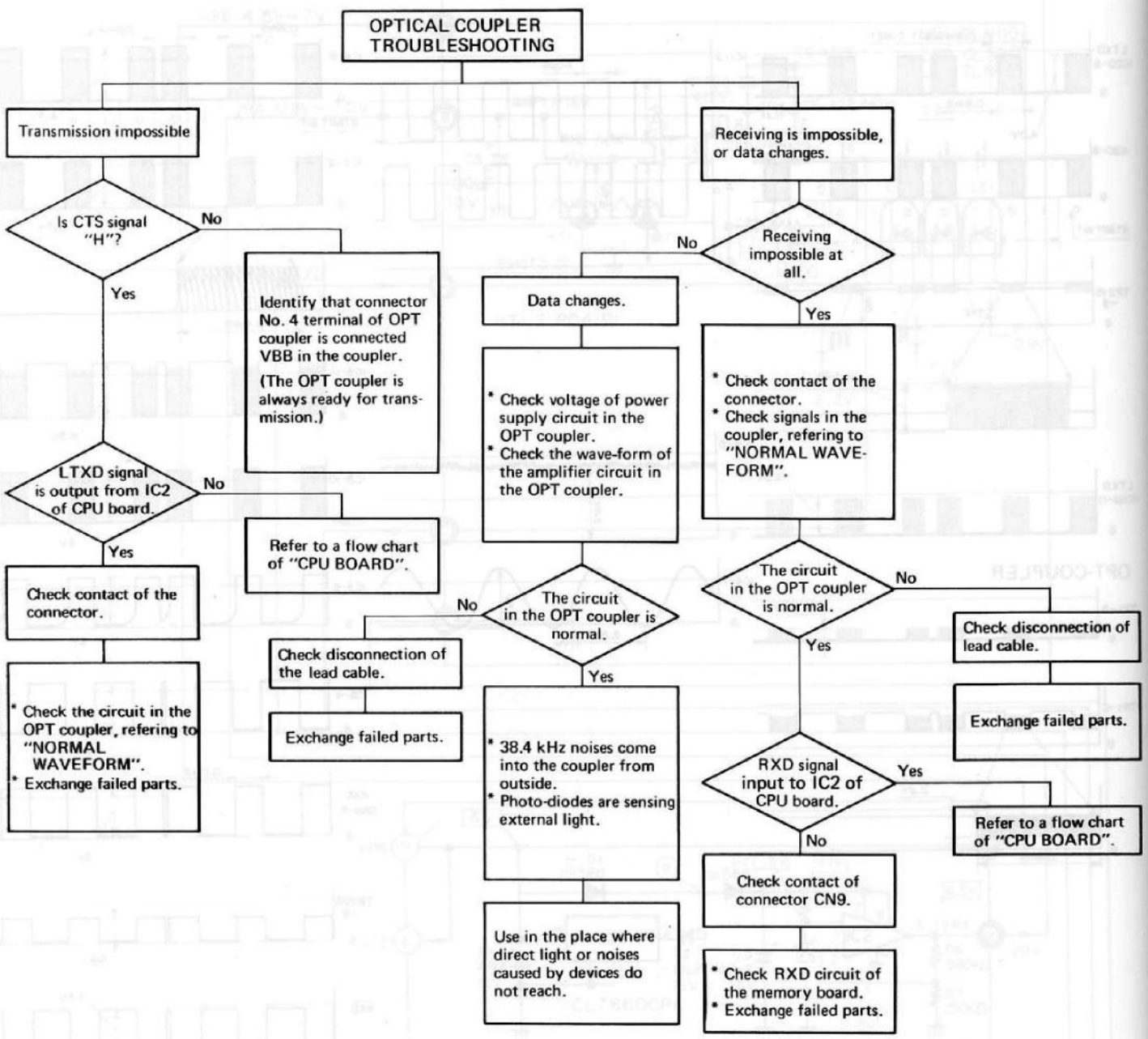
NORMAL WAVEFORM



CARD LAYOUT

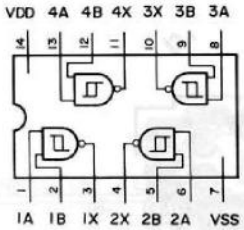


TROUBLESHOOTING

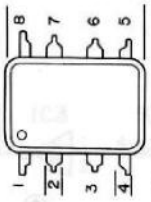


IC TERMINALS

WA3-0982-000
CMOS TC4093BF NAND GATE

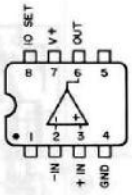


WA4-0260-000
HD44042 PREAMPLIFIER

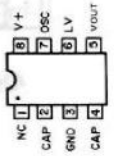


CLOCK FREQUENCY
455 kHz

WA4-0263-000
ICL7611DCPA AMPLIFIER



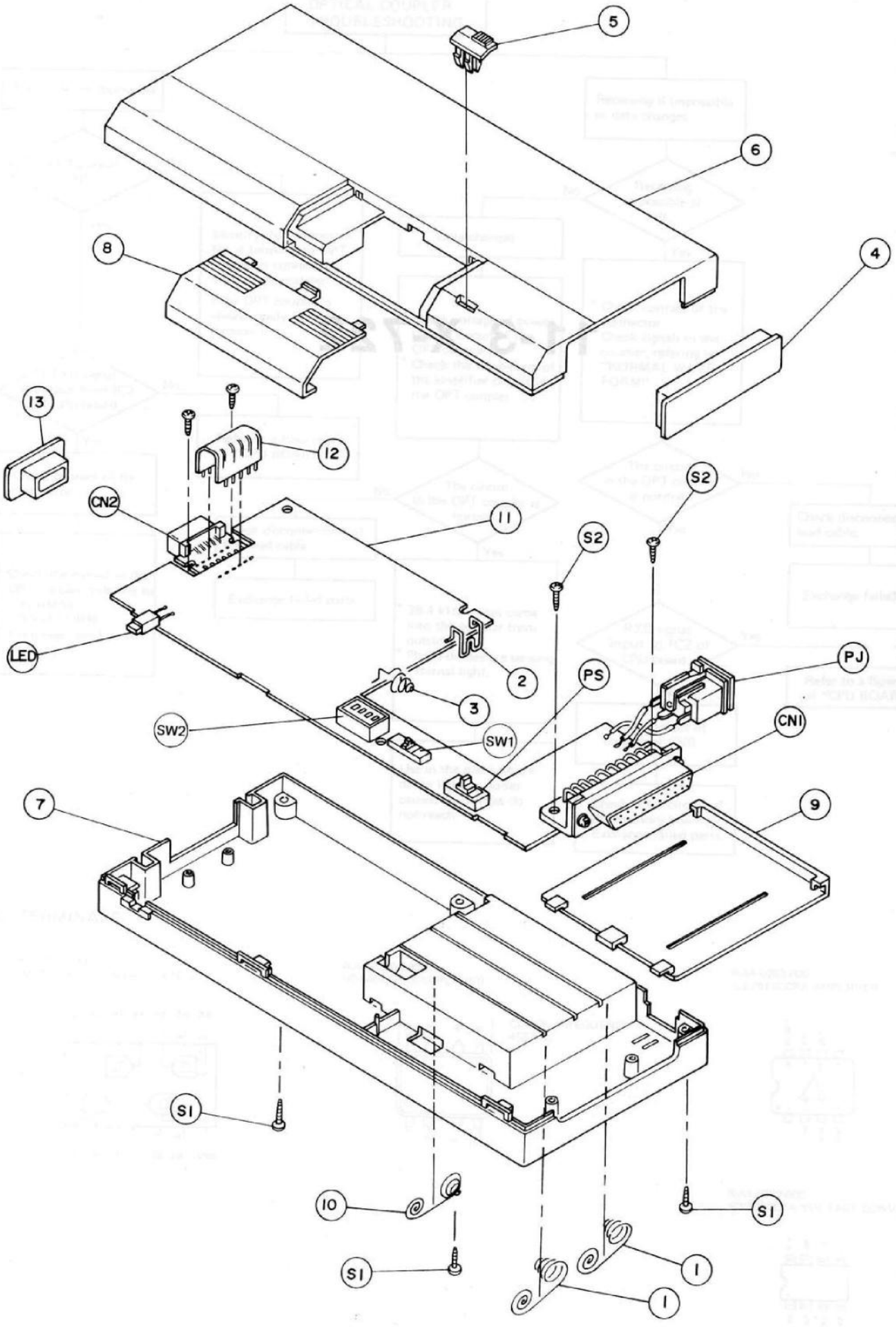
WA4-0262-000
ICL7660CPA VOLTAGE CONVERTER



11-3 X-722

11-3) RS232-C LEVEL CONVERTER X-722

EXPLODED VIEW



N	KEY NO.	PART NO.	Q'TY	DESCRIPTION	REMARKS	PRICE
	1	EA1-3750-000	2	TERMINAL, BATTERY,C1		
	2	EA1-4560-000	1	TERMINAL(+)		
	3	EA1-4561-000	1	" (-)		
	4	EA1-4866-000	1	CAP,CONNECTOR		
*	5	EA1-5061-000	1	KNOB,SWITCH		
*	6	EA1-5075-000	1	UPPER COVER		
*	7	EA1-5076-000	1	LOWER COVER		
*	8	EA1-5077-000	1	COVER, SWITCH		
*	9	EA1-5136-000	1	BATTERY COVER		
*		EA1-5318-000	1	PLATE(C),SHIELD		
	10	EA9-0119-000	1	TERMINAL,BATTERY,C2		
*	11	EH1-0676-000	1	P.C.B.		
*	12	EH2-0398-000	1	FLAT CABLE,8P		
*	13	WS9-0025-000	1	CAP,CONNECTOR		
	C1	VC9-1124-000	1	ELECTROLYTIC CAPACITOR,330uF 16V		
	C2	VC9-0822-000	1	ELECTROLYTIC CAPACITOR,47uF 16V		
	C3	VC9-0834-000	1	" ,33uF 16V		
	C4	VC9-1125-000	1	" ,22uF 16V		
	C5	VC4-3503-222	1	CERAMIC CAPACITOR 2200PF 50V		
*	C6	VC4-3503-101	1	" 100PF 50V		
*	C7	VC4-3503-101	1	CERAMIC CAPACITOR 100PF 50V		
*	C11	VC4-2502-820	1	CERAMIC CAPACITOR 82PF 50V		
	CF	WK2-9005-000	1	CERAMIC FILTER 460.8kHz		
*	CN1	VS1-0158-025	1	CONNECTOR,25P		
*	CN2	WS1-0221-000	1	" , 9P		
	D1	X65-5032-000	1	DIODE,1S1588		
	D2	X65-5032-000	1	" "		
	D3	X65-5032-000	1	" "		
	D4	X65-5032-000	1	" "		
	D5	WA1-0078-000	1	" ,ERB12-01		
	D6	WA1-0204-000	1	DIODE,1S954		
	D7	WA1-0267-000	1	" ,W06B		
	D8	WA1-0204-000	1	" ,1S954		

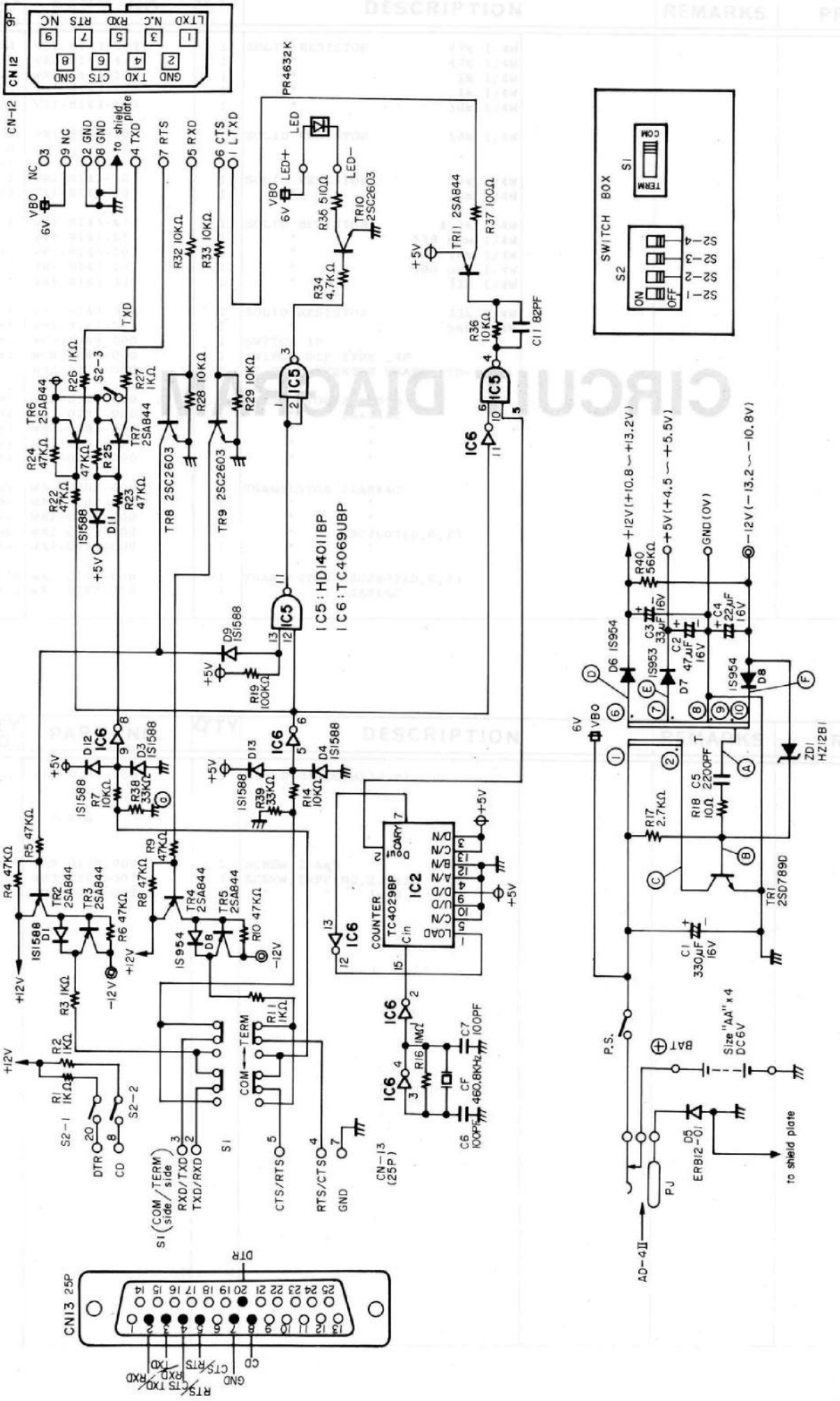
N	KEY NO.	PART NO.	Q'TY	DESCRIPTION	REMARKS	PRICE
	D9	X65-5032-000	1	DIODE,1S1588		
	D11	X65-5032-000	1	" , "		
	D12	X65-5032-000	1	" , "		
	D13	X65-5032-000	1	" , "		
	IC2	WA3-0311-000	1	CMOS IC,TC4029BP		
	IC5	WA3-0633-000	1	CMOS IC,HD14011BP		
	IC6	WA3-0561-000	1	CMOS IC,TC4069BP		
	LED1	WG1-0178-000	1	LED,PR4632K		
	PJ	WS6-9010-000	1	POWER JACK		
	PS	WC3-9013-000	1	SLIDE SWITCH,2P		
	R 1	VR1-8143-102	1	SOLID RESISTOR 1k 1/4W		
	R 2	VR1-8143-102	1	" 1k 1/4W		
	R 3	VR1-8143-102	1	" 1k 1/4W		
	R 4	VR1-8143-473	1	" 47k 1/4W		
	R 5	VR1-8143-473	1	" 47k 1/4W		
	R 6	VR1-8143-473	1	SOLID RESISTOR 47k 1/4W		
	R 7	VR1-8143-103	1	" 10k 1/4W		
	R 8	VR1-8143-473	1	" 47k 1/4W		
	R 9	VR1-8143-473	1	" 47k 1/4W		
	R10	VR1-8143-473	1	" 47k 1/4W		
	R11	VR1-8143-102	1	SOLID RESISTOR 1k 1/4W		
	R12					
	R13					
	R14	VR1-8143-103	1	SOLID RESISTOR 10k 1/4W		
	R15					
	R16	VR1-8143-105	1	SOLID RESISTOR 1M 1/4W		
	R17	VR1-8143-272	1	" 2.7k 1/4W		
	R18	VR1-8143-100	1	" 10 ohm 1/4W		
	R19	VR1-8143-104	1	" 100k 1/4W		
	R20					
	R21					
	R22	VR1-8143-473	1	SOLID RESISTOR 47k 1/4W		
	R23	VR1-8143-473	1	" 47k 1/4W		

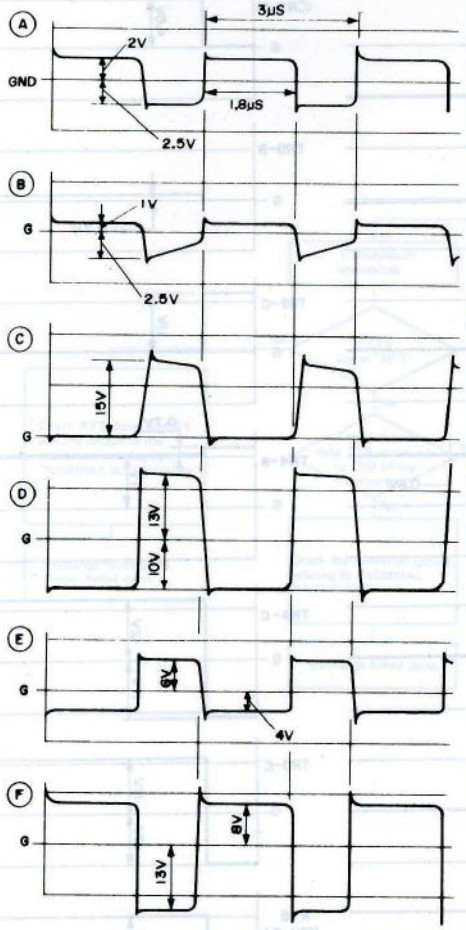
N	KEY NO.	PART NO.	Q'TY	DESCRIPTION	REMARKS	PRICE
	R24	VR1-8143-473	1	SOLID RESISTOR 47k 1/4W		
	R25	VR1-8143-473	1	" 47k 1/4W		
	R26	VR1-8143-102	1	" 1k 1/4W		
	R27	VR1-8143-102	1	" 1k 1/4W		
	R28	VR1-8143-103	1	" 10k 1/4W		
	R29	VR1-8143-103	1	SOLID RESISTOR 10k 1/4W		
	R30					
	R31					
	R32	VR1-8143-103	1	SOLID RESISTOR 10k 1/4W		
	R33	VR1-8143-103	1	" 10k 1/4W		
	R34	VR1-8143-472	1	SOLID RESISTOR 4.7k 1/4W		
	R35	VR1-8143-511	1	" 510 ohm 1/4W		
	R36	VR1-8143-103	1	" 10k 1/4W		
	R37	VR1-8143-101	1	" 100 ohm 1/4W		
	R38	VR1-8143-333	1	" 33k 1/4W		
	R39	VR1-8143-333	1	SOLID RESISTOR 33k 1/4W		
	R40	VR1-8143-563	1	" 56k 1/4W		
*	SW1	WC3-0049-000	1	SWITCH,4P		
*	SW2	WC8-0049-000	1	SWITCH,DIP TYPE ,4P		
	T	EH3-5128-000	1	DC-DC CONVERTOR TRANS. TB-69		
	TR1	WA2-0374-000	1	TRANSISTOR 2SD789D		
	TR2	WA2-0282-000	1	" 2SA844C		
	TR3	WA2-0282-000	1	" "		
	TR3	WA2-0282-000	1	" "		
	TR4	WA2-0282-000	1	" "		
	TR5	WA2-0282-000	1	TRANSISTOR 2SA844C		
	TR6	WA2-0282-000	1	" "		
	TR7	WA2-0282-000	1	" "		
	TR8	WA2-0113-000	1	" 2SC2603(D,E,F)		
	TR9	WA2-0113-000	1	" "		
	TR10	WA2-0113-000	1	TRANSISTOR 2SC2603(D,E,F)		
	TR11	WA2-0282-000	1	" 2SA844C		

N	KEY NO.	PART NO.	Q'TY	DESCRIPTION	REMARKS	PRICE
	ZD1	WA1-0230-000	1	ZENER DIODE,HZ12-B1		
		SCREW				
		XA9-0178-000	1	SCREW,2.6x5		
		XB3-1261-007	3	SCREW,TAPPING,2.6x10		
		XB3-2301-007	2	" " 3x10		

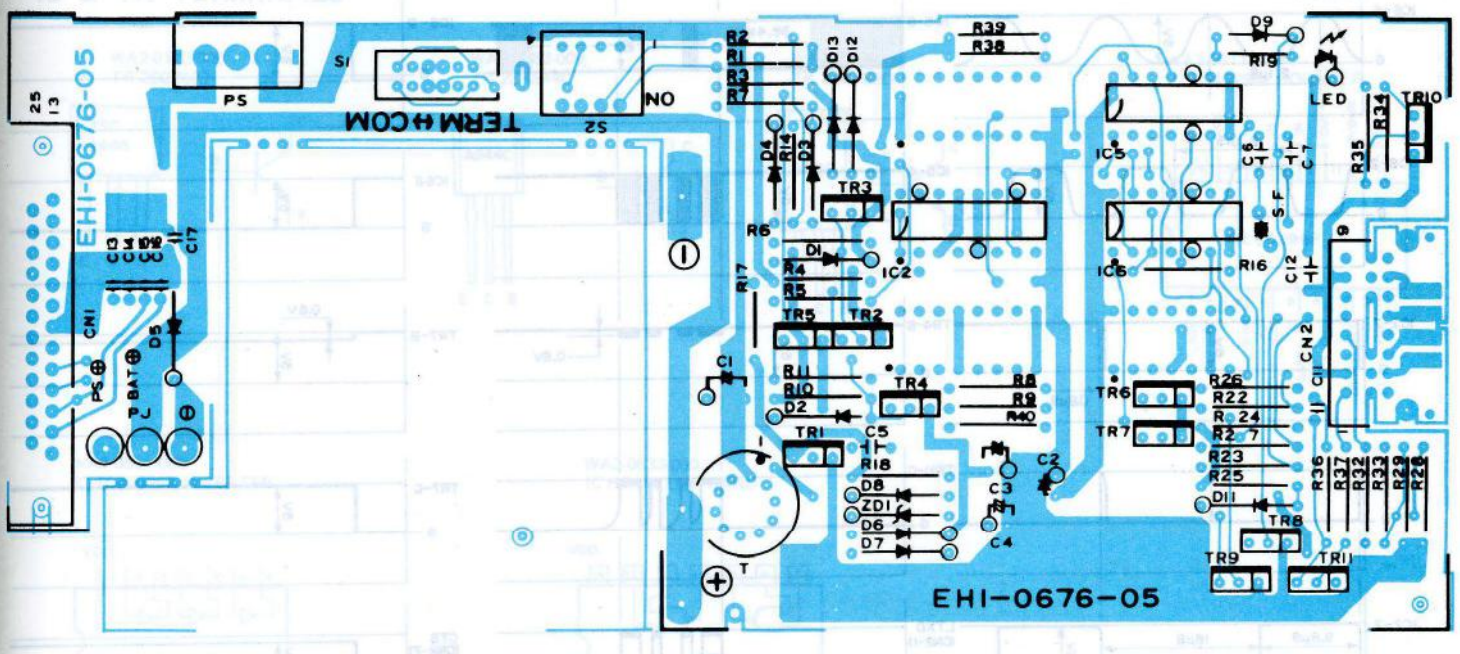
CIRCUIT DIAGRAM

CIRCUIT DIAGRAM



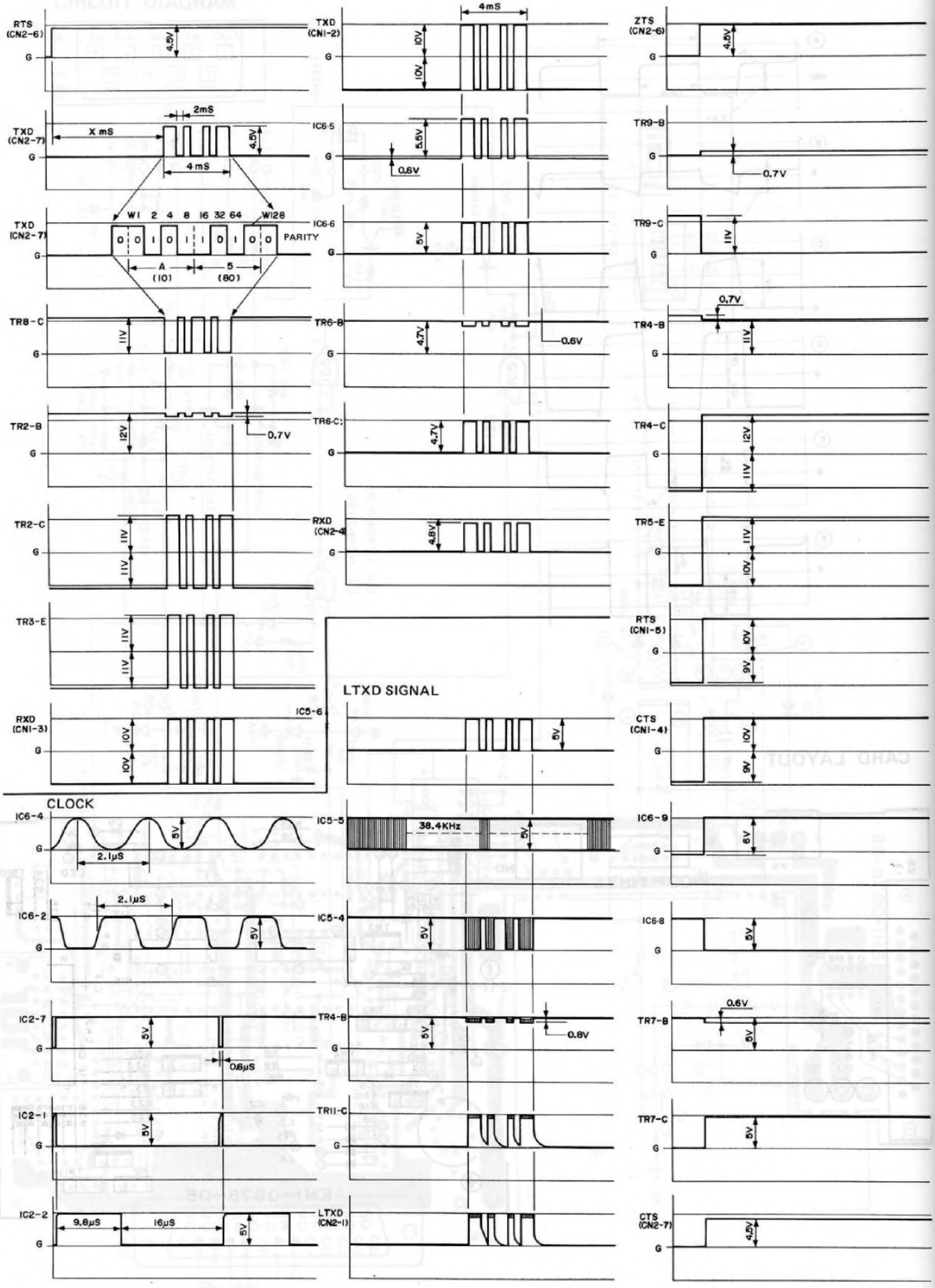

A
3μS
2V
GND
1.8μS
2.5V
B
1V
G
2.5V
C
15V
G
D
13V
10V
G
E
8V
G
4V
F
8V
G
13V

CARD LAYOUT

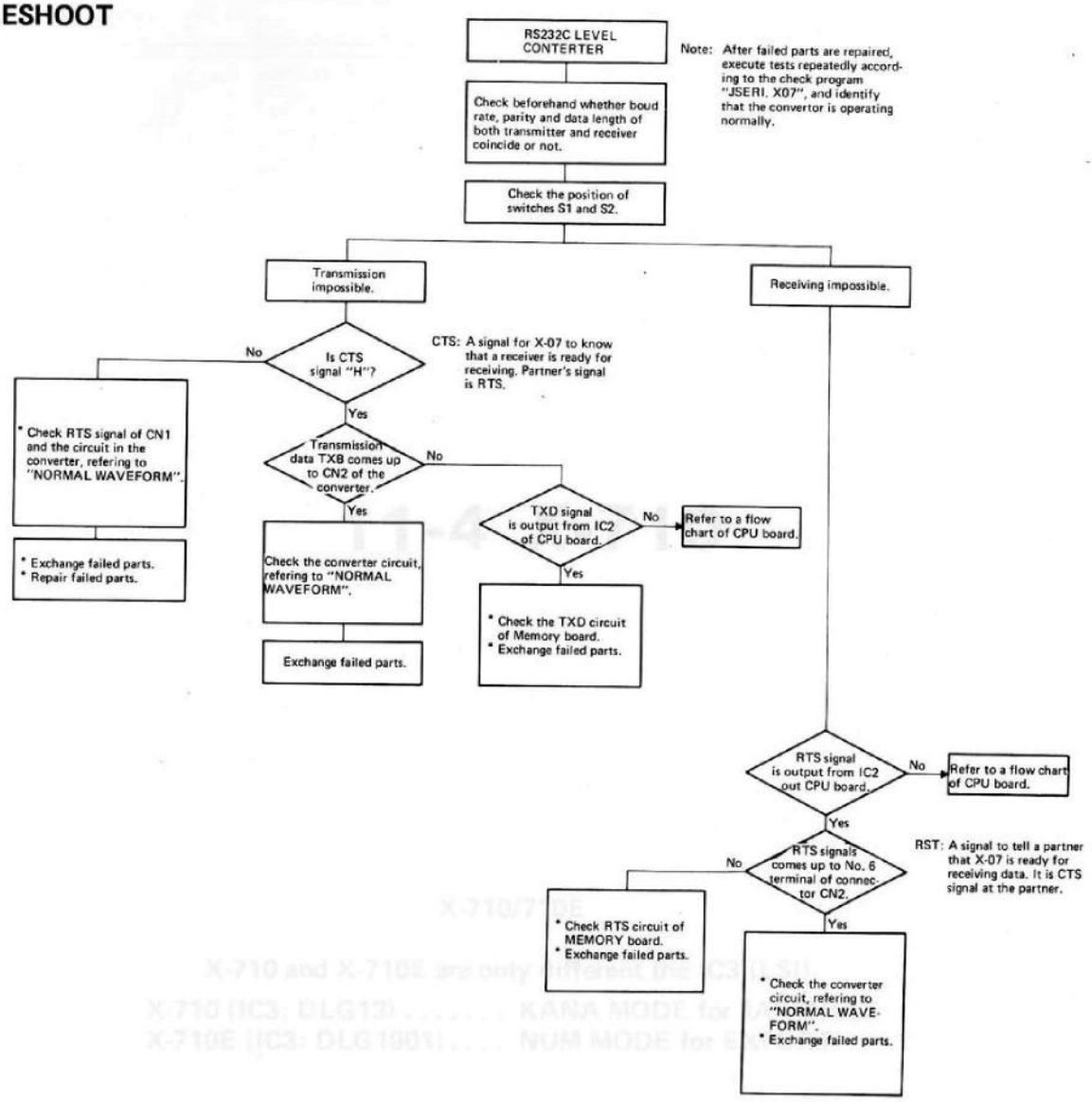

EHI-0676-05
PS
S1
S2
TERM⇔COM
NO
CN1
PS
BAT
J
D5
C17
R2
R1
R3
R7
D13
D12
R39
R38
D9
R19
LED
TR10
R34
R35
IC5
IC6
C6
C7
C12
R16
S-F
D4
R14
D3
TR3
R6
D1
IC2
R4
R5
TR5
TR2
R17
C1
R11
R10
D2
TR4
R8
R9
R40
TR1
C5
R18
D8
ZD1
D6
D7
C2
C3
C4
T
TR6
TR7
R26
R22
R24
R27
R23
R25
D11
TR8
TR9
TR11
R36
R37
R32
R33
R28
R29
CN2
C11
EHI-0676-05

NORMAL WAVEFORM-1

RTS/CTS (Connected No. 4 & No. 5 terminals of CN1)

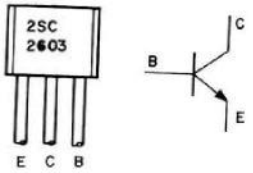


TROUBLESHOOT

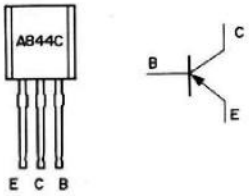


IC & TR TERMINALS

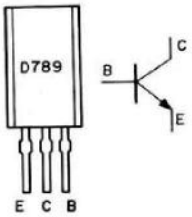
WA2-0113-000
TR 2SC2603



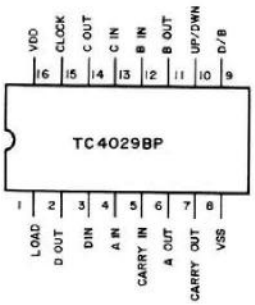
WA2-0282-000
TR 2SA844C



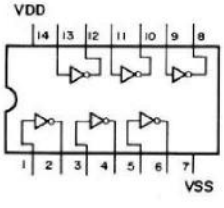
WA2-0374-000
TR 2SD789D



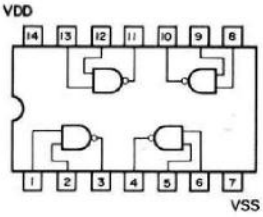
WA3-0311-000
IC TC4029BP COUNTER



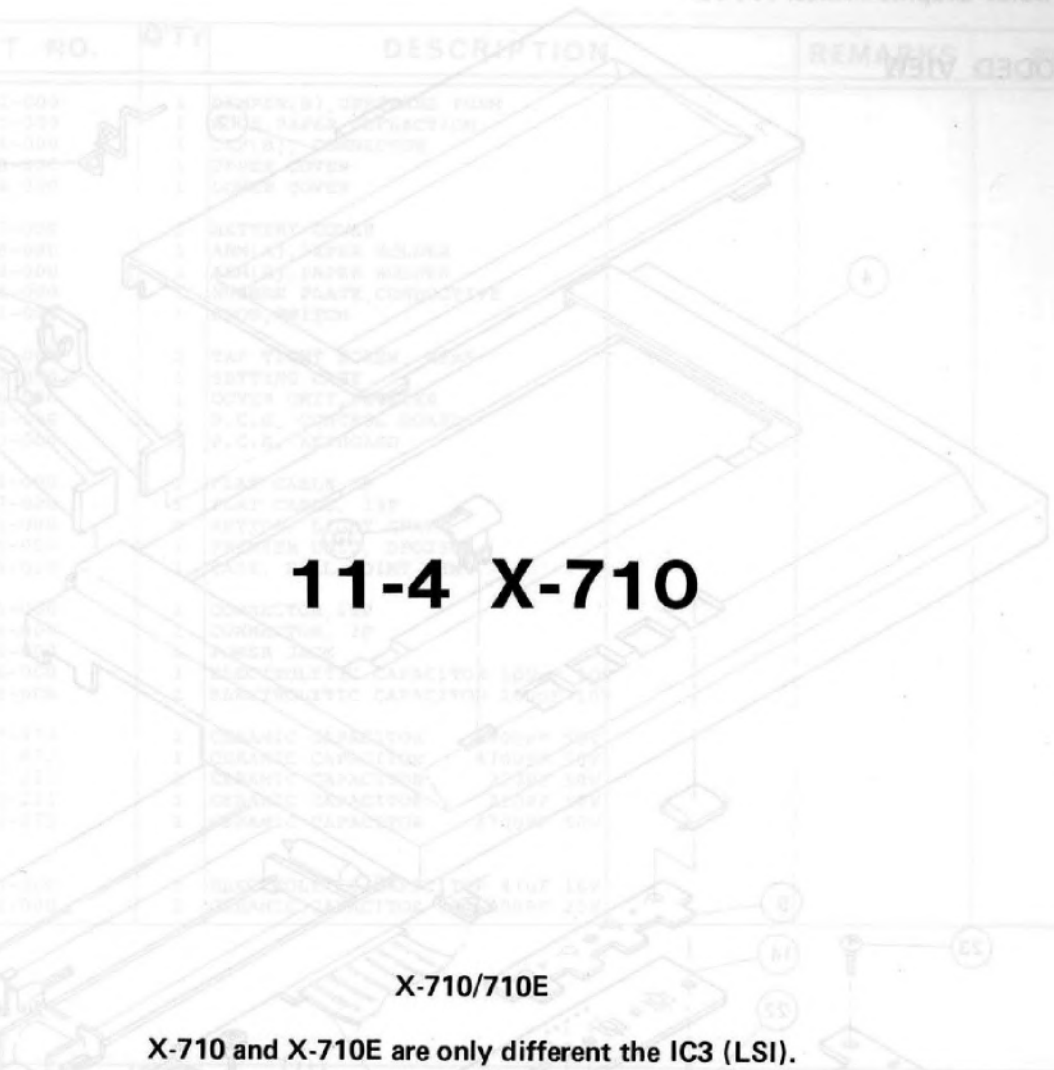
WA3-0561-000
IC TC4069BP INVERTER



WA3-0633-000
IC HD14011BP NAND



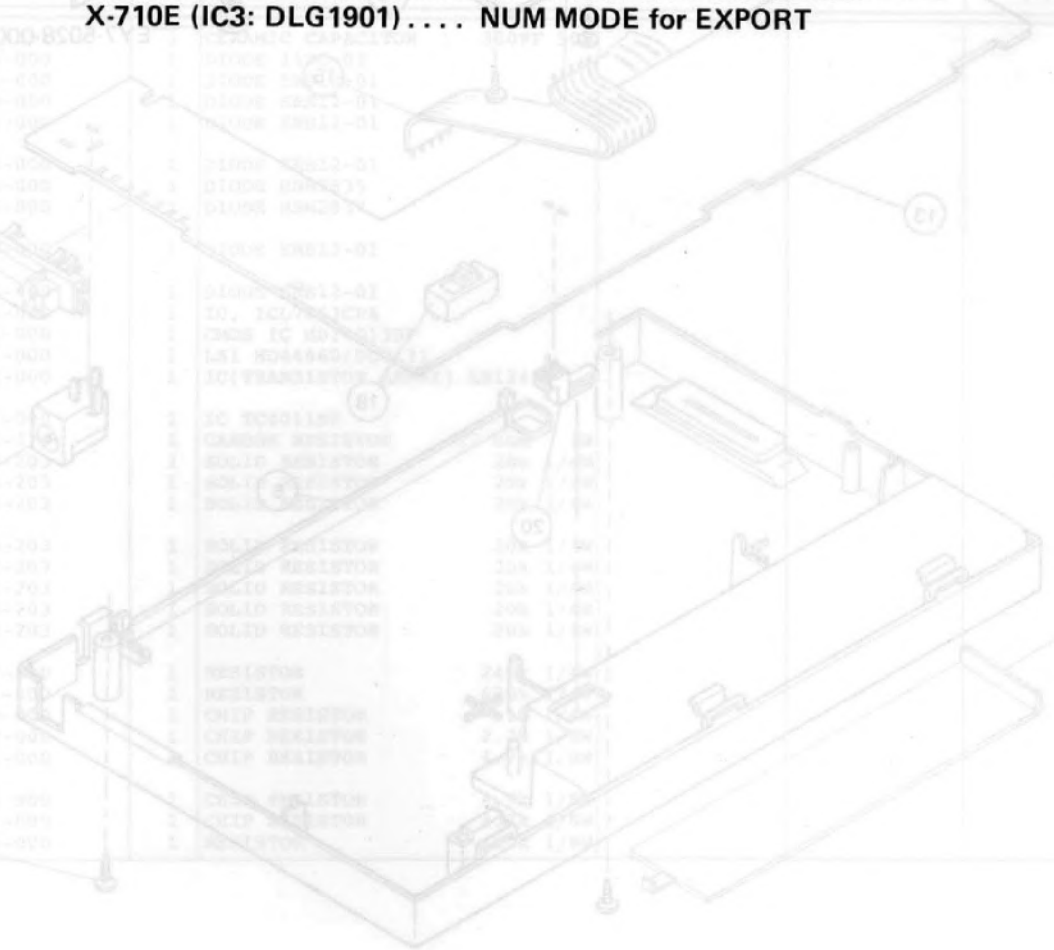
11-4 X-710



X-710/710E

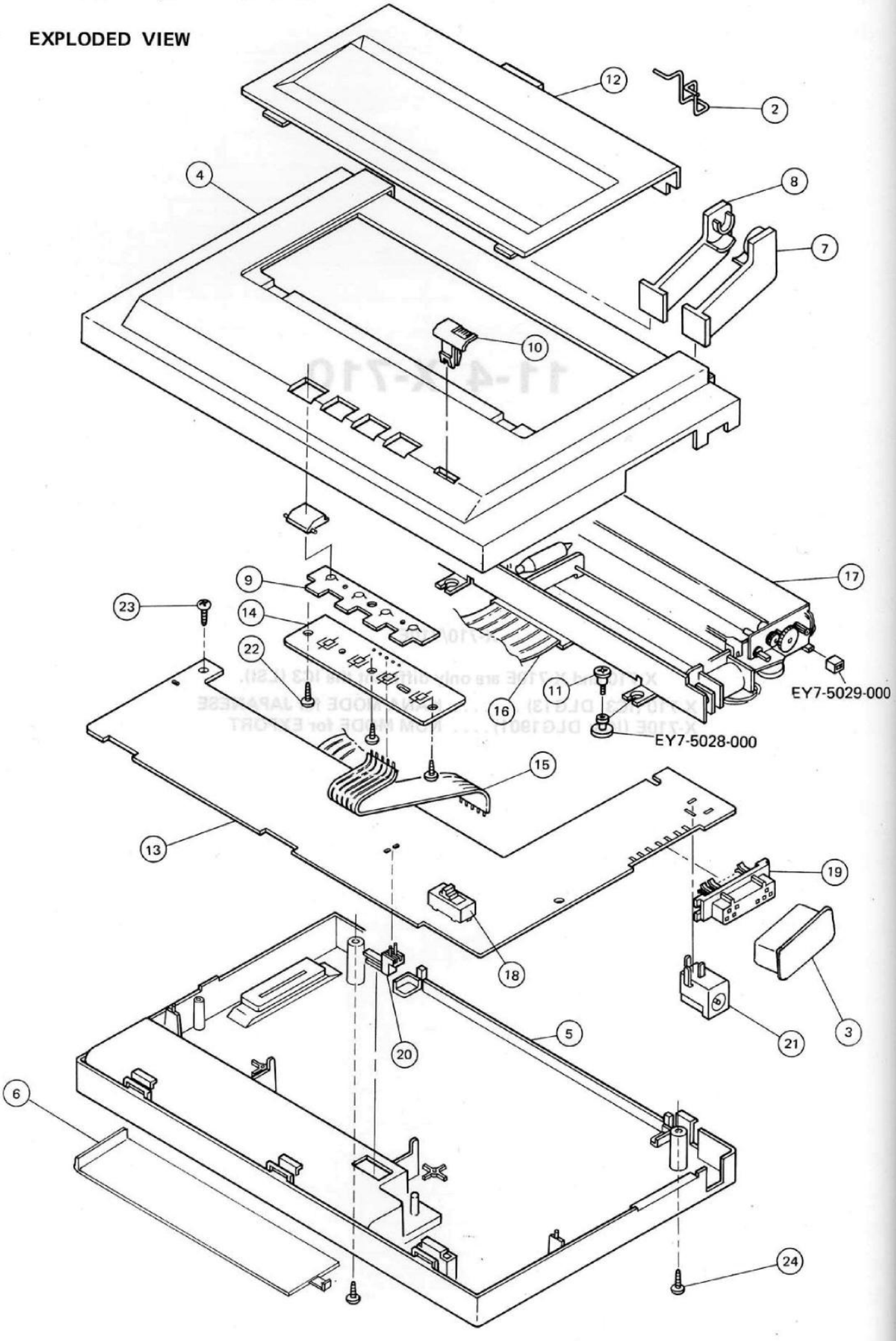
X-710 and X-710E are only different the IC3 (LSI).

X-710 (IC3: DLG13) . . . . . KANA MODE for JAPANESE
X-710E (IC3: DLG1901) . . . . NUM MODE for EXPORT



11-4) Color Graphic Printer X-710

EXPLODED VIEW



N	KEY NO.	PART NO.	Q'TY	DESCRIPTION	REMARKS	PRICE
	1	EA1-2452-000	1	DAMPER(B),URETHANE FORM		
	2	EA1-4905-000	1	HOOK,PAPER DEFLECTION		
*	3	EA1-5046-000	1	CAP(B), CONNECTOR		
*	4	EA1-5053-000	1	UPPER COVER		
*	5	EA1-5054-000	1	LOWER COVER		
*	6	EA1-5057-000	1	BATTERY COVER		
*	7	EA1-5058-000	1	ARM(A),PAPER HOLDER		
*	8	EA1-5059-000	1	ARM(B),PAPER HOLDER		
*	9	EA1-5060-000	1	RUBBER PLATE,CONDUCTIVE		
*	10	EA1-5061-000	1	KNOB,SWITCH		
*	11	EA1-5095-000	2	TAP TIGHT SCREW, M2x8		
*		EA1-8084-000	1	SETTING CASE		
*	12	EF1-0326-000	1	COVER UNIT,PRINTER		
*	13	EH1-0672-000	1	P.C.B, CONTROL BOARD		
*	14	EH1-0673-000	1	P.C.B, KEYBOARD		
*	15	EH2-0394-000	1	FLAT CABLE,5P		
*	16	EH2-0407-000	1	FLAT CABLE, 13P		
*		ES2-2191-000	4	KEYTOP, LIGHT GRAY		
*	17	ES5-0135-000	1	PRINTER UNIT, DPG2308		
*	18	ES5-0139-000	1	CASE, BALL POINT PEN		
*	19	WS1-0205-000	1	CONNECTOR,15P		
*	20	WS3-0476-000	1	CONNECTOR, 2P		
*	21	WS6-0026-000	1	POWER JACK		
	C 1	VC9-1256-000	1	ELECTROLYTIC CAPACITOR 100uF 10V		
	C 2	VC9-1256-000	1	ELECTROLYTIC CAPACITOR 100uF 10V		
	C 3	VC4-3503-472	1	CERAMIC CAPACITOR 4700PF 50V		
	C 4	VC4-3503-472	1	CERAMIC CAPACITOR 4700PF 50V		
	C 5	VC4-3503-221	1	CERAMIC CAPACITOR 220PF 50V		
	C 6	VC4-3503-221	1	CERAMIC CAPACITOR 220PF 50V		
	C 7	VC4-3503-472	1	CERAMIC CAPACITOR 4700PF 50V		
	C 8					
*	C 9	VC9-1210-000	1	ELECTROLYTIC CAPACITOR 47uF 16V		
	C10	VC9-0582-000	1	CERAMIC CAPACITOR 100,000PF 25V		

N	KEY NO.	PART NO.	Q'TY	DESCRIPTION	REMARKS	PRICE
	C11	VC4-3503-331	1	CERAMIC CAPACITOR 330PF 50V		
	D 1	WA1-0355-000	1	DIODE 11DO-03		
	D 2	WA1-0078-000	1	DIODE ERB12-01		
	D 3	WA1-0078-000	1	DIODE ERB12-01		
	D 4	WA1-0078-000	1	DIODE ERB12-01		
	D 5	WA1-0078-000	1	DIODE ERB12-01		
	D 6	WA1-0365-000	1	DIODE HSM2835		
	D 7	WA1-0356-000	1	DIODE HSM2837		
	D 8					
	D 9	WA1-0078-000	1	DIODE ERB12-01		
	D10	WA1-0078-000	1	DIODE ERB12-01		
	IC1	WA4-0261-000	1	IC, ICL7663CPA		
	IC2	WA3-0634-000	1	CMOS IC HD14013BP		
	IC3	WA3-0942-000	1	LSI HD44860(DLG13)		
	IC4	WA2-0382-000	1	IC(TRANSISTOR ARRAY) LB1247		
	IC5	WA3-0974-000	1	IC TC4011BF		
*	R 1	VR1-8013-270	1	CARBON RESISTOR 27 ohm 1W		
	R 2	VR1-8143-203	1	SOLID RESISTOR 20k 1/4W		
	R 3	VR1-8143-203	1	SOLID RESISTOR 20k 1/4W		
	R 4	VR1-8143-203	1	SOLID RESISTOR 20k 1/4W		
	R 5	VR1-8143-203	1	SOLID RESISTOR 20k 1/4W		
	R 6	VR1-8143-203	1	SOLID RESISTOR 20k 1/4W		
	R 7	VR1-8143-203	1	SOLID RESISTOR 20k 1/4W		
	R 8	VR1-8143-203	1	SOLID RESISTOR 20k 1/4W		
	R 9	VR1-8143-203	1	SOLID RESISTOR 20k 1/4W		
	R10	VR9-2777-000	1	RESISTOR 240k 1/4W		
	R11	VR9-1034-000	1	RESISTOR 680k 1/4W		
	R12	VR9-1460-000	1	CHIP RESISTOR 1M 1/4W		
	R13	VR9-2833-000	1	CHIP RESISTOR 2.2k 1/8W		
	R14	VR9-2834-000	1	CHIP RESISTOR 4.7k 1/8W		
	R15	VR9-2834-000	1	CHIP RESISTOR 4.7k 1/8W		
	R16	VR9-2834-000	1	CHIP RESISTOR 4.7k 1/8W		
	R17	VR9-2835-000	1	RESISTOR 20k 1/8W		

N	KEY NO.	PART NO.	Q'TY	DESCRIPTION	REMARKS	PRICE
	R18	VR9-1624-000	2	CHIP RESISTOR 10M 1/8W		
	R19	VR1-8143-104	1	SOLID RESISTOR 100k 1/4W		
	R20	VR1-1143-124	1	CARBON RESISTOR 120k 1/4W		
	R21	VR9-2836-000	1	CHIP RESISTOR 51k 1/8W		
	R22	VR9-2779-000	1	RESISTOR 150k 1/4W		
	R23	VR9-2610-000	1	RESISTOR, METALIC 42.2k 1/4W		
	R24	VR9-0963-000	1	CHIP RESISTOR 150k 1/8W		
	R25	VR9-3088-000	1	CHIP RESISTOR 56k 1/8W		
	R26	VR9-1071-000	1	CHIP RESISTOR 220k 1/8W		
	R27	VR9-1071-000	1	CHIP RESISTOR 220k 1/8W		
	R28	VR9-1624-000	1	CHIP RESISTOR 10k 1/8W		
	R29	VR9-2838-000	1	CHIP RESISTOR 6.8k 1/8W		
	R30	VR9-2838-000	1	CHIP RESISTOR 6.8k 1/8W		
	R31	VR9-1058-000	1	CHIP RESISTOR 100k 1/8W		
	R32	VR9-1058-000	1	CHIP RESISTOR 100k 1/8W		
	R33	VR9-2839-000	1	CHIP RESISTOR 220 ohm 1/8W		
	R34	VR9-2839-000	1	CHIP RESISTOR 220 ohm 1/8W		
	R35	VR1-8143-394	1	SOLID RESISTOR 390k 1/4W		
	R36	VR1-8143-821	1	SOLID RESISTOR 820 ohm 1/4W		
	RA1	VR9-0948-000	1	RESISTOR ARRAY 10k x 8 1/8W		
	SW1	WC3-9013-000	1	SLIDE SWITCH,2P		
	TR1	WA2-0282-000	1	TRANSISTOR 2SA844C		
	TR2	WA2-0282-000	1	TRANSISTOR 2SA844C		
	TR3	WA2-0283-000	1	TRANSISTOR 2SD788		
	TR4	WA2-0283-000	1	TRANSISTOR 2SD788		
	TR5	WA2-0282-000	1	TRANSISTOR 2SA844C		
	TR6	WA2-0367-000	1	TRANSISTOR 2SB739C		
	TR7	WA2-0367-000	1	TRANSISTOR 2SB739C		
	TR8	WA2-0282-000	1	TRANSISTOR 2SA844C		
	TR9	WA2-0113-000	1	TRANSISTOR 2SC2603(D,E,F)		
	TR10	WA2-0195-000	1	TRANSISTOR 2SA844D		
	TR11	WA2-0113-000	1	TRANSISTOR 2SC2603(D,E,F)		
	TR12	WA2-0113-000	1	TRANSISTOR 2SC2603(D,E,F)		

N	KEY NO.	PART NO.	Q'TY	DESCRIPTION	REMARKS	PRICE
	XTAL	WK2-9003-000	1	CERAMIC FILTER 600kHz		
	ZD1	WA1-0351-000	1	ZENER DIODE HZ11A3		
	ZD2	WA1-0412-000	1	ZENER DIODE HZ4LLB		
	ZD3	WA1-0410-000	1	ZENER DIODE HZ3LLA		
		SCREW				
		XA9-0154-000	3	SCREW,TAPPING, 2x4		
		XB3-1260-605	1	SCREW,TAPPING, 2.6x6		
		XB3-1261-007	3	SCREW,TAPPING, 2.6x10		

N	KEY NO.	PART NO.	Q'TY	DESCRIPTION		REMARKS	PRICE	
		For Modified Power Supply Circuit(Without IC1)						
	R10	VR9-3284-000	1	METAL FILM RESISTOR	100K 1/4W			
	R11	VR9-1032-000	1	METAL FILM RESISTOR	33K 1/4W			
	R37	VR1-1141-823	1	CARBON RESISTOR	82K 1/4W	+/- 5%		
	R38	VR1-1141-204	1	CARBON RESISTOR	200K 1/4W	+/- 5%		
	R39	VR1-1141-104	1	CARBON RESISTOR	100K 1/4W	+/- 5%		
	TR10	WA2-0195-000	1	TRANSISTOR 2SA844D				
	TR11	WA2-0113-000	1	TRANSISTOR 2SC2603(D,E,F)				
	TR12	WA2-0113-000	1	TRANSISTOR 2SC2603(D,E,F)				
	ZD2	WA1-0412-000	1	ZENER DIODE HZ4LLB				
	ZD3	WA1-0410-000	1	ZENER DIODE HZ3LLA				

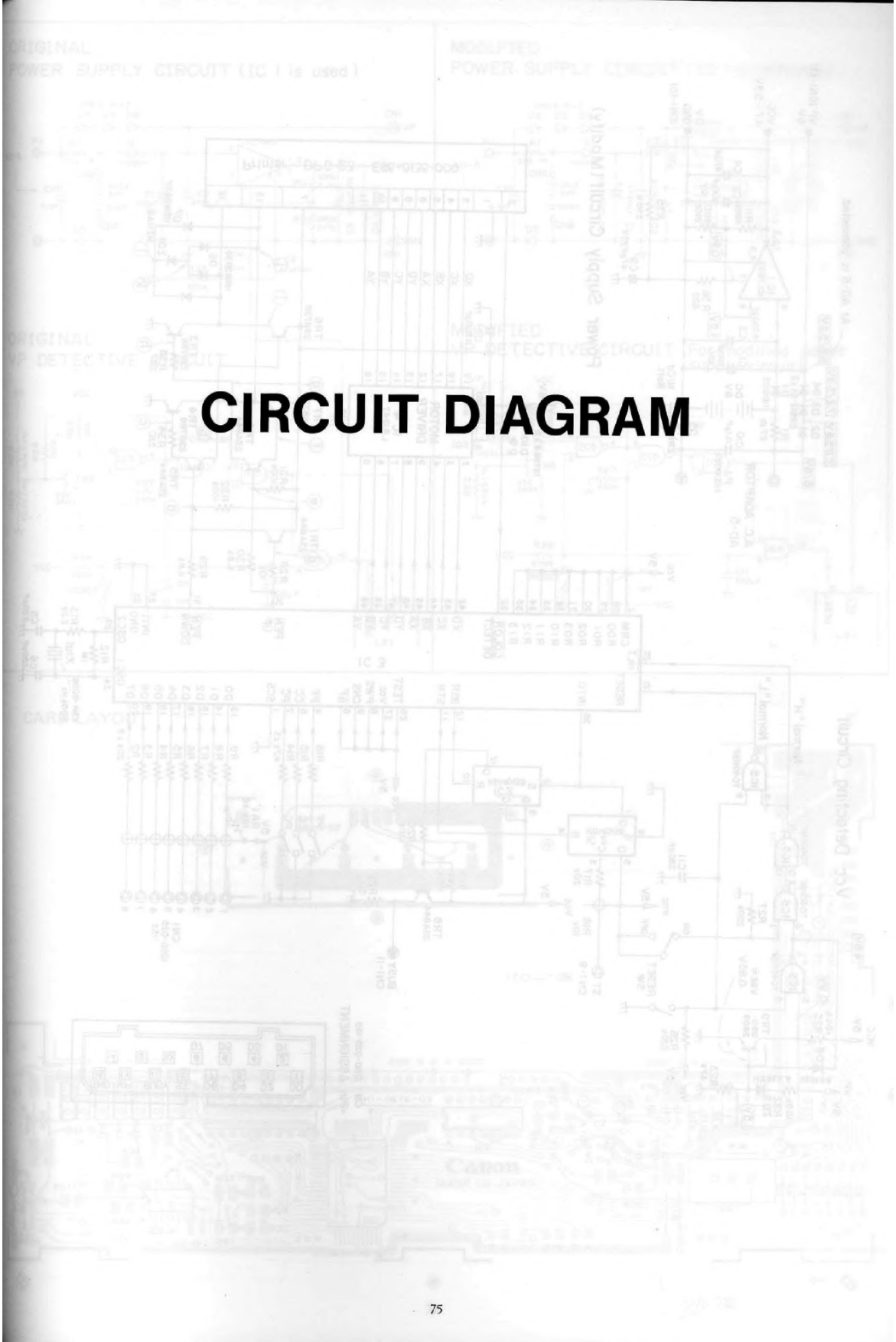
N	KEY NO.	PART NO.	Q'TY	DESCRIPTION		REMARKS	PRICE	
	1	EA1-2452-000	1	DAMPER(B), URETHANE FORM				
	2	EA1-4905-000	1	HOOK, PAPER DEFLECTION				
*	3	EA1-5046-000	1	CAP(B), CONNECTOR				
*	4	EA1-5053-000	1	UPPER COVER				
*	5	EA1-5054-000	1	LOWER COVER				
*	6	EA1-5057-000	1	BATTERY COVER				
*	7	EA1-5058-000	1	ARM(A), PAPER HOLDER				
*	8	EA1-5059-000	1	ARM(B), PAPER HOLDER				
*	9	EA1-5060-000	1	RUBBER PLATE, CONDUCTIVE				
*	10	EA1-5061-000	1	KNOB, SWITCH				
*	11	EA1-5095-000	2	TAPTIGHT SCREW, M2x8				
*		EA1-8084-000	1	SETTING CASE				
*	12	EF1-0326-000	1	COVER UNIT, PRINTER				
*	13	EH1-0672-000	1	P.C.B., CONTROL BOARD				
*	14	EH1-0673-000	1	P.C.B., KEYBOARD				
*	15	EH2-0394-000	1	FLAT CABLE, 5P				
*	16	EH2-0407-000	1	FLAT CABLE, 13P				
*		ES2-2191-000	4	KEYTOP, LIGHT GRAY				
*	17	ES5-0135-000	1	PRINTER UNIT, DPG2308				
*	18	ES5-0139-000	1	SHAFT, ROLL PAPER				
*	19	WS1-0205-000	1	CONNECTOR, 15P				
*	20	WS3-0476-000	1	CONNECTOR, 2P, BATTERY				
*	21	WS6-0026-000	1	POWER JACK, AD-5				
*	C 1	VC9-1256-000	1	ELECTROLYTIC CAPACITOR	100uF	10V		
*	C 2	VC9-1256-000	1	ELECTROLYTIC CAPACITOR	100uF	10V		
	C 3	VC4-3503-472	1	CERAMIC CAPACITOR	4700pF	50V		
	C 4	VC4-3503-472	1	CERAMIC CAPACITOR	4700pF	50V		
	C 5	VC4-3503-221	1	CERAMIC CAPACITOR	220pF	50V		
	C 6	VC4-3503-221	1	CERAMIC CAPACITOR	220pF	50V		
	C 7	VC4-3503-472	1	CERAMIC CAPACITOR	4700pF	50V		
	C 8	----						

N	KEY NO.	PART NO.	Q'TY	DESCRIPTION		REMARKS	PRICE	
	C 9	----						
	C10	VC9-0582-000	1	CERAMIC CAPACITOR	100,000pF	25V		
	C11	VC4-3503-331	1	CERAMIC CAPACITOR	330pF	50V		
	D 1	WA1-0355-000	1	DIODE, 1LDO-03				
	D 2	WA1-0078-000	1	DIODE, ERB12-01				
	D 3	WA1-0078-000	1	DIODE, ERB12-01				
	D 4	WA1-0078-000	1	DIODE, ERB12-01				
	D 5	WA1-0078-000	1	DIODE, ERB12-01				
*	D 6	WA1-0365-000	1	DIODE, HSM2835(BLACK)				
*	D 7	WA1-0356-000	1	DIODE, HSM2837(GREEN)				
	D 8	X65-5032-000	1	DIODE, 1S1588				
	D 9	WA1-0078-000	1	DIODE, ERB12-01				
	D10	WA1-0078-000	1	DIODE, ERB12-01				
	IC1	----	1					
	IC2	WA3-0634-000	1	CMOS IC HD14013BP				
*	IC3	WA3-1151-000	1	CMOS LSI DLG190IE, CONTROLLER				
*	IC4	WA2-0382-000	1	IC LB1247, TRANSISTOR ARRAY				
*	IC5	WA3-0974-000	1	CMOS IC TC4011BF				
*	R 1	VR1-8013-270	1	CARBON RESISTOR	27ohm 1 W			
	R 2	VR1-8143-203	1	SOLID RESISTOR	20k 1/4W			
	R 3	VR1-8143-203	1	SOLID RESISTOR	20k 1/4W			
	R 4	VR1-8143-203	1	SOLID RESISTOR	20k 1/4W			
	R 5	VR1-8143-203	1	SOLID RESISTOR	20k 1/4W			
	R 6	VR1-8143-203	1	SOLID RESISTOR	20k 1/4W			
	R 7	VR1-8143-203	1	SOLID RESISTOR	20k 1/4W			
	R 8	VR1-8143-203	1	SOLID RESISTOR	20k 1/4W			
	R 9	VR1-8143-203	1	SOLID RESISTOR	20k 1/4W			
*	R10	VR9-3284-000	1	RESISTOR, METALIC	100k 1/4W			
	R11	VR9-1201-000	1	RESISTOR, METALIC	33k 1/4W			
*	R12	VR9-3100-000	1	RESISTOR, CHIP TYPE	1M 1/8W			
*	R13	VR9-3077-000	1	RESISTOR, CHIP TYPE	2.2k 1/8W			

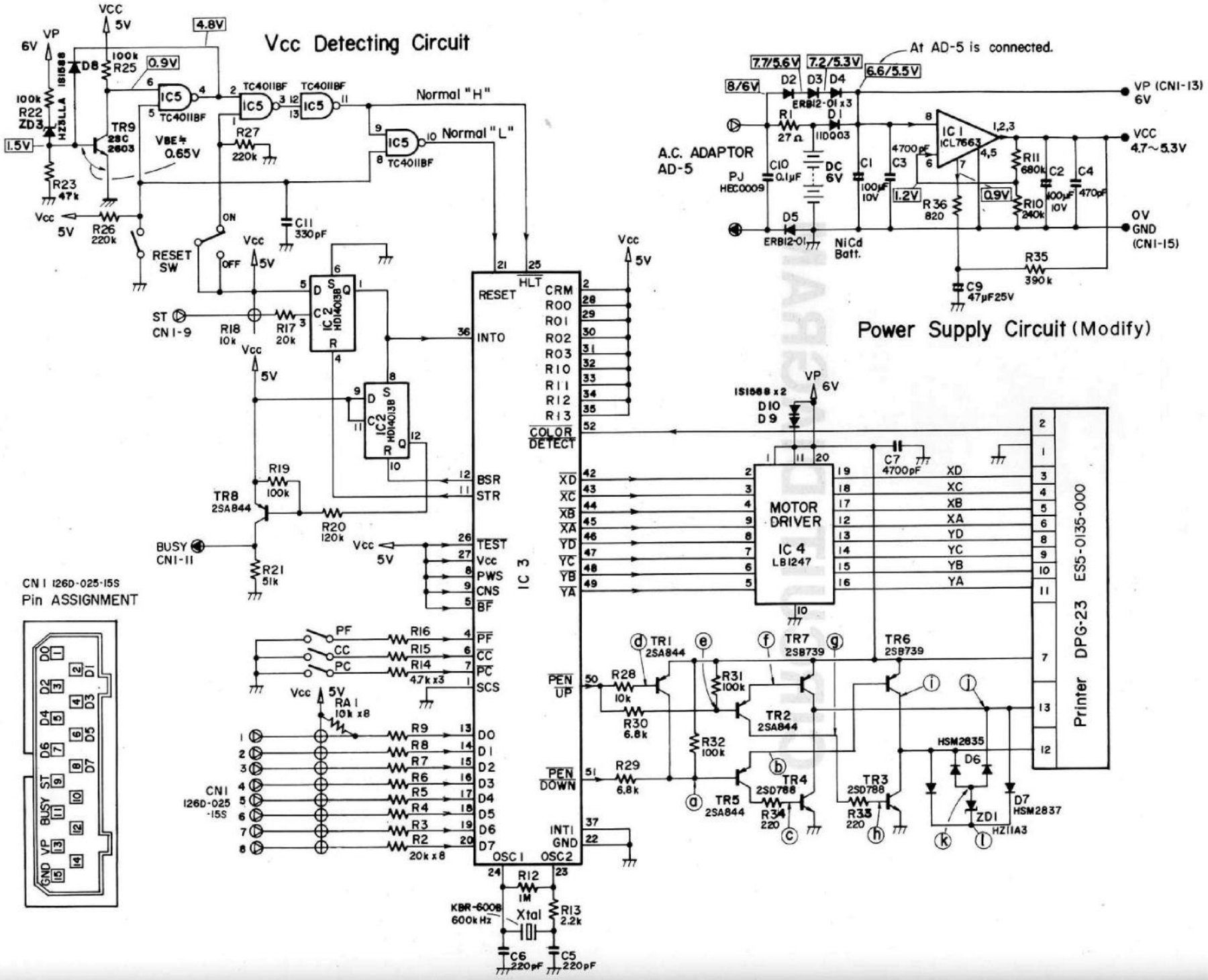
N	KEY NO.	PART NO.	Q'TY	DESCRIPTION		REMARKS	PRICE	
*	R14	VR9-3079-000	1	RESISTOR, CHIP TYPE	4.7k 1/8W			
*	R15	VR9-3079-000	1	RESISTOR, CHIP TYPE	4.7k 1/8W			
*	R16	VR9-3079-000	1	RESISTOR, CHIP TYPE	4.7k 1/8W			
*	R17	VR9-3083-000	1	RESISTOR, CHIP TYPE	20k 1/8W			
*	R18	VR9-3081-000	1	RESISTOR, CHIP TYPE	10k 1/8W			
*	R19	VR1-1143-104	1	CARBON RESISTOR	100k 1/4W			
*	R20	VR1-1143-124	1	CARBON RESISTOR	120k 1/4W			
*	R21	VR9-3087-000	1	RESISTOR, CHIP TYPE	51k 1/8W			
*	R22	VR9-3284-000	1	RESISTOR, METALIC	100k 1/4W	+/- 1%		
*	R23	VR9-2778-000	1	RESISTOR, METALIC	47k 1/4W	+/- 1%		
	R24	----						
	R25	VR9-2141-000	1	RESISTOR, CHIP TYPE	100k 1/8W			
*	R26	VR9-3093-000	1	RESISTOR, CHIP TYPE	220k 1/8W			
*	R27	VR9-3093-000	1	RESISTOR, CHIP TYPE	220k 1/8W			
*	R28	VR9-3081-000	1	RESISTOR, CHIP TYPE	10k 1/8W			
*	R29	VR9-3080-000	1	RESISTOR, CHIP TYPE	6.8k 1/8W			
*	R30	VR9-3080-000	1	RESISTOR, CHIP TYPE	6.8k 1/8W			
	R31	VR9-2141-000	1	RESISTOR, CHIP TYPE	100k 1/8W			
	R32	VR9-2141-000	1	RESISTOR, CHIP TYPE	100k 1/8W			
*	R33	VR9-3075-000	1	RESISTOR, CHIP TYPE	220ohm1/8W			
*	R34	VR9-3075-000	1	RESISTOR, CHIP TYPE	220ohm1/8W			
	R35	----						
	R36	----						
	R37	VR1-8143-823	1	SOLID RESISTOR	82k 1/4W			
*	R38	VR1-1143-204	1	CARBON RESISTOR	200k 1/4W			
*	R39	VR1-1143-104	1	CARBON RESISTOR	100k 1/4W			
*	R40	VR1-1143-204	1	CARBON RESISTOR	200k 1/4W			
	RA1	VR9-0948-000	1	RESISTOR ARRAY	10k x 8 1/8W			

CIRCUIT DIAGRAM

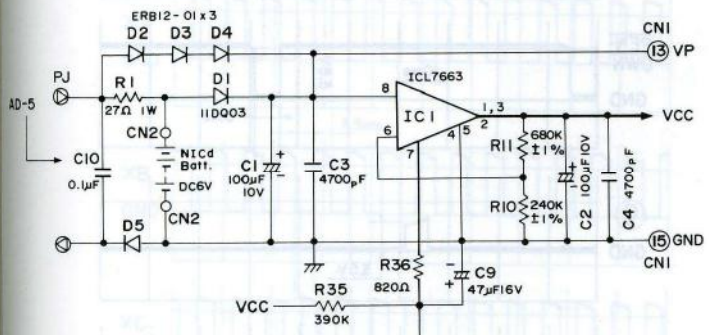
N	KEY NO.	PART NO.	Q'TY	DESCRIPTION		REMARKS	PRICE	
	SW1	WC3-9013-000	1	SLIDE SWITCH, 2P, POWER				
	TR1	WA2-0282-000	1	TRANSISTOR 2SA844(C,D)				
	TR2	WA2-0282-000	1	TRANSISTOR 2SA844(C,D)				
*	TR3	WA2-0283-000	1	TRANSISTOR 2SD788				
*	TR4	WA2-0283-000	1	TRANSISTOR 2SD788				
	TR5	WA2-0282-000	1	TRANSISTOR 2SA844(C,D)				
*	TR6	WA2-0367-000	1	TRANSISTOR 2SB739				
*	TR7	WA2-0367-000	1	TRANSISTOR 2SB739				
	TR8	WA2-0282-000	1	TRANSISTOR 2SA844(C,D)				
	TR9	WA2-0113-000	1	TRANSISTOR 2SC2603(D,E,F)				
	TR10	WA2-0195-000	1	TRANSISTOR 2SA844(D)				
	TR11	WA2-0113-000	1	TRANSISTOR 2SC2603(D,E,F)				
	TR12	WA2-0113-000	1	TRANSISTOR 2SC2603(D,E,F)				
*	XTAL	WK2-9003-000	1	CERAMIC FILTER 600kHz				
*	ZD1	WA1-0351-000	1	ZENER DIODE HZ11A3				
*	ZD2	WA1-0412-000	1	ZENER DIODE HZ4LLB				
*	ZD3	WA1-0410-000	1	ZENER DIODE HZ3LLA				
		SCREWS						
*		EA1-5095-000	2	TAPTIGHT SCREW M2 x 8, PRINTER				
		XA9-0154-000	3	SCREW, TAPPING M2 x 4, P.C.B.,	KEYBOARD			
		XB3-1260-605	1	SCREW, TAPPING 2.6x 6, P.C.B.,	CONTROLL BOARD			
		XB3-1261-007	3	SCREW, TAPPING 2.6x10, COVER				



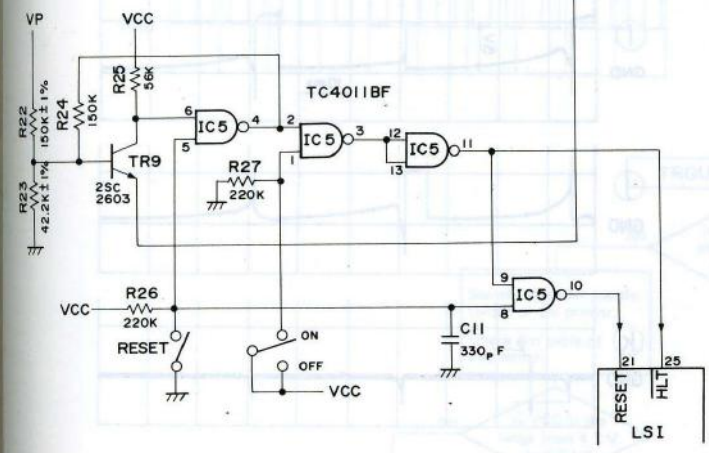
CIRCUIT DIAGRAM


Vcc Detecting Circuit
Power Supply Circuit (Modify)
At AD-5 is connected.
A.C. ADAPTOR AD-5
NiCd Batt.
VP (CN1-13) 6V
VCC 4.7~5.3V
OV GND (CN1-15)
Normal "H"
Normal "L"
RESET SW
ON
OFF
ST CN1-9
BUSY CN1-11
TR8 2SA844
TR1 2SA844
TR2 2SA844
TR3 2SD788
TR4 2SD788
TR5 2SA844
TR6 2SB739
TR7 2SB739
IC 3
IC 4 LB1247
MOTOR DRIVER
Printer DPG-23 ES5-0135-000
CN1 126D-025-15S
Pin ASSIGNMENT
Xtal
KBR-600B
600kHz

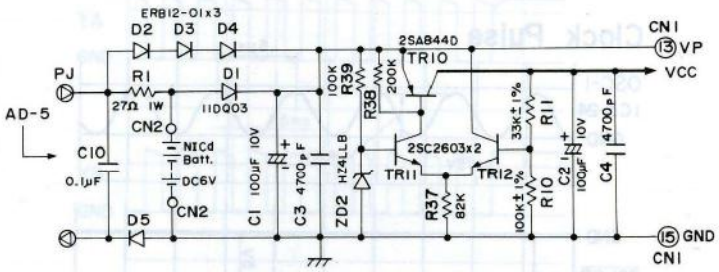
ORIGINAL
POWER SUPPLY CIRCUIT (IC 1 is used)



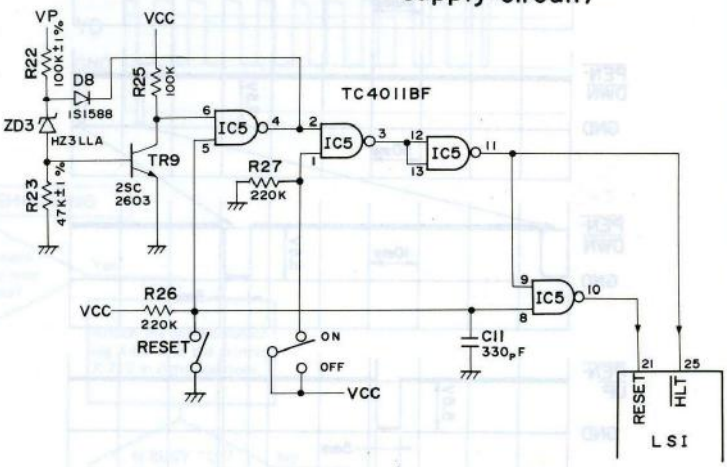
ORIGINAL
VP DETECTIVE CIRCUIT



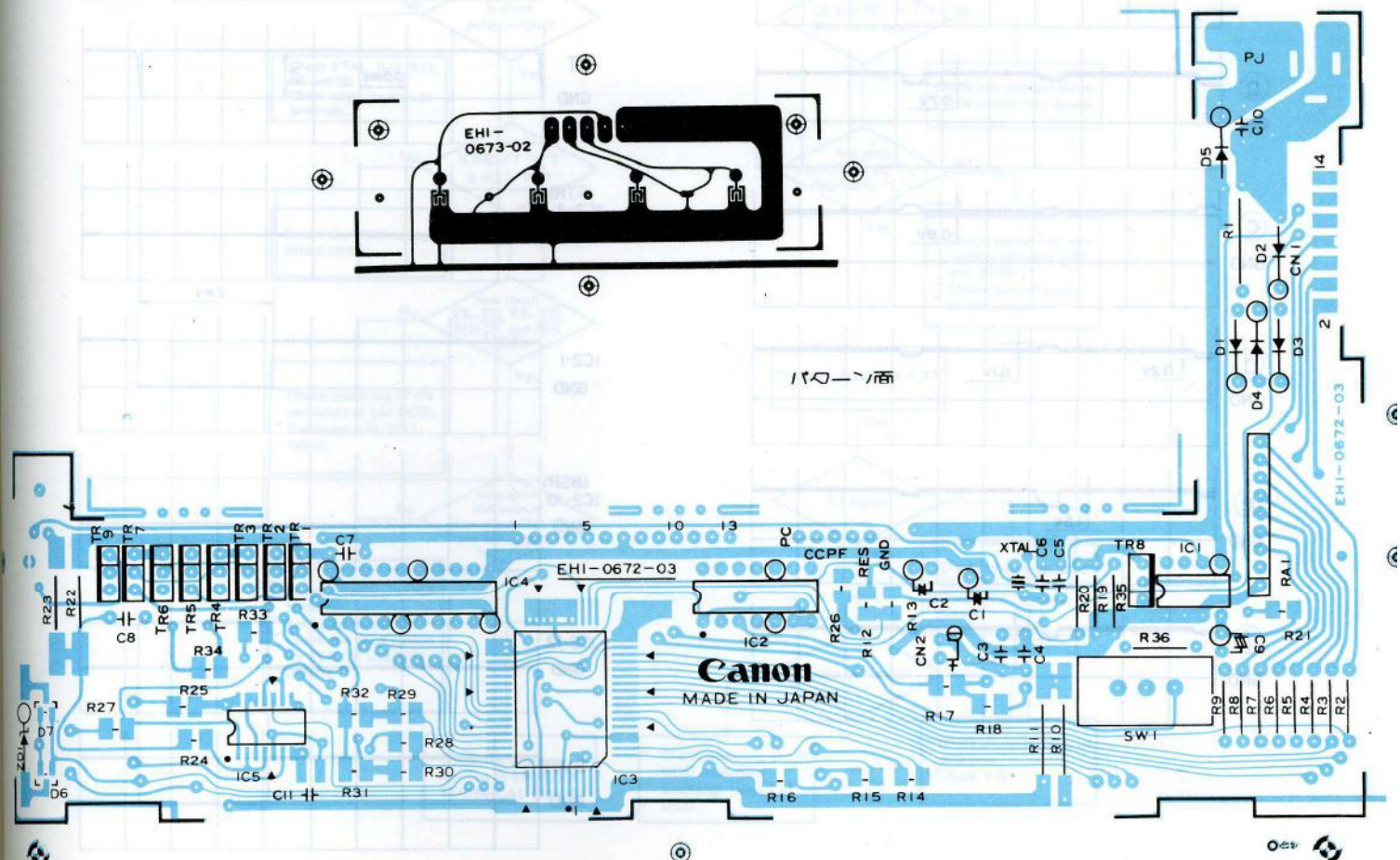
MODIFIED
POWER SUPPLY CIRCUIT (IC 1 is removed)



MODIFIED
VP DETECTIVE CIRCUIT (For modified power supply circuit)

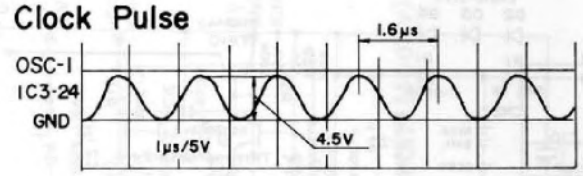


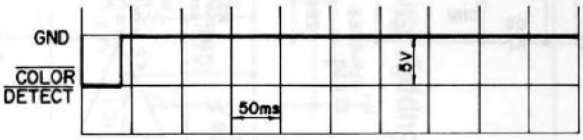
CARD LAYOUT

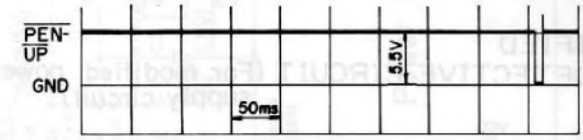


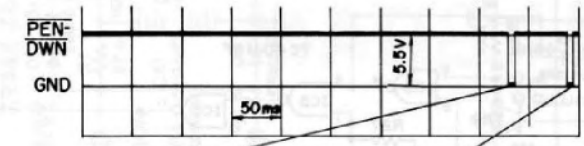
NORMAL WAVEFORM

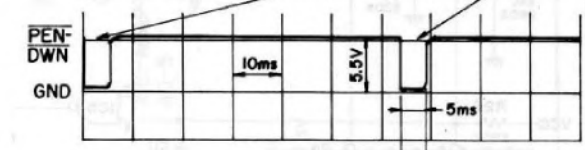
Clock Pulse

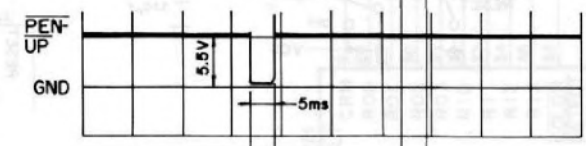


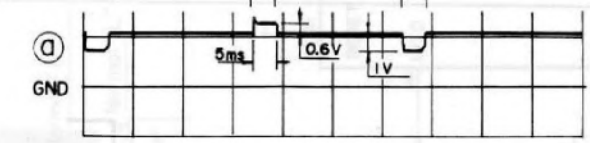


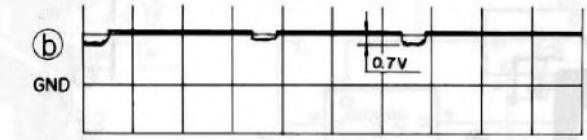


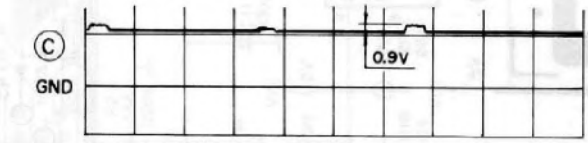


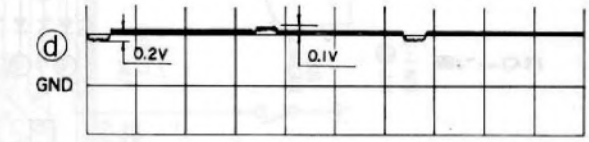


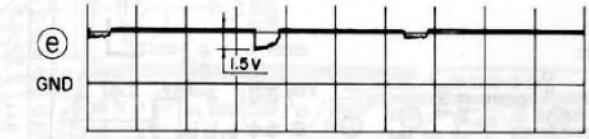


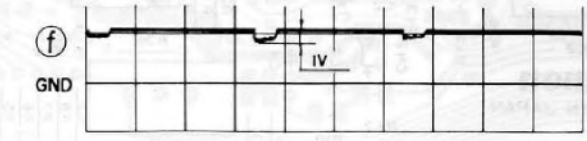


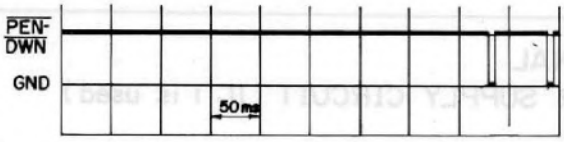


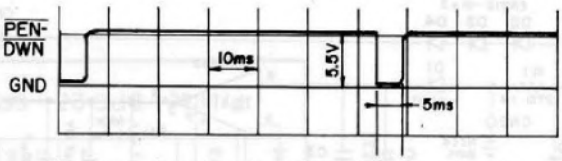


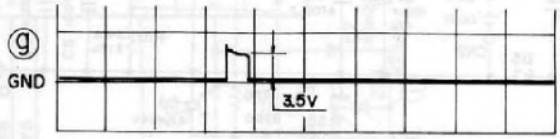


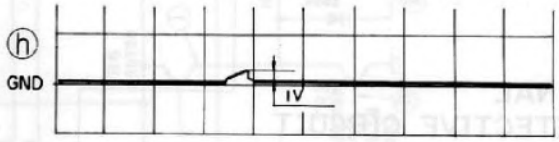


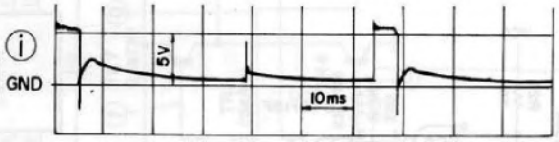


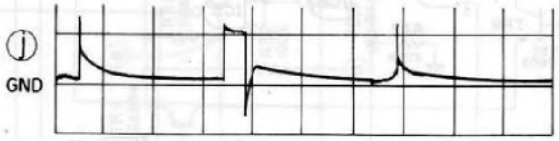


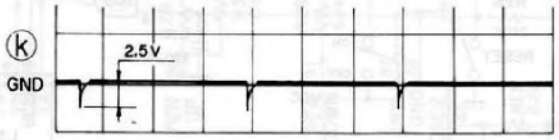


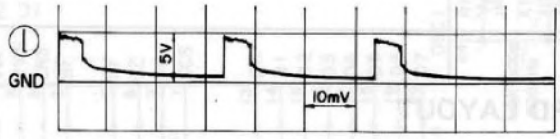


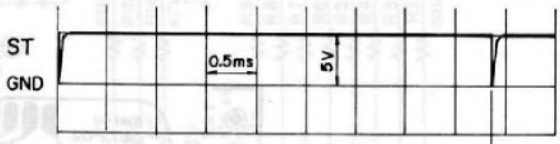


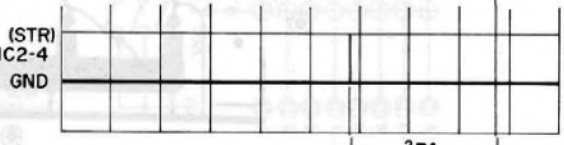


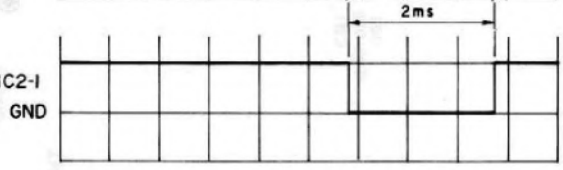


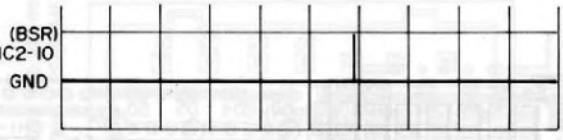


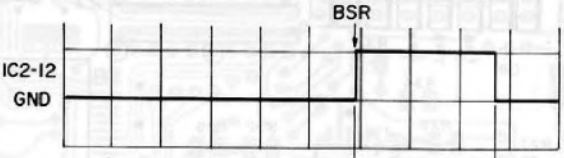


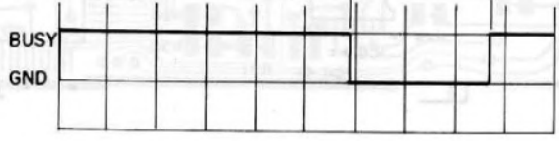




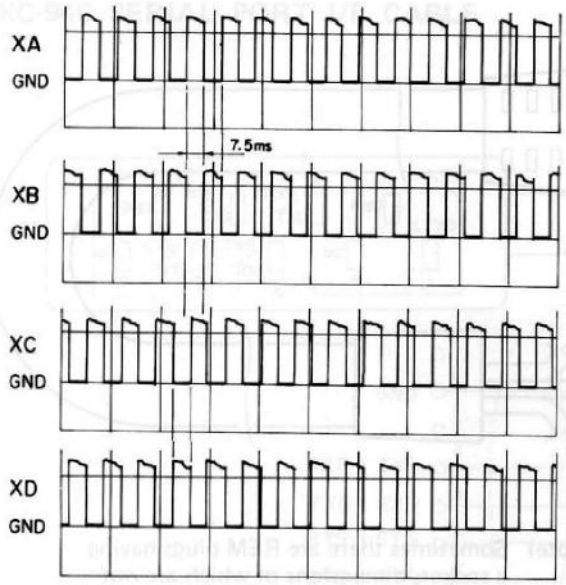




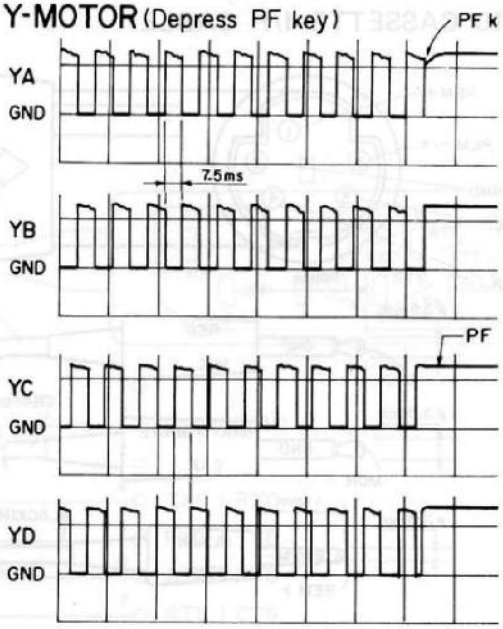




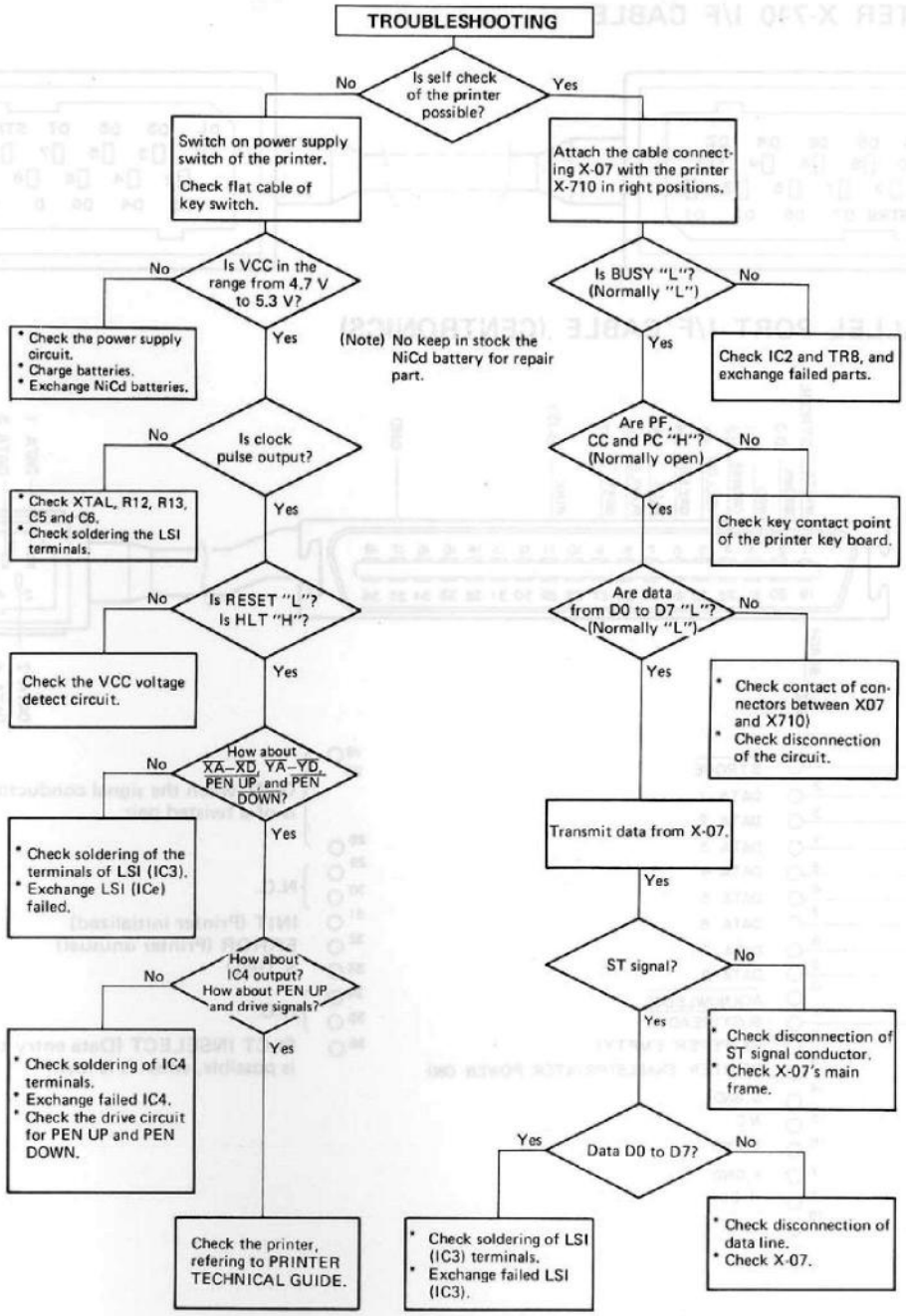
X-MOTOR



Y-MOTOR (Depress PF key)

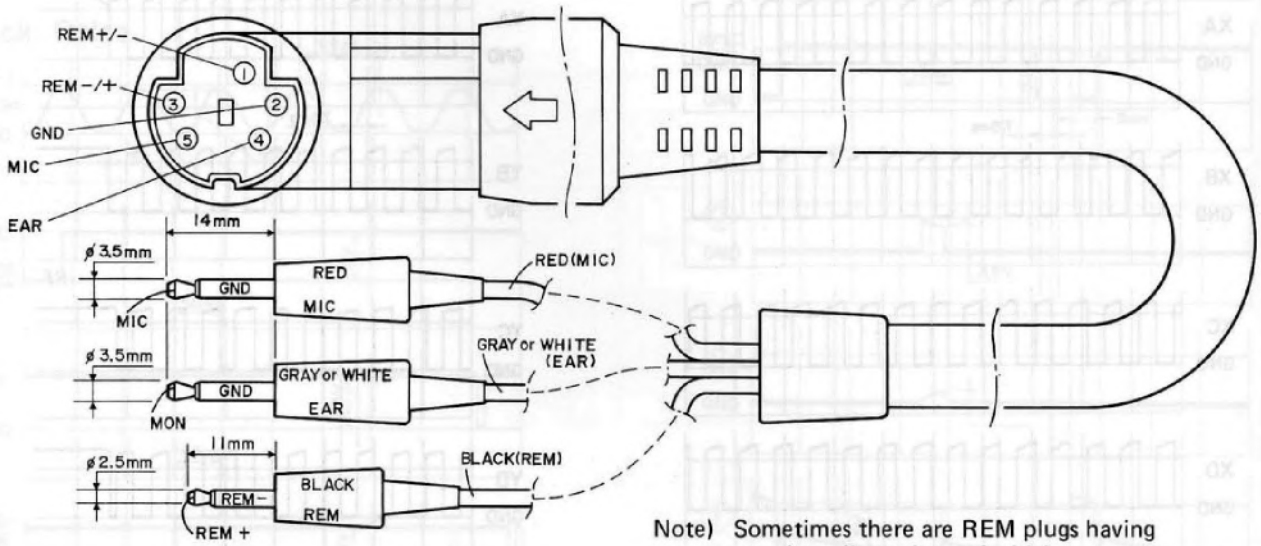


TROUBLESHOOTING



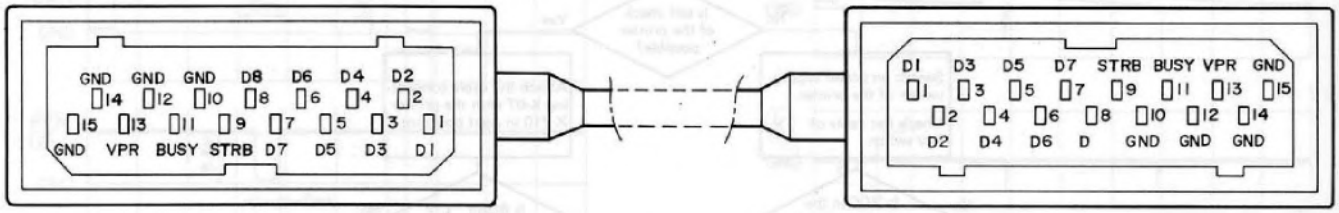
12. CABLE FOR OPTION

XC-910 CASSETTE I/F CABLE

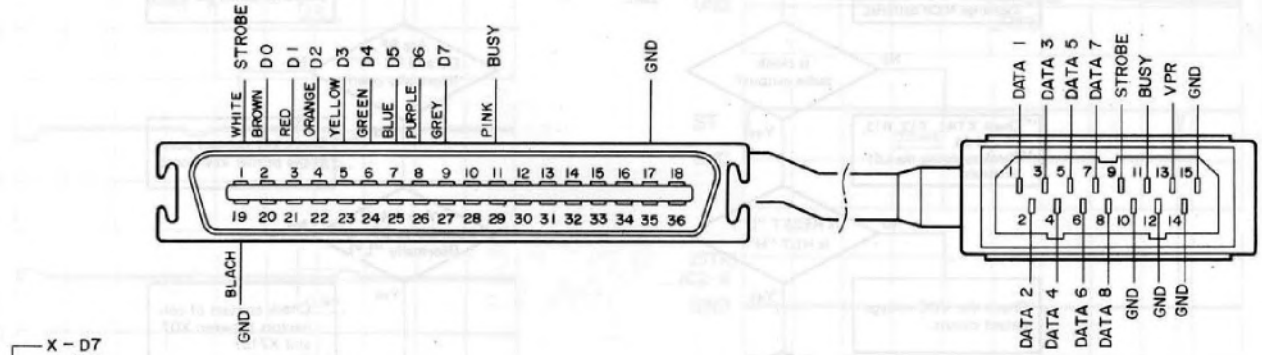


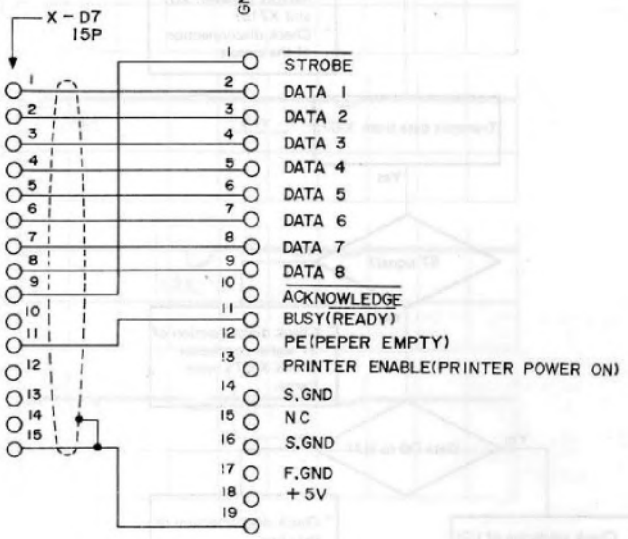
Note) Sometimes there are REM plugs having a socket, dimensions of which are not suitable for tape recorders manufactured by several makers.

XC-920 PRINTER X-710 I/F CABLE



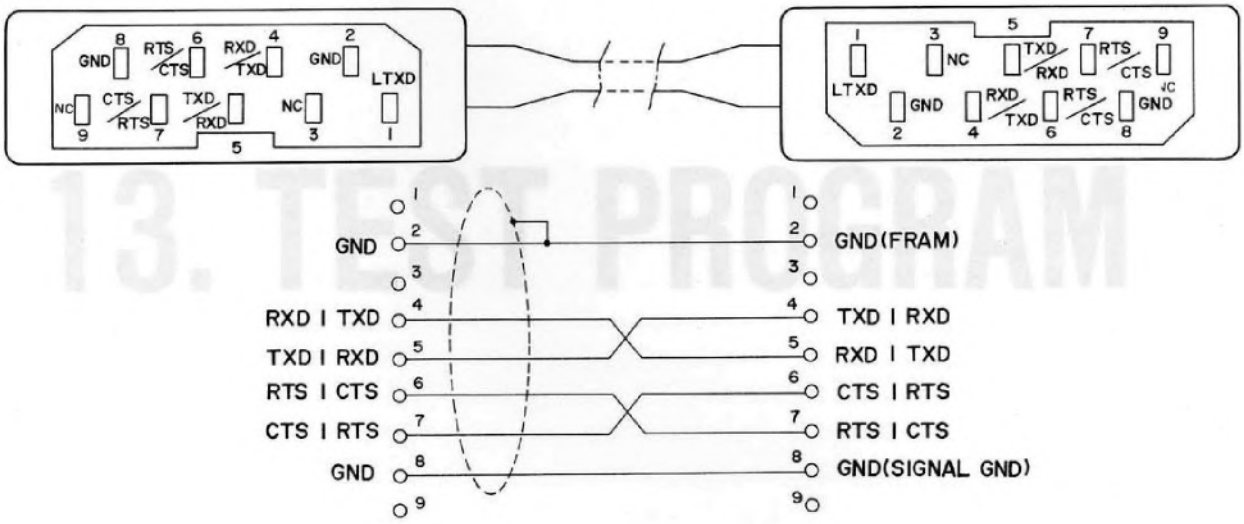
XC-930 PARALLEL PORT I/F CABLE (CENTRONICS)





19, 20 GND, when the signal conductor is of a twisted pair.
28
29, 30 N.C.
31 INIT (Printer initialized)
32 ERROR (Printer unusual)
33 S. GND
34, 35 N.C.
36 SLCT INSELECT (Data entry to the printer is possible, when it is low.)

XC-940 SERIAL PORT I/F CABLE



13. TEST PROGRAM

13. TEST PROGRAM

X-07 TEST PROGRAM SPECIFICATIONS

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1. INTRODUCTION

The X-07 test program is stored in the X-07 TEST in an AS-100 floppy disk (8-inch diameter) and is supplied to the user.

The user transfers the X-07 test program to the X-07 by using one of the AS-100 RS232C ports (COM2). (See Section 4 for details.)

Then, the transferred X-07 test program is stored in the pertinent RAM card. The program stored in the RAM card is used to test an X-07 to be checked.

(1) X07 test programs

1.	JMEM. X07	RAM card and 8k byte optional RAM read/write check
2.	JOPRAM. X07	RAM card and 8k byte optional RAM galloping check
3.	JUSER. X07	RAM card and 8k byte optional RAM read/write check
4.	JDUMP1. X07	Memory dump
5.	JSERI. X07	COM and OPT check
6.	JPRINT. X07	GPR check
7.	JKEY. X07	Keyboard check
8.	JCAS. X07	Cassette check
9.	JSBRAM. X07	Sub-CPU RAM check
10.	JCUNIT. X07	Sub-CPU check
11.	JRAM. X07	X-07 RAM, and ROM check support program
12.	CARD1. MID	X-07 RAM and ROM check program
13.	JRAM2. X07	RAM and ROM check support program for other than X-07
14.	CARD2. MID	RAM and ROM check program for other than X-07

(2) X-07 ⇄ AS-100, X-07 ⇄ X-07 RAM card transfer program (X-07 side)

1.	JCARDS. X07	RAM card data transmission
2.	JCARDR. X07	RAM card data reception

(3) AS-100 program utility

1.	JTRANSA. BAS	ASCII data transmission and reception
2.	JCARDR. BAS	RAM card data reception
3.	JCARDS. BAS	RAM card data transmission
4.	JFDUMP. BAS	File dump
5.	X07. BAS	X-07 command output program
6.	TTY3D. SUB	Intermediate code data reception RSINIT and intermediate code data transmission RSINIT
7.	TTY3DN. SUB	ASCII data reception RSINIT and ASCII data transmission RSINIT

2. X-07 TEST PROGRAM OPERATIONS

2.1 JMEM. X07

Purpose: Checks read/write operations or only the read operation in the RAM card and optional RAM spaces.

Requirements: The X-07 must be normal.
The test area must be reserved with an FSET command before JMEM. X07 execution.

(Output message/operation)	(Explanation)
* JMEM V1.00	Start message
RW−−>0 R−−>1	Read/write or read check specification 0 : Read check after write operation 1 : Read and compare check
START (HEX) =	Test start address (hexadecimal) (Note 1)
STOP (HEX) =	Test end address
INIT (HEX) =	Data at the test start address (Note 2)
ADD (HEX) =	Data at the address (test start address + 1)
LOOP (DEC) =	Test count (decimal)
ST = xxxx SP = xxxx IV = xx AV = xx LP = xx * OK (Y/N)	Confimation message ST for start and SP for stop IV for INIT, AV for add, and LP for loop
* DISP (Y/N)	Displayed and monitored on the screen? (Note 3)
*** CHECK OK !!	Normal end
* ERROR !! * ADR = xxxx * RD = xx TST = xx	Error message ADR: Error address RD for read data and TST for calculation value

Notes: 1. HEX: Hexadecimal data

2. The data string is as follows (for ST =2000, SP = 27FF, IV = 00, and AV = 01):

MEMORY	(LOOP = 1)	(LOOP = 2)	(LOOP = 3)
M	2000H → 00H (IV)	2000H → 01H (IV+AV)	2000H → 02H (IV+AV+AV)
E	2001H → 01H (IV+AV)	2001H → 02H (IV+AV+AV)	2001H → 03H (IV+AV+AV+AV)
M	⋮	⋮	⋮
O	27FEM → FEH	27FEM → FFH	27FEH → 00H
R	27FFH → FFH	27FFH → 00H	27FFH → 01H
Y			

3. Monitored data to be displayed:

```
* WRITE  LP = x
* xxxx   TST = xx
    ↑       ↑
    Address  Write data

* READ  LP = x
* xxxx   RD = xx   TST = xx
    ↑       ↑         ↑
    Address  Read data  Calculation data (correct data)
```

Other

Processing time (time period required for one loop operation)

Area size	With display		Without display	
	R	RW	R	RW
2k bytes	1.3 m	2.3 m	13 m	23.5 m
4k bytes	1.3 m x 2	Same as in the left column.	Same as in the left column.	Same as in the left column.
8k bytes	2.3 m x 4	Same as in the left column.	Same as in the left column.	Same as in the left column.

2.2 JOPRAM. X07

Purpose: Performs RAM card and optional RAM galloping checks.
(The RAM card is checked in 2k byte units, whereas the optional RAM is checked in 8k byte units.)

Requirements: The X-07 must be normal.
To stop the check being executed, press the RESET switch.

(Output message/operation)	(Explanation)
* GAL-OPTION, CARD V1.00	Start message
2000–2FFF (CARD) 2000–3FFF (OPTION) 4000–4FFF (CARD) 1. OPTION 2. CARD (1/2)	Checks whether the RAM is mounted, then outputs this message for the mounted address space. Select one of the RAM cards and optional RAMs, if both are mounted.
START?	Press the RETURN key to start the test.
* END	Test end message
2000–2FFF (CARD) 2000–3FFF (OPTION) 4000–4FFF (CARD)	Displays the tested area's contents
OK!	Normal end
ERROR! ADR = xxxx	Error message

Processing time

	Processing time
RAM card (4k bytes)	3 hours (approx.)
Optional RAM	23.5 hours (approx.)

Purpose: Tests the RAM card and optional RAM. (User-oriented simplified memory test program)

Requirements: Start address = H'2000', End address = H'2FFF', Data pattern = H'55', 'AA'
These items are fixed. The user may modify these items if necessary.
The test area must be reserved with an FSET command.

Modifications:

1. Address modification
   1 Optional RAM (8k byte) test
   Line No. 20, Line No. 50 } &H2FFF → &H3FFF
   2 RAM card test when the optional RAM is mounted:
   Line No. 20, Line No. 50 } &H2000 → &H4000, &H2FFF → &H4FFF
2. Data modification
   Example: H'55' → H'00', H'AA' → H'FF'
   Line No. 10 &H55 → &H00
   Line No. 80 &HAA → &HFF
   Line No. 90 &HAA → &HFF

Output message/operation	Explanation
DATA = 5	
* OK	55 pattern check OK
DATA = AA	
* OK	AA pattern check OK
* END	End message
ERR xxxx xx	Error message

xxxx: Error address
xx: Error data (AA or 55)

Processing time

	Processing time
RAM card (4k bytes)	2.6 m
Optional RAM (8k bytes)	5.2 m

Note: To check error data, key in PRINT HEX$ (PEEK (I)) immediately after the error message is output.

2.4 JDUMP1. X07

Purpose: Outputs the dump list of an arbitrary address space to the GPR.

Requirements: The X-07 must be normal.

Output message/operation	Explanation
*** DUMP LIST ***	Start message
START (hex) =	DUMP start address input (hexadecimal)
STOP (hex) =	DUMP end address input (hexadecimal)
* xxxx	First address of dumped 16-byte area

GPR output message example: START = 2000, STOP = 2015

```
*** DUMP LIST ***
@ START 2000 --> 2015 STOP
*2000 00  01  16  24  25  4A  41  43    20  20  50  02  0F  00  00  00
*2010 00  00  00  5E  06  0A  0D  BE    4A  43  41  53  00  79  05  14
```

Note: The GPR output is performed according to the [1, 1] specification for reduced character and blue attributes.

2.5 JSERI. X07

Purpose: RS232C, level converter, and optocoupler transmission and reception check and RS232C and level converter loop check

Requirements: The BASIC must operate normally.

Output message/operation	Explanation
* SERIAL V1.00	Start message
COM -> OPT -> 1	Select COM or OPT.
BAUD RATE =	For COM selection: 100 ~ 8000 (increment: 100) For OPT selection: 100 ~ 2400 (increment: 100)
MODE (A -> H)	(see table below)
SEND, RECV, SELF -> (0/1/2)	Select transmission, reception, or SELF for COM.
DATA INIT (HEX) = ADD (HEX) =	Note 1

MODE	EP	PEN	CBL
A	0	0	0
B	0	0	1
C	0	1	0
D	0	1	1
E	1	0	0
F	1	0	1
G	1	1	0
H	1	1	1

EP: Even parity
1: Even
2: ODD

PEN: Parity enable
1: Enable
0: Disenable

CBL: Character Bit length
1: 8 bits
0: 7 bits

COM/OPT = AAA BAUD = BBB MODE = CCC DDD IV = EE AV = FF OK (Y/N)	Confirmation message AAA : COM or OPT BBB : BAUD RATE DDD : SELF, RECEIVE, SEND EE : INIT value FF : ADD value
DISP (Y/N)?	Monitor output assignment (This message is not output for the OPT test. Y is assumed to be specified for transmission and N is assumed to be specified for reception.)

Messages for SELF/SEND/RECEIVE specifications are as follows.

(1) SELF specification

* SELF	For positions indicated by an asterisk, an asterisk is displayed for transmission; SPACE is displayed for reception.
HIT RETURN	The check starts when the RETURN key is pressed.
* SEND = xx * RECV = xx	Output for DISP = Y.

(2) SEND specification

* SEND	The asterisk blinks for each transmission.
HIT RETURN 2'ND	Press the RETURN key for the reception side first.
* SEND = xx	Output for DISP = Y. This is the case for the OPT specification.

(3) RECEIVE specification

* RECEIVE	The asterisk blinks for each reception.
HIT RETURN 1'st	Press the RETURN key first.
* RECEIV = xx TST = YY	XX: Reception data, YY: Expected value
	Output for DISP = Y. This is not the case for the OPT specification.

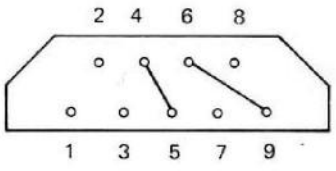
Error message

ERROR !! BAUD = xxx MODE = xx RECV = xx TST = xx	An I/O error message may be output on the BASIC side.

Others

Arrange pins as follows to perform the loop test with the SELF specification:

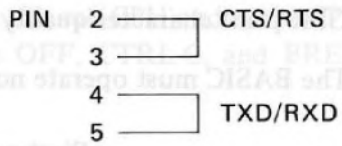
○ When looping back with the main frame RS-232C connector



PIN 4 (TXD) — 5 (RXD)

PIN 6 (CTS) — 9 (VCC)

○ When signals are returned at the level converter:



Note: Assume that the first position of the send/receive data is IV and the nth data is An, then

$An = IV + (n - 1) \times AV$

However, the value of An cannot be confined in the 0 to 255 range, the upper limit is specified as 127 (H'7F') for 7-bit data transfer operations and 255 (H'FF') for 8-bit data transfer operations. If the value exceeds the limit, it is set at −128 and −256 respectively.

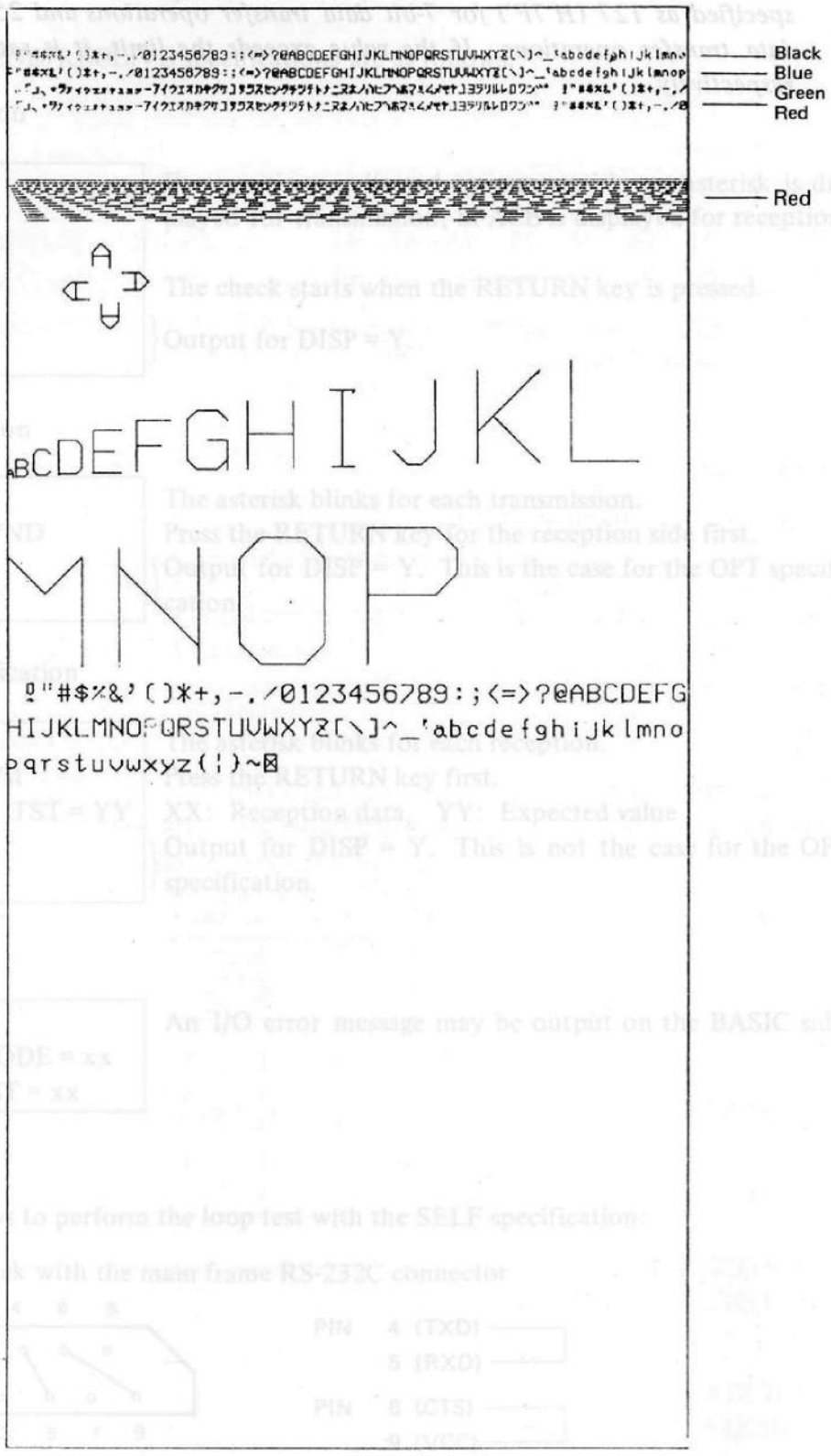
2.6 JPRINT. X07

Purpose: GPR print character quality and function test

Requirements: The BASIC must operate normally.

Output message	Explanation
* PRINT CHECK V1.00	Start message

Printout example



2.7 JKEY. X07

Purpose: Checks interfaces between the sub-CPU and main CPU by inputting data from the keyboard. (The OFF, CTRL-C, and BREAK keys cannot be used.)

Requirements: The BASIC must operate normally.

Output message/operation	Explanation
*** KEY V1.00	Start message
* INPUT KEY	Press any key. (The OFF, CTRL-C, and BREAK keys cannot be used.)
KEY = x	Input key echo back
CODE = xx	Input key code (ASCII code)

2.8 JCAS. X-07

Purpose: Cassette test program.
Test is performed by CSAVE and CLOAD verify operations for a cassette recorder.

Requirements: The BASIC must operate normally.

Output message/operation	Explanation
* CASSETTE V1.00	Start message
WAIT	Wait for data area sum check completion.
* I'm OK !!	DATA AREA SUM CHECK IS OK
* I'm NG!!, A = xx	Data area sum check is not OK: Sum check value = XX (expected value = 00)
PUSH REC. HIT RETURN	Set the cassette deck to the record mode and press the RETURN key, then the CSAVE operation starts.
CSAVE END	Save operation on the cassette has been completed.
REWIND & PUSH PLAY	Rewind the cassette tape and set the play mode.
HIT RETURN	CLOAD verify operation begins.
* CASSET END	Verify end
Error message Bad	Verify error
Or, the tape is fed to the end without the following message:	The program name is missing.
FOUND: JCAS	

2.9 JSBRAM. X07

Purpose: Sub-CPU RAM check

Requirements: The BASIC must operate normally.

Output message/operation

SUB CPU RAM V1.00	Start message
OK	Normal end
* END	
ERROR	Error message
ADR = xxxx	
* END	

Processing time: One minute (approx.)

Note: An error may occur if keys are input continuously at random while the test is being executed.

2.10 JCUNIT. X07

Purpose: Sub-CPU function test

Requirements: The BASIC must operate normally.

Output message	Explanation
* CCUNIT CHECK V1.00	Start message
HIT RETURN	The test starts when the RETURN key is pressed.
IF BUZZ ON, CTRL-S	An error is assumed if the buzzer does not sound.
AND HIT "Y"	An error is assumed if the buzzer does not stop sounding when the CTRL-S is input. Input Y.
!"#$%&' ( ) *+, –./01234 56789:;< = > ?@ABCDEFGH IJKLMNOPQRSTUVWXYZ[¥]∧_' a b c d e f g h i j k l m n o	Displayed approximately two seconds.
Empty screen	Displayed approximately two seconds.
————————	Displayed approximately four seconds.
HIT "F1"	Input F1.
PRINT "X-07 is NO. 1	Displayed approximately one second.
PUSH "< –"	Input arrow mark (◂).
HIT F6	Input F.
HIT SPACE KEY	Input a blank character by pressing the SPACE key.
KEEP PUSHING "U" KEY UNTIL OFF MESSAGE	Continue pressing the U key.

UUUナナナÜÜÜ 444 Same as above.

TAKE YOUR FINGER OFF Release the key.

■♠ ◯ Displayed approximately one second.

◆tim◆cld◆loc◆1st◆run

Beep sounded (do, re, mi, fa, sol, la, and si, do).

Error message

ERROR 1 to ERROR 13: See the following explanations:

1. An error is found in the compare check after data is set in TIME$.
2. An error is found in the compare check after data is set in DATE$.
3. Horizontal cursor position error
4. Vertical cursor position error
5. Other than B is found in character 3 of line 0 on the screen.
6. Character string ! to 0 was output and the data was read and compared, but a compare error has occurred.
7. The data read after the screen clear operation contains other than blank characters.
8. The data read after dots are output to the screen contains other than dots.
9. The data read after clearing dots on the screen contains dots.
10. Same as error 8.
11. Same as error 8.
12. Character variable initial value error
13. FUNCTION key read error

Others:
An error is assumed if the pertinent screen or sound is not obtained according to operations. For example, an error is assumed if the specified buzzer does not sound.

2.11 JRAM. X07

Purpose: Passes parameters to CARD1. MID (Section 2.12) and outputs reports containing execution results to the GPR. This program does not check data selected according to the following operations, that is, the selected data is passed to CARD1. MID
JRAM. X07 is not the test program for the X07.

Requirements: The BASIC must operate normally.
The ROM/RAM switch must be set to ROM.
The RAM card (CARD1) must be mounted.
RAM check is performed in a range other than 2000 to 2FFF.

Output message/ operation	Explanation
* SET CARD1 V1.00	Start message
REPORT (Y/N)?	Reports to be output to GPR after CARD1 execution.
* REPORT *	Header message (Note 1)
!! NO REPORT !!	No data is found for the report.
* REPORT END *	Reporting end
INIT. (Y/N)	Initializing CARD1?
* INITIALIZE *	Initialize operation starts. (Note 2)
* INIT END *	Initialize operation end
ROM SUM (Y/N)	Sum check by using CARD1.
*** ROM SUM ***	ROM sum check start message
DEFAULT ADR B000 –> BFFF C000 –> FFFF	Sum check area after initialization
ADD SUM (Y/N)	Add other area for sum check.
START ADR =	ROM sum check area start address
AREA SIZE =	Area size (bytes) to be added
OK (Y/N)	Confirmation
NEXT (Y/N)	Other additional data? (Three additional areas may be specified, that is, a total of five areas.)
* ROM SUM END	Sum check area address setting completed
RAM R/W (Y/N)	RAM read/write?
* RAM R/W *	
DEFAULT ADR 0000 –> 1FFF	Check area after initialization
ADR MODIFY (Y/N)	Check area to be modified?
START ADR =	Check start address
END ADR =	Check end address
OK (Y/N)	

PW. ON–OFF (Y/N)	Perform check after power off?
GALLOP (Y/N)	Galloping check?
* GALLOP *	
DEFALT ADR	Check area after initialization
0000 –> 1FFF	
ADR MODIFY (Y/N)	Check area to be modified?
START ADR =	Check start address
END ADR =	Check end address
OK (Y/N)	
*** SET RAM END ***	End message

Card 1 generation end. Turn power off, remove card 1, install it on the X-07 to be checked, turn power on, then the program begins running.

Notes: 1. In addition to the message, the question is whether deletion is to be performed after the report is output to the GPR.
See Section 2.12 for report contents.

2. *The contents of initialized card 1*
   1. *Address 2800 contains "love".*
   2. *The following are checked:*
      RAM read/write (0000 to 1FFF)
      ROM sum check (B000 to BFFF and C000 to FFFF)
      Power on/off (0400 to 1FFF)
      Galloping check (0000 to 1FFF)
3. *The report area is deleted and initialized.*
4. *Display error flag is initialized.*

Others: The RAM read/write and galloping tests use the same start and end address areas, that is, the same address space is checked. The last specified start and end addresses are used.

2.12 CARD1. MID (See Section 4.4.1 for Preparations.)

Purposes: When the BASIC does not operate normally, this program allows operations to be performed with a minimum of resources under control by using the keyword love.
The system RAM is checked and the system ROM sum check value is obtained. In this case, the RAM area other than the space from address H'2000' to address H'2FFF' can be checked.

Requirements: Control must be passed by using the keyword love.
The ROM/RAM switch must be set to ROM.

Functions

1. Obtains the contents of interface chip registers saved before control is passed to the BASIC. (See Table 2.12.1, Interface Chip Registers for details.)
2. Performs sum check for memory devices such as the system ROM.
3. Performs read/write check for memory devices such as the system RAM (other than the range from 2000 to 2FFF).
4. Data holding check after:
   Power off → on → off → on
5. Performs system RAM galloping check.
6. Performs reporting (See Section 2.12 (Report Output Example) for details).

Note: Tests for items 2–5 are specified by support program JRAM. X07.
Areas must be specified for items 2, 3, and 5.
For item 6, data can be output to the GPR by using JRAM. X07.

Output message/operation	Explanation
* C. RAM-CARD V1.00	Start message
* REG 00, 02, A2, E3, * 80, C3, 05, A5	Contents of interface chip registers 0 ~ 7
–ROM SUM START B000H –> BFFFH = 24H C000H –> FFFFH = AEH –ROM SUM END	Sum check value display
–RAM R/W START ST = 0000, SP = 1FFF DATA (1) END DATA (2) END –RAM R/W END	Start and end addresses Test order: DATA (1): 00, 01, ..., FF Test order: DATA (2): FF, FE, ..., 00 Read/write operations by incrementing addresses
–P ON/OFF START (1) 400H –– > 1FFFH	Address fixed: H'400' to H'1FFF' Writes data pattern 00, 01, ..., FF in this order by incrementing addresses. After this message is output, the X-07 system is set to the off state. Press the ON key when necessary.
–P ON/OFF END (1) RESET !	After the ON key is pressed, the read check is performed. If the check is OK, this message is output. Press the RESET switch.

Start message
REG message

–P ON/OFF START (2)	Writes data pattern FF, FE, ..., 00.
400H –– > 1FFFH	Press the ON key when necessary.
–P ON/OFF END (2)	If the test is OK, the message is output.
–GALLOP START	
ST = 0000, SP = 1FFF	
LOOP = 00	Loop test starts with the initial value 00.
DATA (00)	The test is carried out with data items set to 00 except for the target address.
DATA (FF)	The test is carried out with data items set to FF except for the target address.
	The loop has no end point. Press the RESET to stop the test.

Note: The same start and end address areas are used for the RAM read/write and galloping check; hence, the same values are used respectively.

Error messages

! RAM R/W ERR
E. ADR = xxxx
E. DAT = xx T. DAT = xx

Report output example

```
*** REPORT ***

*C.RAM-CARD V1.00
* REG  00,02,A2,C3,
*      80,C3,05,0A
-ROM SUM START
B000H->BFFFH = 24H
C000H->FFFFH = AEH
-ROM SUM END
-RAM R/W START
ST=0000,SP=1FFF
DATA(1) END
DATA(2) END
-RAM R/W END
-P ON/OFF START(1)
400H-->1FFFH
*C.RAM-CARD V1.00
* REG  00,02,A2,C3,
*      80,C3,05,0A
-P ON/OFF END(1)
RESET !
*C.RAM-CARD V1.00
* REG  00,02,A2,C3,
*      80,C3,05,0A
-P ON/OFF START(2)
400H-->1FFFH
*C.RAM-CARD V1.00
* REG  00,02,A2,C3,
*      80,C3,05,0A
-P ON/OFF END(2)
-GALLOP START
ST=0000,SP=1FFF
LOOP=00
DATA(00)
*** REPORT END ***
```

Report output message

This is output when the display output message cannot be read or display output becomes impossible because of X-07 failure.
Generally, the display output messages are used.
The following are unique as report output messages:

! DISP ERROR X

X = 1: Report output is impossible because data is being output from the sub-CPU.
X = 2: Report output is impossible because the data to be output to the sub-CPU is not received by the sub-CPU.

* C. RAM CARD END

This message is additionally output when the galloping check is not performed.

Table 2.12.1 Interface Chip Registers

Bit name / Name (Address)	D7	D6	D5	D4	D3	D2	D1	D0	
R0 (F0)	(TS0) TRQ1	(TS1) TRQ0	BIE	ALMIE	XTIE	XRIE	TIE	RIE	* Interrupt control REG
R1 (F1)	$\overline{TB7}$	$\overline{TB6}$	$\overline{TB5}$	$\overline{TB4}$	$\overline{TB3}$	$\overline{TB2}$	$\overline{TB1}$	$\overline{TB0}$	* External BUS receive REG
R2 (F2)	CBSY	ALRM	AUX1	AUX2	0	BGI	XTRE	XRRF	* Flag REG
R3 (F3)	Meaningless								
R4 (F4)	FCAS	BRGST	CNTR	LEO	MD1	MD0	BZON	REM	* Mode REG
R5 (F5)	Meaningless								
R6 (F6)	$\overline{CTS}$	0	FE	OE	PE	TXE	RXRDY	TXRDY	* UART status REG
R7 (F7)	D7	D6	D5	D4	D3	D2	D1	D0	* UART data REG

Report message correspondence

Example ——— R0 ~ R3

* REG 00, 02, A2, C3,

* 80, C3, 05, 0A

——— R4 ~ R7

Processing time

Galloping

Area size	Time	Explanation
2k bytes	1 hour	30 minutes for each DATA (00) and DATA (FF)
8k bytes	16 hours	8 hours for each DATA (00) and DATA (FF)

Purpose: Passes parameters to CARD2. MID (Section 2.14).
This program only sets data selected according to the following operations in CARD2. MID. JRAM2. X07 is not the test program for the X07.

Requirements: The BASIC must operate normally.
The ROM/RAM switch must be set to ROM.
The RAM card (card 2) must be installed.
RAM check range: Mounted range other than that from 1000 to 1FFF.

Output message/operation	Explanation
* SET CARD2 V1.00	Start message
ROM SUM (Y/N)	Performs the sum check on card 2.
*** ROM SUM ***	ROM sum check set start message
DEFAULT ADR 4000 –> 5FFF 6000 –> 7FFF A000 –> AFFF	The sum check value is output unconditionally for each of these areas.
ADD SUM (Y/N)	Add other area.
START ADR =	ROM sum check start address
AREA SIZE =	Area size (bytes) to be added
OK (Y/N)	Confirmation
NEXT (Y/N)	Other additional areas? (Up to two areas may be added, a total of five areas.)
* ROM SUM END	Sum check area set end
RAM R/W (Y/N)	Performs RAM read/write operations.
* RAM R/W *	
DEFAULT ADR 8000 –> 97FF	
ADR MODIFY (Y/N)	Check area to be modified?
START ADR =	Check start address
END ADR =	Check end address
OK (Y/N)	
PW. ON–OFF (Y/N)	To be checked after power is turned off?
* POWER ON –> OFF	
DEFAULT ADR 8000 –> 97FF	
ADR MODIFY (Y/N)	
START ADR =	
END ADR =	
OK (Y/N)	

```
GALLOP (Y/N)      Galloping check to be carried out?
* GALLOP *
DEFAULT ADR
8000 -> 97FF
ADR MODIFY (Y/N)

START ADR =
END ADR =
OK (Y/N)

*** SET RAM END ***
```

Notes: 1. Since the same start and end addresses are used for the RAM read/write, power on/off, and galloping checks, the same address space is tested.
The last specified start and end addresses are used.
Card 2 is automatically initialized when this program is executed.

2. *Card 2 contents after initialization*
   1. *RAM read/write (8000 to 97FF)*
      ROM sum check (4000 to 5FFF, 6000 to 7FFF, and A000 to AFFF)
      Power on/off check (8000 to 97FF)
      Galloping check (8000 to 97FF)
   2. *Display error flag is initialized.*
   3. *Printer error flag is initialized.*

2.14 CARD2.MID (See Section 4.4.2 for Preparations)

Purpose: Performs RAM and ROM sum check for other than the X-07. These tests can be carried out by receiving control by means of the keywore love even if "love" has been written at an address greater than 2000 (on the FFFF side). If the printer is connected, the message is output to the printer and display device.

Requirements: Control must be passed by means of the keyword love.
The X-07 must be normal.

Functions:

1. Performs ROM sum check.
2. Performs RAM read/write check (other than the range 1000 to 1FFF).
3. Data holding check in other than the range 1000 to 1FFF under the following condition:
   Power off → on → off → on
4. Performs a RAM galloping check (other than the range 1000 to 1FFF).

Note: Tests 1 ~ 4 can be specified by support program JRAM2. X07.
Test areas can be specified for tests 1 ~ 4.

Output message/operation	Explanation
*OPTION MEM V1.00	
-ROM SUM START	
4000H->5FFFH = 00H	} Output in any case.
6000H->7FFFH = 00H	
A000H->AFFFH = 00H	
B000H->BFFFH = 24H	} Additional registration in JRAM2. X07.
C000H->FFFFH = AEH	
-ROM SUM END	
-RAM R/W START	
ST=8000,SP=97FF	Check range
!RAM R/W ERR.	} Error message
E.ADR=8000	
E.DAT=FF T.DAT=00	
DATA(1) END	Set 00, 01, 02, ..., FF beginning from the start address.
!RAM R/W ERR	} Error message
E.ADR=8001	
E.DAT=FF T.DAT=FE	
DATA(2) END	Set FF, FE, FD, ..., 00 beginning from the start address.
-RAM R/W END	
-P ON/OFF START(1)	
8000->97FF	Check range
*OPTION MEM V1.00	Turn power off.
!RAM R/W ERR	} Error message
E.ADR=8000	
E.DAT=FF T.DAT=00	
-P ON/OFF END(1)	Set 00, 01, 02, ..., FF beginning from the start address.
RESET !	Press the RESET switch.
*OPTION MEM V1.00	
-P ON/OFF START(2)	Set FF, FE, FD, ..., 00 beginning from the start address.
8000->97FF	
*OPTION MEM V1.00	
!RAM R/W ERR	
E.ADR=8001	
E.DAT=FF T.DAT=FE	
-P ON/OFF END(2)	
-GALLOP START	
ST=8000,SP=97FF	
LOOP=00	Output the loop count (00 to FF).
DATA(00)	Set 00 in area except for the target address.
!GALLOP ERR	
E.ADR=8001	
E.DAT=FF T.DAT=00	

Note: *E. ADR:* *Error address*
E. DAT: *Error data*
T. DAT: *Write data*

Other messages	Explanation
! DISP ERR X	X = 1: Output is impossible because data is being output from the sub-CPU. X = 2: Output is impossible because the data to be output to the sub-CPU cannot be received by the sub-CPU. The message is output to the printer.
* OPTION MEM END	This message is output when the galloping check is not carried out.
! GPR ERR	Output is impossible because the GPR is busy. This message is output to the display device.

Processing time

Galloping

Area size	Time	Explanation
2k bytes	1 hour	30 minutes for each DATA (00) and DATA (FF)
8k bytes	16 hours	8 hours for each DATA (00) and DATA (FF)

2.15 JCARDS.X-07

Purpose: Transfers the RAM card contents to the AS-100 without alteration.

Requirements: The ROM/RAM switch must be set to ROM.

Operation procedures: The following must be set by the AS-100 before it is executed:

```
A > RSHND
A > SUBMIT  TTY3D or SUBMIT TTY3DN
A > BASIC   JCARDR
```

Connect the X-07 and AS-100 COM2 via level converter RS-232C and input the commands listed above in the AS-100.
Specify the file to be received and storing file to be used on the AS-100 side by using JCARDR, then execute the program.

Output message

** CARD SEND END	Transmission end message

2.16 JCARDR.X-07

Purpose: Transfers data from the AS-100 to the X-07 without alteration.

Requirements: The ROM/RAM switch must be set to ROM.

Operation procedures: The following must be set on the AS-100 side before program execution:

```
A > RSHND
A > SUBMIT  TTY3D
A > BASIC   JCARDS
```

Connect the X-07 and AS-100 COM2 via level converter RS-232C and input the commands listed above.
Specify the file to be transferred and its destination on the AS-100 side by using JCARDS, then execute the program.

Output message

** CARD RECEIVE

xxxx Monitor and output the number received bytes.

** RECEIVE END End message

3. AS-100 PROGRAM OPERATIONS

3.1 JTRANSA. BAS

Purpose: ASCII data transmission and reception

Operation procedures: Operate according to the following messages:

```
***** TRANSFER ASCII DATA *****

* IF YOU GET ASCII DATA FROM TTY: , USE TTY3DN
* IF YOU PUT ASCII DATA  TO  TTY: , USE TTY3D OR TTY3DN

 TTY3DN : RSINIT TTY: 300  8OS BUFSIZ(4096,4096) NOWAIT
 TTY3D  : RSINIT TTY: 300  8OS BUFSIZ(4096,4096) WAIT NOAUTOX

* IF YOU USE X07 AS SOURCE OR DESTINATION
 X07 OPERATION
      DATA --> X07
         : INIT 05,"CON:",300,"D"
         : EXEC &HEE1F
      DATA <-- X07
         : INIT 01,"CON:",300,"D"
         : LIST 01

* SOURCE      (TTY:/XXXXXX.XXX)           = ?JCARDS.X07
* DESTINATION (TTY:/LST:/CRT:/XXXXXX.XXX) = ?CRT:

!!  OK (Y/N)     ?Y


********** START GET &PUT ASCII DATA **********

 SOURCE      FILE = JCARDS.X07

 DESTINATON FILE = CRT:


10 REM CARD->AS-100
20 INIT 01,"COM:",300,"D"
30 I=&H2000
40 FORJ=I TO &H2FFF
50  A=PEEK(J)
60  OUT 01,A
70 NEXT J
100 PRINT"** CARD SEND END"
110 END


**********  END  TRANSFER  **********

 TRANSFER SIZE     =  163

 SOURCE      FILE = JCARDS.X07

 DESTINATION FILE = CRT:
```

Note: Do not execute this program with the following specifications: Input = TTY and output = LST.

Purpose: Stores the card-unit information received from the X07 in a XXXXXX.MID file. Or X07's program unit data may be stored.

Operation procedures: Operate according to the following messages:

```
****** RECEIVE MID-DATA FROM " X07 "

* AS-100 : RSHND
  TTY3D  : RSINIT TTY: 300 8OS BUFSIZ(4096,4096) WAIT NOAUTOX

* X07    : JCARDS / SAVE "COM:",300,"D"

* RECEIVE DATA-FILE                = ?TEST
* 1. DATA-SIZE    2. 13-NULL (1/2) ?1
* RECEIVE DATA-NUM.                = ?4096
      OK (Y/N)                      ?Y



********** START RECEIVING **********

 DATA FILE NAME = TEST

 TRANSFER DATA  = 4096

**********  END  RECEIVING **********
 MID-DATA  SIZE =  4096

 DATA-FILE NAME = TEST.MID
```

Note: For program-unit data reception, select 2.13-NULL.
Mid is automatically appended, so do not specify MID in the XXXXXX.MID.

Purpose: Transmits the file received and stored with JCARDR. BAS to the X-07.

Operating procedures: Operate according to the following messages:

```
***** SEND MID-DATA TO " X-07 "

* AS-100 : RSHMD
 TTY3DM  : RSINIT TTY: 300 8BS BUFSIZ(4096,4096) NOWAIT
 TTY3D   : RSINIT TTY: 300 8BS BUFSIZ(4096,4096) WAIT NOAUTOX

* X07    : JCARDR / LOAD "COM:",300,"D"

* SEND MID-DATA FILE              = ?TEST
* 1. DATA-SIZE   2. 13-NULL (1/2) ?1
* SEND DATA-NUM.                  = ?4096
    OK (Y/N)                      ?Y
```

```
********** START SENDING **********

 MID-DATA NAME = TEST

 TRANSFER DATA  = 4096

**********  END  SENDING **********

 PROGRAM SIZE =  4096
 MID DATA NAME = TEST.MID
```

Note: For program-unit data reception, select 2.13-NULL.
For card-unit data reception, select 1. DATA-SIZE.
MID is automatically appended, so do not specify MID in XXXXXX. MID.

3.4 JFDUMP.BAS

Purpose: Edits contents of each file in the list format and outputs the resultant data. Use this program to check whether the contents of XXXXXX.MID received from the X-07 is correct.

Operation procedures: Operate according to the following messages:

```
***** FILE DUMP START *****

* IF YOU GET DUMP DATA FROM TTY: ,USE TTY3D
  TTY3D  : RSINIT TTY: 300 8OS BUFSIZ(4096,4096) WAIT NOAUTOX

* IF YOU USE X07 AS SOURCE
 X07 OPERATION
      X07 (RAM-CARD) --> MID-CODE DATA
         : * USE PROGRAM JCARDS IN X07
      X07'S PROGRAM
         : * SAVE "COM:",300,"D"


 SOURCE       FILE  (TTY:/XXXXXX.XXX)       = ?JCARDS.X07
 DESTINATION  FILE  (CRT:/LST:/XXXXXX.XXX) = ?CRT:

 DUMP     ADDR = ?0
 1.DATA SIZE   2. 13-NULL  (1/2) ?1
 DUMP     SIZE = ?200

!!  OK (Y/N)      ?Y
```

```
*00000    0D 0D 0A 31 30 20 52 45    4D 20 43 41 52 44 2D 3E
*00010    41 53 20 31 30 30 0D 0A    32 30 20 49 4E 49 54 20
*00020    23 31 2C 22 43 4F 4D 3A    22 2C 33 30 30 2C 22 44
*00030    22 0D 0A 33 30 20 49 3D    26 48 32 30 30 30 0D 0A
*00040    34 30 20 46 4F 52 4A 3D    49 20 54 4F 20 26 48 32
*00050    46 46 46 0D 0A 35 30 20    20 41 3D 50 45 45 4B 20
*00060    4A 29 0D 0A 36 30 20 20    4F 55 54 20 23 31 2C 41
*00070    0D 0A 37 30 20 4E 45 58    54 20 4A 0D 0A 31 30 30
*00080    20 50 52 49 4E 54 22 2A    2A 20 43 41 52 44 20 53
*00090    45 4E 44 20 45 4E 44 22    0D 0A 31 31 30 20 45 4E
*000A0    44 0D 0A 00 00 00 00 00    00 00 00 00 00 00 00 00
*000B0    00 00 00 00 00 00 00 00    00 00 00 00 00 00 00 00
*000C0    00 00 00 00 00 00 00 00    00 00 00 00 00 00 00 00


*** FILE DUMP END ***
```

Note: DUMP ADR: Dump start address
It is advisable to select 2.13-NULL to check XXXXXX.MID stored in program units.
Do not use this program with specifications input = TTY and output = LST.

3.5 X07. BAS

Purpose: This is prepared to use AS-100 keyboard instead of X-07.

Operation procedures: No special instructions to be given.
Execute after the following:
RSHND
SUBMIT TTY3D or SUBMIT TTY3DN
X-07 preparation: INIT#5, "COM:", 300, "D"
EXEC & HEEIF

Note: Press the CANCEL key to terminate the program.

```
********** AS-100 TO X07 **********

• SET MESSAGE TO X07
?10 REM TEST
?20 ?"••• CANON •••"
?30 ?"    X-07"
?40 LOCATE 0,0:?"&&";
?50 GOTO 20
?RUN
?EXEC &HEE33
?LPRINT[0,1]
?
```

3.6 TTY3D. SUB

```
rsinit tty: 300 8os bufsiz(4096,4096) wait noautox
```

3.7 TTY3DN. SUB

```
rsinit tty: 300 8os bufsiz(4096,4096) nowait
```

Note: RSINIT command (RS232C port assignment) is used for these two programs, where:

tty: *COM2 is used.*

300: *Baud rate*

80S: *Equivalent to mode D of X-07*

BUFSIZ: *Send/receive buffer size*

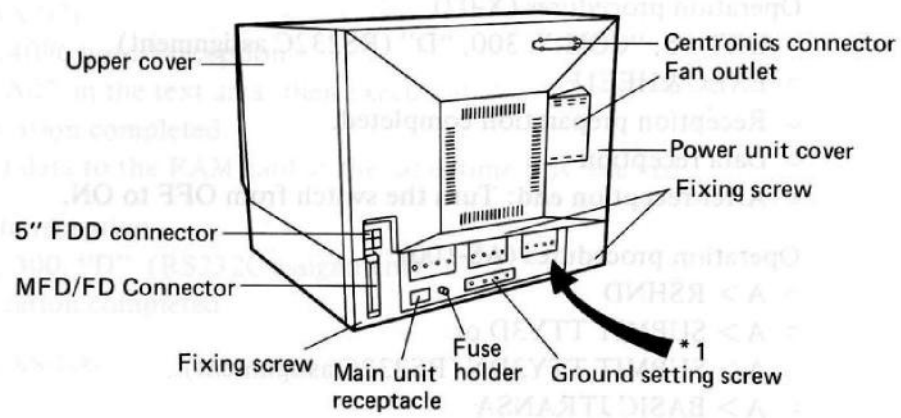
WAIT/NOWAIT: *Specifies whether null codes are stored in the buffer when there is no reception data, when data transmission to the AS-100 takes.*
WAIT: Waits for a reception interrupt.

NOAUTOX: *Specifies whether the &H11 and &H13 codes (for the select and deselect assignment) are to be ignored. This is valid only for reception.*
NOAUTOX: &H11 and &H13 are treated as data items.

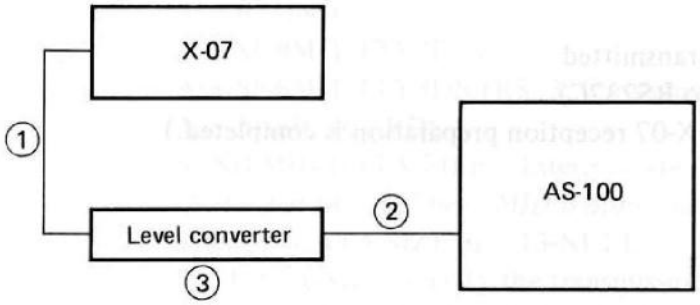
Others: For faster transfer or for 8ES transmission, specify the respective parameters in the X-07 and AS-100.

4. OTHERS

4.1 RS232C Cables and Connectors



*1: Install connector (2) in this port.

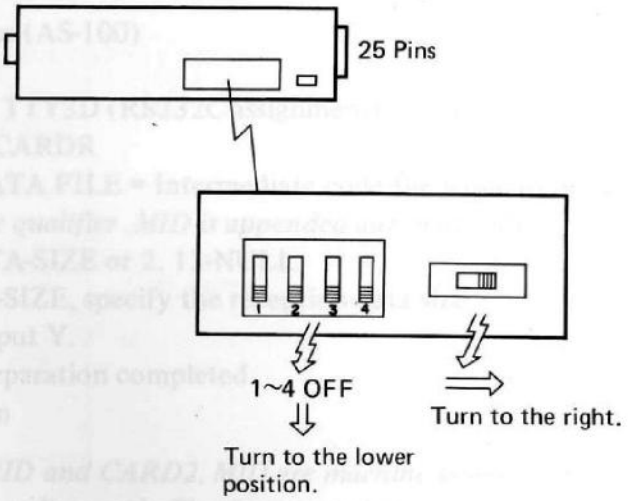


(1) X-07 associated RS232C cable (9 pins)

(2) RS232C cable used in AS-100 (25 pins)

X-07		AS-100	
1	○—○	1	PROTECTIVE GROUNDS
2	○—○	2	TRANSMITTED DATA
3	○—○	3	RECEIVED DATA
4	○—○	4	REQUEST TO SEND
5	○—○	5	CLEAR TO SEND
6	○—○	6	DATA SET READY
7	○—○	7	SIGNAL GROUND
8	○—○	8	RECEIVED LINE SIGNAL DETECTOR
20	○—○	20	DATA TERMINAL READY

(3) Switches on level converter panel



① AS-100 → X-07

Operation procedures (X-07)
- INIT #5, "COM:", 300, "D" (RS232C assignment)
- EXEC &HEE1F
- Reception preparation completed.
- Data reception
- After reception end: Turn the switch from OFF to ON.

Operation procedures (AS-100)
- A > RSHND
- A > SUBMIT TTY3D or
  A > SUBMIT TTY3DN (RS232C assignment)
- A > BASIC JTRANSA
- SOURCE = ASCII file name to be transmitted
- DESTINATION = TTY: (Output to RS232C)
- OK (Y/N) Input Y. (Input Y after X-07 reception preparation is completed.)
- Data transmission

② X-07 → AS-100

Operation procedures (X-07)
- Load the program in the text area.
- INIT #1, "COM:", 300, "D" (RS232C assignment)
- LIST #1 (Execute after AS-100 reception preparation is completed.)
- Data transmission

Operation procedures (AS-100)
- A > RSHND
- A > SUBMIT TTY3DN (RS232C assignment)
- A > BASIC JTRANSA
- SOURCE = TTY: (Input from RS232C)
- DESTINATION = ASCII file name to be received
- OK (Y/N) Input Y.
- Reception preparation completed.
- Data reception

Note: Start X-07 data transmission about in less than 10 seconds after AS-100 reception preparation is completed.

① AS-100 → X-07

Operation procedures (X-07)

①-1 RAM card data (4096 bytes) reception

- Load JCARDR. X07 in the text area, then execute it.
- Reception preparation completed.
- Transfer received data to the RAM card at the same time it is received.

①-2 Program-unit data reception

- LOAD "COM:", 300, "D" (RS232C assignment)
- Reception preparation completed.

Operation procedures (AS-100)

- A > RSHND
- A > SUBMIT TTY3D or
  A > SUBMIT TTY3DN (RS232C assignment)
- A > BASIC JCARDS
- SEND MID-DATA FILE = Intermediate code file name to be transmitted
  (Note that the qualifier. .MID is appended automatically.)
- Select 1. DATA-SIZE or 2.13-NULL.
- For DATA-SIZE, specify the transmission data size.
- OK (Y/N) Input Y. (Input after the X-07 reception preparation is completed.)
- Data transmission

Note: Select 1. DATA-SIZE for card-unit data transmission.
Select 2. 13-NULL for program-unit data transmission.

② X-07 → AS-100

Operation procedures (X-07)

②-1 RAM card data (4096 bytes) transmission

- Load JCARDS. X07 in the text area, then execute it.
  (Execute after AS-100 reception preparation is completed.)
- Data transmission

②-2 Transmitting a program from the text area

- SAVE "COM:", 300, "D" (RS232C assignment)
  (Execute after the AS-100 reception preparation is completed.)
- Data transmission

Operation procedures (AS-100)

- A > RSHND
- A > SUBMIT TTY3D (RS232C assignment)
- A > BASIC JCARDR
- RECEIVE DATA FILE = Intermediate code file name to be received
  (Note that the qualifier .MID is appended automatically.)
- Select 1. DATA-SIZE or 2. 13-NULL.
- For 1. DATA-SIZE, specify the reception data size.
- OK (Y/N) Input Y.
- Reception preparation completed.
- Data reception

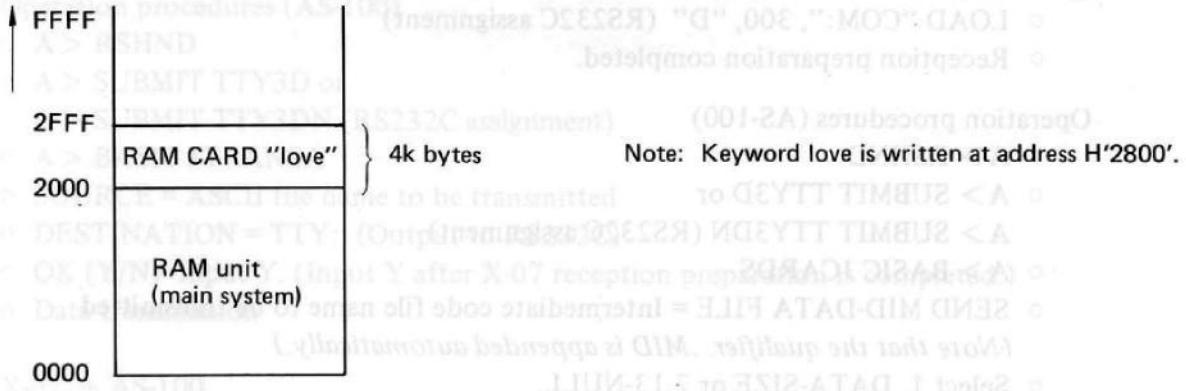
Note: CARD1. MID and CARD2. MID are machine language programs and are referred to as intermediate code files for convenience.

4.4 CARD1. MID, CARD2. MID Preparations and Procedures

Store CARD1.MID or CARD2.MID in the RAM card (with 4k byte backup) by using JCARDR.X07 on the X-07 side with the ROM/RAM switch set to ROM.

4.4.1 CARD1. MID outline

This program (machine language) is written in the 4k byte RAM card to check devices such as the RAM.
Before control is transferred to the BASIC, execute this program by use of the love keyword.



4.4.2 CARD1. MID generation

1. Load the JCARDR. X07 in the normal X07, then execute the program. (ROM/RAM switch must be set to ROM.)
2. Transfer the CARD1. MID from the AS-100. (See Figure 1 for details.)
3. Initialize the text area by using the NEW command, then load the JRAM. X07 in the text area execute it.
   (When the JRAM. X07 is executed, parameters are written in the CARD1. MID and keyword love is prepared. (See Figure 2 for details.)

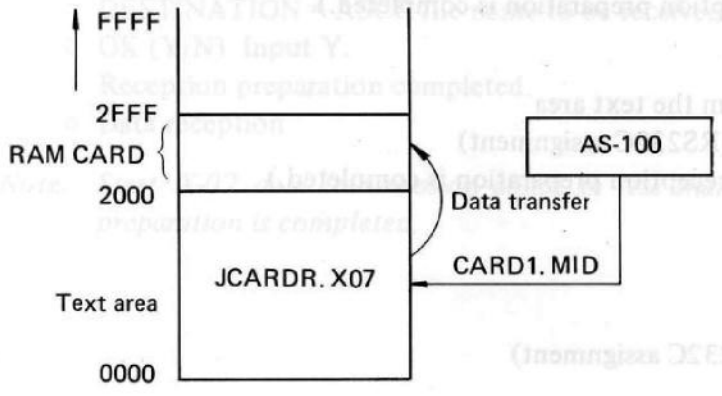


Fig. 1

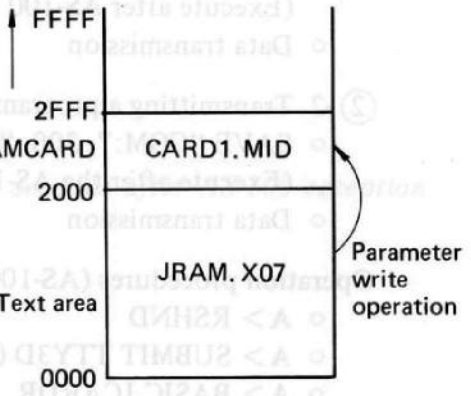


Fig. 2

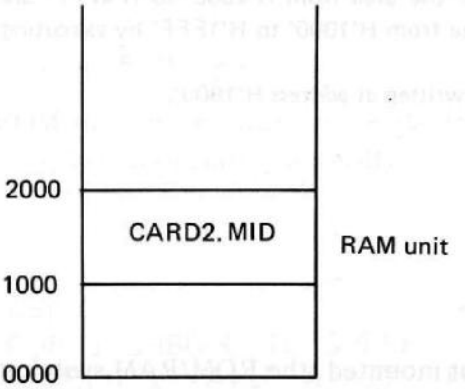
4.4.3 CARD1. MID execution

1. Install the RAM card generated as explained in Section 4.4.2 in the X-07 to be checked. The ROM/RAM switch of the X-07 must be set to ROM.
2. Turn power on, then control is passed to the CARD1. MID in the RAM card through keyword love.
3. To terminate the program, press the RESET switch.
4. To output a report, remove the RAM card from the X-07 to be checked, then install it in the normal X-07 and execute the JRAM. X07.

Note: Execute the FSET4096 before generating the CARD1. MID.

4.4.4 CARD2. MID outline

This program (machine language) is written in the RAM to check the RAM card and 8k byte option chip.
Execute this program by use of keyword love before control is passed to the BASIC.



Note: Keyword love is written at address H'1800'.

4.4.5 CARD2. MID generation

1. Load the JCARDR. X07 in the normal X-07 and execute the program. (The ROM/RAM switch must be set to ROM.)
2. Transfer the CARD2. MID from the AS-100. (See Figure 3 for details.)
3. Initialize the text area by using the NEW command, then load the JRAM2. X07 in the text area and execute it.
   (When the program is executed, parameters are written in the CARD2. MID in the RAM card. Keyword love has not been written yet.) (See Figure 4 for details.)

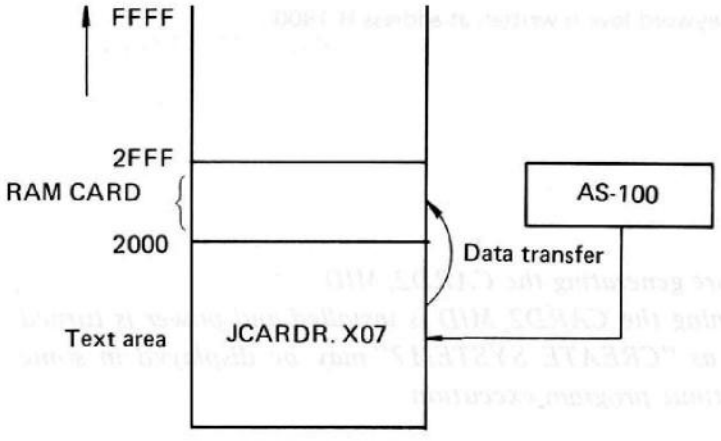


Fig. 3

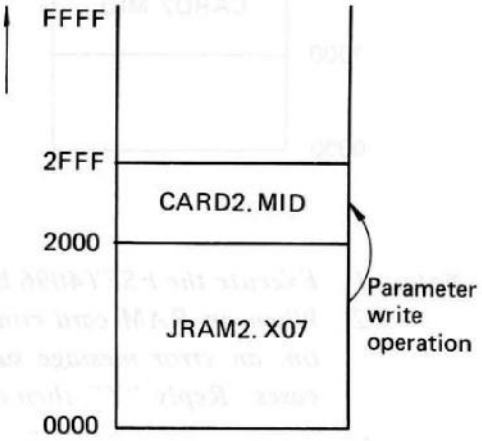


Fig. 4

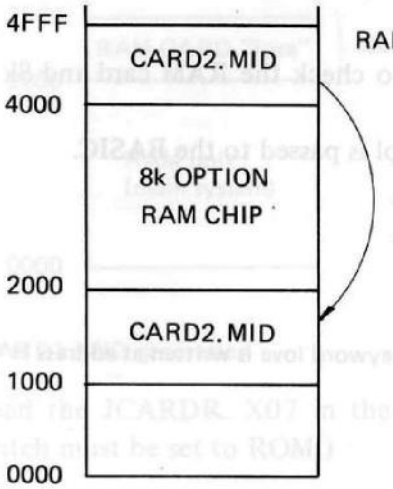
4.4.6 CARD2.MID execution

1. Install the RAM card generated as explained in Section 4.4.5 in the X-07 to be checked. The ROM/RAM switch of the X-07 to be checked may be set to ROM or RAM.

2. The following two operations are used depending on whether the 8k byte option RAM chip is mounted or not.

   2.1 When the 8k byte option RAM chip is mounted (the ROM/RAM switch must be set to RAM)

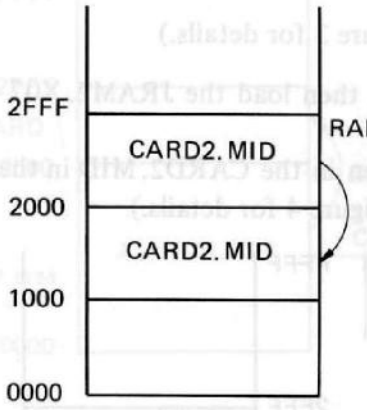
   Since the RAM card address ranges from H'4000' to H'4FFF', execute the EXEC &H4F40 and load the CARD2.MID from the RAM card in the RAM of the main system beginning from an address greater than H'1000'.



The contents of the area from H'4000' to H'4FFF' are transferred to the area from H'1000' to H'1FFF' by executing EXEC &4F40.
Keyword love is written at address H'1800'.

   2.2 When the 8k byte option RAM chip is not mounted (the ROM/RAM switch must be set to ROM.)

   Since the RAM card address ranges from H'2000' to H'2FFF', execute EXEC &H2F20 to load the CARD2.MID from the RAM card in the RAM of the main system beginning from an address greater than H'1000'.



The contents of the area from H'2000' to H'2FFF' are transferred to the area from H'2000' to H'2FFF' by executing EXEC &H2F20.
Keyword love is written at address H'1800'.

Notes: 1. Execute the FSET4096 before generating the CARD2.MID.
2. When an RAM card containing the CARD2.MID is installed and power is turned on, an error message such as "CREATE SYSTEM?" may be displayed in some cases. Reply "Y", then continue program execution.

5. SIMPLIFIED CHECKS

5.1 Simplified RAM Check (JUSER)

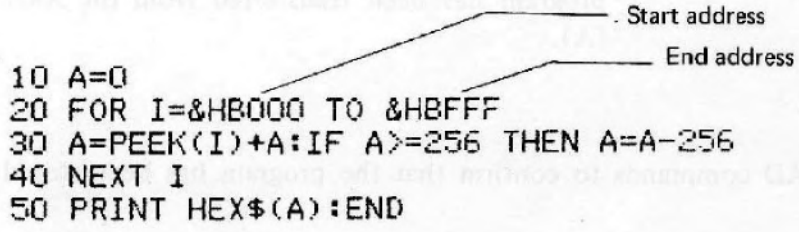
RAM read/write operations can be performed by inputting and executing the following program when the X-07 is operating normally:

```
10 CLS:PRINT"RAM CHECK":A=&H55:PRINT"DATA=";HEX$(A)
20 FOR I=&H2000 TO &H2FFF
30 POKE I,A
40 NEXT I
50 FOR I=&H2000 TO &H2FFF
60 IF A<>PEEK(I)THEN PRINT"ERR ";HEX$(I);" ";HEX$(A):END
70 NEXT I
75 PRINT"* OK"
80 IF A=&HAA THEN PRINT "*END":END
90 A=&HAA:PRINT"DATA=";HEX$(A):GOTO 20
```

See Section 2.3 JUSER.X07 for details.

5.2 Simplified ROM Check

The ROM sum check value can be obtained by inputting and executing the following program when the X-07 is operating normally:




```
10 A=0
20 FOR I=&HB000 TO &HBFFF
30 A=PEEK(I)+A:IF A>=256 THEN A=A-256
40 NEXT I
50 PRINT HEX$(A):END
```

5.3 Simplified Parallel Port Check

Connect the color graphic printer (X-710), then operate as follows:

(1) LPRINT [*1, *2] RETURN

where: *1 is the character size (1~16).
*2 is the color assignment (0 for black, 1 for blue, 2 for green, and 3 for red).

(2) Load or generate the pertinent program in the text area.

(3) LLIST RETURN

5.4 Simplified Serial Port Check

① Loop test

Arrange pin connections for the loop test in which signals are returned in the serial port or level converter in the system. (See Section 2.5 JSERI.X07 for details.)

Execute the following program:

```
10 INIT #1, "COM:", 300, 'D'
20 FOR I = 1 TO 10                    Input/output count is 10.
30 OUT #1, &H55                       Data value is 55 in hexadecimal notation.
40 PRINT HEX$ (INP (#1)); I
50 NEXT I
60 END
```

② Communication check

Connect two X-07s, then operate as follows:

X-07 (A)	X-07 (B)
1. Load or generate the program in the text area.	1. NEW RETURN
2. Execute SAVE"COM:" or SAVE"OPT:"	2. Execute LOAD "COM:" or LOAD "OPT:"
	3. Execute a LIST command to confirm that the program has been transferred from the X-07 (A).

5.5 Simplified Cassette Check

Execute the CSAVE and CLOAD commands to confirm that the program has been stored normally in the tape cassette.

1. Load or generate the program in the text area.
2. Prepare the cassette recorder for recording.
3. CSAVE"TST" RETURN
4. After SAVE is completed, rewind the cassette tape, then CLOAD? "TST" RETURN
5. If normal, the FOUND message is output and the cursor blinks.
   If abnormal or if an error occurs, the FOUND message is not output. A BAD message is output in this case.

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